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Essays on the information content of CEO letters in annual reports

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Thesis submitted for the degree of Doctor of Philosophy in Finance and Management

SOAS University of London
School of Finance and Management

May 2026

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Signed: Maysam Altarawneh

Date: 29/05/2026

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Abstract

CEO letters in annual reports are important vehicles for CEOs to communicate with shareholders and an important source of information for stakeholders, including investors, when making investment decisions. This thesis investigates the informative, predictive, and rhetorical roles of the language used in CEO letters to understand better how CEO discourse can signal underlying firm performance and influence reader interpretation. Drawing on both quantitative and qualitative approaches, this thesis contributes to the literature on corporate narrative disclosures by conducting three interconnected studies that thoroughly investigate the relationship between the information content of CEO letters and financial performance. In the first study, I analyse the association between the tone of CEO letters and financial performance and investigate whether CEO power moderates this relationship. Using a panel regression analysis, the results reveal that the tone of CEO letters is significantly and positively related to firms' current financial performance. However, this relationship becomes less pronounced as CEO power increases, suggesting that greater CEO power provides stronger support for the impression management theory.

In the second study, I investigate whether textual features derived from CEO letters can enhance the understanding of the role of textual data in assessing multi-horizon bankruptcy risks. Using a two-layer stacked Long Short-Term Memory (LSTM) model, I create three configurations that share the same architecture but differ in their input features: the first includes only financial data, the second includes only textual data, and the third integrates both. The results reveal that combining both financial and textual data yields the strongest predictive performance, suggesting that the information content of CEO letters adds incremental predictive value to the bankruptcy prediction model. More importantly, I find that although all textual features extracted from CEO letters appear to enhance predictive performance, the informativeness and individual contribution of each textual feature vary with the prediction horizon. Thus, among all textual features, the readability of CEO letters appears to contribute most to improving the detection of imminent bankruptcy risks,

representative topics for medium-term bankruptcy risks, and CEO turnover for longer-term bankruptcy risks.

In the third study, I examine whether the corpora of CEO letters from failed and active firms differ in their use of interactional metadiscourse features. The results show that the use of stance and engagement markers in CEO letters differs between failed and active firms. The communicative functions of these markers also appear to vary across both corpora. In particular, CEOs of failed firms seem to employ reactive impression strategies to restore legitimacy, while those of active firms tend to use proactive impression management strategies to maintain legitimacy. Collectively, these findings underscore the importance of the information content of CEO letters.

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Chapter 1

Introduction

1.1 Background

For decades, financial reporting and financial statements, in particular, have been an important source of information for various stakeholders, including investors and creditors (Amiram et al., 2018). Such primary sources of data on the economic activities of public firms have been widely analysed and used to assess firms' past and current financial performance, evaluate managerial performance, and make investment decisions (Palepu et al., 2021, pp. 3-11). Seminal studies, such as those of Beaver (1966), Altman (1968), and Ohlson (1980), have therefore relied heavily on financial ratios derived from financial statement items to detect financial difficulties and assess firms' likelihood of bankruptcy. However, financial statements have been argued to offer incomplete information regarding the reporting entity (Yang et al., 2018). Thus, a growing body of research suggests that although financial factors are informative about firms' financial health, relying solely on them is insufficient to provide a complete picture of firms' performance, their leadership, strategies, or prospects, as financial statement numbers cannot capture such aspects (e.g., Lev, 2018; Tang et al., 2020). In addition, quantitative information alone has been reported to limit investors' ability to assess and predict future performance, as it lacks qualitative explanations of firms' outlooks and expectations for the future economic environment (Huang et al., 2018). Therefore, there appears to be a broad consensus in the literature that quantitative data alone offer investors an incomplete picture and a partial understanding of a firm's financial performance.

It is in this context that textual analysis in accounting and financial reporting, particularly in narrative disclosures, has attracted greater attention in the accounting and finance literature in the last decade (Ataullah et al., 2018; Bochkay et al., 2023). Thus, there has been growing interest in the qualitative information disclosed by firms, which managers use as a means of communication, known as annual report narratives (Fisher et al., 2019; Hasan, 2020; Nguyen et al., 2025). Annual report narratives, which supplement financial statements, provide managers with a channel to convey contextual information about their firm's performance and position to external users, bridging the gap between the firm's financial statement figures and their underlying business principles (Merkley, 2014).

The importance of narrative disclosures stems from multiple factors. First, the scope of narrative disclosures extends beyond what is conveyed by the financial statements, which assists diverse stakeholder groups in interpreting, contextualising, and evaluating firms' financial performance (Baginski et al., 2018; Michelon et al., 2022). In addition, the information content of various forms of narrative disclosures has consistently been shown to be particularly informative for predicting future performance, assessing bankruptcy risks, and thus providing incremental information beyond the reported financials (e.g., Boudt et al., 2018; Elsayed and Elshandidy, 2020; Huang et al., 2023). Moreover, annual report narratives have attracted considerable interest not only in academic research and literature but also from professional bodies. For instance, in the UK context, the Financial Reporting Council (FRC) first developed and published guidance on strategic reports in 2014, then consistently revisited and updated it over the years, especially in 2018 and 2022, to include guidance on narrative reporting. The guidance aims to advise firms on the best approaches to narrative disclosures, ensuring shareholders receive a holistic and meaningful picture of the firm's performance, future prospects, business model, strategies, and risks (FRC, 2022). Therefore, it is evident that annual report narratives are particularly important for conveying material information to external users, helping them allocate resources more effectively, and assess directors' stewardship. This is further supported by various scholars who suggest that

firms routinely use narrative disclosures to supplement financial reports and meet investors' informational needs regarding underlying financial performance and other non-financial aspects, which in turn investors use as material, influencing their decision-making (e.g., Merkley, 2014; Michelon et al., 2022; Yekini et al., 2016).

CEO letters in annual reports, in particular, are among the most prominent narrative sections, which are persuasive in nature and play a key role in building credibility, instilling confidence among stakeholders, and convincing them that the firm is implementing effective strategies (Hyland, 1998). Hence, they are crucial corporate narrative documents that contain public and personal statements with significant rhetorical importance in defining how performance will be measured and evaluated; communicating corporate information, including strategic objectives; and constructing leader identity (Amernic et al., 2010, pp. 31-32; Craig and Amernic, 2019). However, such direct communications to shareholders are voluntary, and are unaudited, giving CEOs the flexibility to control the information disclosed in their letters to shareholders, which may be subject to impression management, making them interesting documents for examination (Boudt and Thewissen, 2019).

CEO letters to shareholders are considered important for various reasons. First, CEO letters are among the most widely read sections, usually positioned at the beginning of corporate annual reports and, therefore, set the tone for the entire report (Mäkelä and Laine, 2011). In addition, investors can rely on CEO letters because they are easy to read, usually short, and bear the CEO's signature, which may lead investors to consider the information contained in these letters credible (Kiattikulwattana, 2019). Hence, the information contained in CEO letters has consistently been reported to be used by investors as material, consequently, influencing their capital allocation and investment decisions (e.g., Conaway and Wardrope, 2010; Hooghiemstra, 2010). CEO letters are also essential sources of information for financial analysts, as CEOs use them to communicate corporate information, including the firm's business model, vision, and strategies, thus reinforcing management confidence (Yan et al., 2019). Moreover, CEO letters are highly rhetorical products, crafted to galvanise

support through metadiscourse features that aim to persuade readers and guide their understanding of a text (Hyland, 2005, p. 63). Therefore, the information content of CEO letters is important and influential for stakeholders.

The rest of this introduction chapter is organised as follows. Section 1.2 presents the research motivation and the main gaps identified in the literature on narrative disclosures. Section 1.3 discusses research objectives. Section 1.4 discusses research findings and contributions to the literature. Finally, section 1.5 explains the structure of the current thesis.

1.2 Research motivation and gaps

CEO letters in annual reports are of great importance, especially for their usefulness in decision-making. The information content of narrative disclosures, including CEO letters, plays a key role in how various stakeholders interpret, assess and react to corporate performance (Michelon et al., 2022). For instance, the information conveyed in these letters is generally regarded as influential for investors, who often use them in their decision-making (Hooghiemstra, 2010; Kiattikulwattana, 2019). Several studies demonstrate that CEO letters contain useful information for predicting firms' future performance (e.g., Boudt and Thewissen, 2019; Patelli and Pedrini, 2013). In addition, various scholars suggest that the explanations CEOs provide in their letters regarding firm performance and position affect analysts' forecasts (e.g., Yan et al., 2019), and have the potential to reveal how CEO mindsets operate during periods of global boom and expansion, which can improve the assessment of risk management policies and practices (e.g., Amernic et al., 2010, p. 6). Hence, an improved understanding of such matters can be relevant to various stakeholders seeking early insights into potential firms' future financial issues. Therefore, the practical significance and importance of CEO letters motivate me to examine the information they contain, as an enhanced understanding of the informative, predictive, and persuasive value

of such narratives would be of great benefit to stakeholders who use CEO letters in their decision-making.

Moreover, CEO letters are unaudited, which, to a substantial extent, gives CEOs some control over the information disclosed, allowing them to shape the communicated messages as they see fit (Boudt and Thewissen, 2019). This, in particular, motivates me to examine whether the information contained in CEO letters serves to inform external users of corporate performance and other important organisational aspects, or is used strategically to persuade them and influence their perceptions of performance, or a mix of both. A substantial body of research identifies and documents evidence of various impression management strategies employed in annual report narratives. These include the concealment of negative outcomes through rhetorical manipulation (e.g., Aerts and Yan, 2017), minimising disclosures on poor performance and emphasising positive outcomes (e.g., Leung et al., 2015; Yang and Liu, 2017), manipulating the tone and readability of disclosures (e.g., Asay et al., 2018a; Fisher et al., 2019), or performance attribution, through selectivity in the presentation of information (e.g., Edgar et al., 2018), defensive framing (e.g., Berkin et al., 2025; Talbot and Boiral, 2018) and self-serving bias (e.g., Moreno et al., 2019). Therefore, a closer examination and analysis of the information contained within CEO letters would facilitate a deeper understanding of CEOs' communication practices and the purpose that such narratives serve.

Furthermore, although prior studies have examined the association between CEO letters' content and firm performance, especially focusing on textual aspects such as tone (e.g., Alshorman and Shanahan, 2022; Boudt and Thewissen, 2019), thematic content (e.g., Conaway and Wardrope, 2010), rhetorical profiles (e.g., Aerts and Yan, 2017), and other factors, including CEO characteristics (e.g., Liu and Nguyen, 2020; Park et al., 2018), they have not been generally investigated as much as other narrative disclosures. The literature, thus, contains substantially more studies investigating the informativeness and the textual characteristics of other narrative outlets, such as 10-K filings (e.g., Bakarich et al., 2019;

Lopatta et al., 2017), management discussions and analysis (MD&A) sections (e.g., Bochkay and Levine, 2019; Huang et al., 2023), earnings conference calls (e.g., Druz et al., 2020; Heinrichs et al., 2019) and earnings press releases (e.g., Arslan-Ayaydin et al., 2016; Boudt et al., 2018; Huang et al., 2018). In addition, most existing research on CEO letters has largely focused on non-UK-based firms (e.g., Abrahamson and Amir, 2006; Conaway and Wardrope, 2010; Hooghiemstra, 2010; Kiattikulwattana, 2019; Yan et al., 2019). Therefore, given that CEO letters in annual reports are an important vehicle for managers to communicate their attitudes, values and the firm's performance, which deserve a much closer examination than has been conducted previously (Amernic et al., 2010, p. 5), these reasons collectively motivate me to examine the information content of CEO letters of UK firms.

After reviewing prior research on annual report narratives, particularly CEO letters to shareholders, a few gaps in the accounting and financial reporting literature are identified as follows. First, numerous studies in the literature have investigated the language used in narrative disclosures, particularly focusing on disclosure tone, and whether it is associated with firm performance (e.g., Boudt et al., 2018; Du and Yu, 2021; Rahman, 2019). In addition, the role of CEO-specific characteristics, such as CEO power, has been argued to influence tone at the top significantly (e.g., Amernic et al., 2010, pp. 31-33) and, thus, shape how managers interpret firm performance and express tone in their communications (e.g., Ning et al., 2025). However, although several studies investigate the relationship between CEO power and financial performance (e.g., Veprauskaitė and Adams, 2013), as well as the association between CEO power and narrative disclosure tones (e.g., DeBoskey et al., 2019), examining whether CEO power moderates the relationship between the tone of CEO letters and firm performance remains limited, especially in UK-based studies. Such an investigation is particularly informative, as it helps clarify whether CEOs' tones serve a communicative and informative purpose or merely pursue a strategic objective, depending on CEOs' power.

Second, prior studies have consistently reported that discretionary narrative disclosures contain useful information beyond the numbers reported in financial statements, provide explanatory power in predicting corporate failure up to a few years prior to failure, and thus improve the accuracy of bankruptcy prediction models when integrated with financial data (e.g., Elsayed and Elshandidy, 2020; Huang et al., 2023; Mai et al., 2019). However, an important question remains, leaving a significant gap in understanding the extent to which textual features derived from narrative disclosures complement financial data and improve predictive performance, particularly over multiple horizons. In other words, it remains unclear whether all textual features extracted from narrative disclosures contribute equally to predictive performance, or whether certain textual factors are more informative than others as the prediction horizon changes. This is crucial as it improves the understanding of the role of textual data in assessing multi-horizon bankruptcy risks.

Third, despite the well-established importance of CEO letters in the literature and the clear relevance of interactional metadiscourse features to narrative disclosures (Hyland, 2005, p. 14), there remains a critical gap in understanding how CEOs employ stance and engagement markers in their discourse, especially in contexts of organisational financial distress and stability. While several studies have investigated CEO letters from a metadiscourse perspective (e.g., Aerts and Yan, 2017; Huang and Rose, 2018; Lee, 2020), few have examined whether the use of stance and engagement markers differs between CEO letters from failed and active firms. This focus is especially important given Hyland's (1998) assertion that within the domain of corporate financial reporting, it would be "rewarding to examine the use of CEOs' metadiscourse in high and low performance companies, good and bad years, or different commercial sectors". Therefore, an improved understanding of the differences in the use of interactional metadiscourse markers across CEO letters of failed and active firms can provide important insights into how CEOs' communicative and linguistic styles evolve in contexts of organisational success and failure.

1.3 Research objectives

This thesis aims to investigate the information content of CEO letters in the annual reports of UK firms. To fulfil this aim, this thesis, therefore, focuses on six primary objectives. First, by controlling for firm-specific characteristics, CEO personality traits, and corporate governance (CG) mechanisms, it aims to examine the association between the tone of CEO letters and financial performance. Specifically, it aims to improve understanding of whether the tone conveyed in CEO letters is consistent with firms' current performance or serves as an impression management tool aimed at shaping and influencing stakeholders' perceptions. Second, motivated by prior research findings that CEO-specific characteristics may shape how CEOs interpret firms' performance and express tone in their communications, it aims to investigate the moderating effect of CEO power on the relationship between CEOs' tones and financial performance. Hence, it aims to examine whether CEO power influences CEOs' choice of language and tone in their letters to shareholders.

Third, it aims to investigate whether the information content of CEO letters, encompassing various textual features, provides incremental and valuable cues beyond financial ratios for predicting multi-horizon bankruptcy risks. Fourth, it aims to investigate the role and individual contribution of each textual feature derived from CEO letters to predictive performance over multiple prediction horizons. In particular, it aims to improve the understanding of whether all textual features extracted from CEO letters contribute equally to predictive performance, or whether the informativeness and predictive value of features differ across prediction horizons, with some features being more informative than others.

Fifth, it aims to examine whether there are statistically significant differences in the use of interactional metadiscourse markers across CEO letters of failed and active firms.

Specifically, it aims to investigate whether CEOs' choice of interactional devices is shaped by firms' financial performance and organisational pressures. Sixth, it aims to analyse whether the rhetorical and communicative functions of the identified interactional metadiscourse

resources employed in CEO letters from failed firms differ from those of their active counterparts. Thus, it aims to investigate whether the same interactional devices used by CEOs of both failed and active firms serve different or similar rhetorical and communicative purposes.

1.4 Findings and contributions

Several intriguing findings have emerged across the chapters of this thesis, contributing to the accounting and financial reporting literature in general, and to the literature on CEO letters, in particular. In this section, I will summarise the findings of each empirical chapter and discuss how these findings contribute to the extant literature.

In the first study, Chapter 2, I employ a random effects (RE) panel regression model to examine the association between the tone of CEO letters and the financial performance of UK FTSE 350 non-financial firms from 2013 to 2021, controlling for firm-specific characteristics, CEO personality traits and corporate governance (CG) mechanisms. I also investigate whether CEO power moderates this relationship. The findings reveal that the tone of CEO letters and financial performance are positively related, suggesting that the tone of CEO letters reflects firms' actual financial performance. I also find that CEO power is positively and significantly related to the tone of CEO letters. More importantly, I find that the interaction term between firm performance and CEO power is significantly and negatively related to the tone of CEO letters, suggesting that as CEO power increases, the association between CEO letter tones and firm performance becomes less pronounced. This finding, therefore, indicates that as CEO power increases, there is greater support for the impression management theory, suggesting that the tone of CEO letters may be more subject to impression management when CEOs are powerful.

In addition, I find that certain firm-specific characteristics, personality traits and corporate governance (CG) mechanisms are significantly related to the tone of CEO letters. In particular, I find that larger firms and those with higher betas are significantly and positively associated with the tone of CEO letters. Meanwhile, leverage appears to be negatively but insignificantly related to the tone of CEO letters, indicating that while highly leveraged firms are associated with more negative tones, the association is not statistically significant. Regarding CEO personality traits, I find a significant and positive association between CEO openness to experience and the tone of CEO letters. In terms of corporate governance mechanisms, I find that the number of board meetings and the proportion of non-executive directors on the board are significantly and negatively associated with the tone of CEO letters. Notably, I find a significant positive relationship between the proportion of female directors on boards and the tone of CEO letters, which may be due to the relatively low representation of females on UK boards. Lastly, although audit committees' independence and the tone of CEO letters appear to be negatively related, the relationship is not significant. This finding suggests that although a more independent audit committee may exert pressure for less positive disclosure tones, the effect is not statistically significant in this context.

Together, these findings expand the existing empirical literature, especially regarding the conflicting position in the literature on whether tones expressed in CEO letters pursue strategic objectives, such as persuading stakeholders and influencing their perceptions of firms' performance, or communicating the firms' underlying performance. This study's findings thus demonstrate that CEO power influences how CEOs express tone in their letters, suggesting that the tone of CEO letters is more likely to be subject to impression management when CEOs possess greater power.

In the second study, Chapter 3, I investigate the impact of integrating accounting-based, market-based and macroeconomic indicators with text-based features derived from CEO letters on the classification accuracy and power of a multi-horizon bankruptcy prediction

model. I use a sample of UK FTSE 350 non-financial firms from 2002 to 2022, encompassing firms that have failed due to bankruptcy (filed for insolvency) and those that remained solvent and active. I develop a two-layer stacked Long Short-Term Memory (LSTM) model over a three-year rolling window to capture sequential and temporal dependencies in corporate failure patterns and to predict whether a firm is likely to remain active or fail within one, two, or three years. To investigate the predictive value of the information contained in CEO letters, I create three configurations of the two-layer stacked LSTM model that share the same architecture but differ in their input features: the first uses only financial data, the second only textual data, and the third integrates both. I also conduct a series of feature ablation tests to investigate how each textual feature contributes to improving predictive performance by omitting one textual variable at a time while keeping all other features unchanged.

The findings demonstrate that integrating textual data with financial data significantly enhances predictive performance, underscoring the incremental explanatory power of the textual information contained in CEO letters. More importantly, although all textual features extracted from CEO letters appear to enhance predictive performance, I find that the informativeness and importance of textual features vary with the prediction horizon. For instance, among all textual features, the readability of CEO letters appears to contribute the most to improving the detection of imminent bankruptcy risks (one year before failure). Meanwhile, representative topics embedded in CEO letters emerge as the most informative textual variable for enhancing the identification of medium-term bankruptcy risks (two years before failure). Whereas CEO turnover appears to be the most useful text-based feature for improving the detection of longer-term bankruptcy risks (three years before failure).

Overall, this study contributes to the extant literature on corporate failure prediction and, particularly, to the literature on annual report narratives in several ways. First, empirically, it demonstrates that the information content of CEO letters is an important factor in assessing multi-horizon bankruptcy risks. In particular, the findings indicate that the information

contained in CEO letters, when integrated with financial data, can be used to predict corporate failure with high accuracy up to three years before the firm's actual failure. Second, it extends the empirical literature on the usefulness of textual information conveyed in narrative disclosures by demonstrating that the predictive value and informativeness of textual features extracted from CEO letters differ across financial distress stages and prediction horizons. Specifically, this study's findings clarify and improve the understanding of the role of multiple textual dimensions of CEO letters in assessing multi-horizon bankruptcy risks. In practice, an improved understanding of such matters facilitates more informed decision-making for various stakeholders, allowing them to better prioritise and include specific textual aspects of CEO letters in their evaluations, depending on the prediction horizon of their interest. Lastly, this study also contributes to the bankruptcy prediction literature by demonstrating the effectiveness of LSTM networks for multi-horizon corporate failure prediction and their ability to capture long-term dependencies in sequential data.

Finally, in the third study, Chapter 4, using a comparative corpus-based research methodology, I examine the use of interactional metadiscourse markers in CEO letters, comparing those of UK FTSE 350 non-financial failed and active firms between 2018 and 2022. The findings reveal that CEOs of failed firms tend to rely more heavily on personalising their discourse and engaging stakeholders through excessive self-mentions and engagement markers. In contrast, the CEOs of active firms appear to use hedges, boosters and attitude markers more frequently in their discourse. Interestingly, although CEOs of both failed and active seem to draw on similar interactional devices, the rhetorical and communicative functions that these resources serve vary across the two corpora. For instance, the CEOs of failed firms seem to use interactional metadiscourse features reactively to persuade stakeholders and restore organisational legitimacy. Whereas the CEOs of active firms tend to employ them proactively to gain stakeholders' support, project credibility, and maintain legitimacy. Collectively, these findings underscore that CEO letters

are highly rhetorical products that include non-propositional material, which CEOs appear to use strategically to fulfil different rhetorical objectives and purposes.

This study contributes to the literature on corporate narrative disclosures, particularly CEO letters, in several ways. First, it fills an underexplored gap in the literature and contributes to understanding how financial performance and organisational pressures shape CEOs' rhetorical and communicative practices in their letters to shareholders. It also contributes to the extant literature on legitimacy and impression management theories by demonstrating how CEOs manage stakeholders' perceptions during periods characterised by different financial performance and organisational pressures. In particular, the findings reveal that although CEOs of both failed and active firms may use the same interactional metadiscourse devices, these resources serve different rhetorical purposes across the two corpora, depending on whether CEOs aim to restore or maintain legitimacy through the text. This study, therefore, offers stakeholders, including investors and regulators, important insights into how managerial communication styles evolve in contexts of organisational success and failure.

1.5 Structure of the thesis

This section presents the structure of the current thesis, which comprises five chapters.

Chapter 2 investigates the relationship between the tone of CEO letters and the financial performance of UK firms and examines whether this association is moderated by CEO power. This chapter also reviews the relevant literature on annual report narratives and disclosure tone, identifies gaps in the literature, formulates the research hypotheses, outlines the research methodology, presents the results, and discusses the research findings and their implications. Chapter 3 investigates the explanatory power and role of the textual features derived from CEO letters in assessing multi-horizon bankruptcy risks. This chapter also discusses theoretical considerations and relevant prior literature on narrative disclosures and corporate failure prediction, and identifies the gaps within the extant literature. It also outlines the research methodology and the design of the two-layer stacked LSTM model, and presents the evaluation metrics used to assess predictive performance. Lastly, this chapter discusses the research findings and their implications. Chapter 4 analyses the use and communicative functions of the interactional metadiscourse resources employed in CEO letters, comparing those of failed firms and active firms. This chapter also discusses the analytical framework, identifies the gaps within the literature, outlines the research methodology, presents the results, and discusses the research findings and their implications. Lastly, chapter 5 summarises the findings of this thesis, discusses the practical implications, highlights the limitations and provides suggestions for future research.

Chapter 2

The moderating role of CEO power in the relationship between CEO letter tone and firm performance: UK evidence

2.1 Introduction

Over the years, there has been a growing interest in textual analysis in the accounting and financial reporting literature (Ataullah et al., 2018). In particular, there has been an increased focus on examining firms' disclosures and how their choices of financial communication impact investors and market outcomes (Blankespoor, 2018). Therefore, various studies have investigated firms' qualitative disclosures, referred to as annual report narratives, which managers use to communicate with shareholders and provide information beyond that presented in financial statements (e.g., Asay et al., 2018a; Bassyouny et al., 2020; Merkley, 2014). Narrative disclosures play an essential role in communicating managerial insights to investors, helping them better understand firms' performance, position, and future prospects (Michelon et al., 2022). Corporate narratives, therefore, provide investors with valuable information, as quantitative data alone does not offer a complete picture of firms' financial performance and future direction (Arslan-Ayaydin et al., 2016).

CEO letters to shareholders are among the most widely read narrative disclosures and have consistently been recognised as an essential source of information for institutional and private investors when making investment decisions (Kiattikulwattana, 2019; Patelli and Pedrini, 2013). Such narrative sections of annual reports constitute a set of complex communicative acts that carry multiple overtones, in which the language used serves a strategic form of "sense-making" (Amernic et al., 2010, p. 15). CEO letters allow CEOs to communicate important information regarding organisational performance, future direction, business strategy, and to rationalise corporate events (Segars and Kohut, 2001). However,

they are unaudited, allowing CEOs to shape the messages communicated using the language they see fit, which could lead to biased reporting (Boudt and Thewissen, 2019). Therefore, CEO letters may serve as a strategic vehicle for communicating information that could influence readers' perceptions of the firm's underlying performance.

The use of managerial tone in annual report narratives, in particular, has been the subject of various accounting research studies (Ning et al., 2025). Narratives' tone refers to the language used in corporate communications, and the use of positive and negative words (Henry and Leone, 2016). According to González et al. (2021), the tone of corporate communications is as important as the amount of information disclosed to the markets. The tones that managers employ in their narrative disclosures can, thus, serve different purposes. For instance, it may offer additional information that mitigates the information asymmetry problem (Martikainen et al., 2023), while also serving as a strategic tool aimed at influencing stakeholders' perceptions of firms' performance (Boudt and Thewissen, 2019). Several studies investigate the tone conveyed in narrative disclosures, shedding light on how managers manipulate it (e.g., Huang et al., 2014) and on the factors that drive it (e.g., Fisher et al., 2019). For instance, Boudt and Thewissen (2019) examine CEO letters and report that managers manipulate tone by strategically positioning positive and negative words within the letter to foster a more positive perception of the underlying message. Caserio et al. (2019) investigate the tone employed in MD&As and find that managers of unhealthy firms use more overly positive and forward-looking words to manage stakeholders' opinions than managers of healthy firms. In addition, studies analysing the tone of earnings press releases, such as those by Arslan-Ayaydin et al. (2016) and Arslan-Ayaydin et al. (2020), find that equity-based incentives and career concerns motivate managers to adopt a more positive tone to convey an optimistic picture of firms' financial performance.

Moreover, considering CEO-specific characteristics, Marquez-Illescas et al. (2019) explore the influence of CEO personality traits on corporate disclosures by analysing the tone of earnings announcements and show that more narcissistic CEOs are more inclined to exhibit

a positive bias when assessing and describing performance. Similarly, Liu and Nguyen (2020) investigate whether CEOs' characteristics influence their choice of language in their letters and document that overconfident CEOs tend to employ an overly positive tone inconsistent with current financial performance. Further, DeBoskey et al. (2019) examine the impact of CEO power on the tone of earnings announcements and find that powerful CEOs use a more positive and aggressive tone. Thus, CEO-specific characteristics, and CEO power in particular, have been shown to influence tone at the top (Amernic et al., 2010, pp. 31-33) and, thus, motivate managers to adopt a more positive tone (Cao et al., 2022; González et al., 2021).

The present study expands upon and contributes to the vast body of research on annual report narratives by examining the relationship between CEO letter tones and firms' financial performance and whether this relationship is moderated by CEO power. Two primary rationales drive this investigation. First, while there are studies that investigate the association between CEO power and disclosure tone (e.g., DeBoskey et al., 2019) as well as CEO power and financial performance (e.g., Veprauskaitė and Adams, 2013), limited attention has been given to whether CEO power moderates the relationship between CEO letter tones and financial performance. Such an examination would provide valuable insights into whether CEOs' tones serve a communicative and informative purpose or merely pursue a strategic objective, and whether CEO power plays a role in understanding that.

Second, several studies investigating the relationship between managerial power and disclosure tone mainly focus on non-UK firms (e.g., Cao et al., 2022; DeBoskey et al., 2019), with a few studies on the relationship between CEO letter tones and financial performance, and whether this relationship is moderated by CEO power, using UK firms. This gap further motivates my investigation of this relationship in UK firms, which operate under different corporate governance systems than those in other countries.

Therefore, in the present study, using a sample of FTSE 350 non-financial UK firms, I examine the association between the tone of CEO letters and firms' financial performance,

and whether CEO power moderates this association. To capture CEO power, I employ principal component analysis (PCA) to construct a CEO power index from five CEO power-related variables, while controlling for firm-specific characteristics, CEO personality traits, and corporate governance mechanisms.

The findings reveal that the tone of CEO letters is significantly and positively related to firm performance, suggesting that the tone conveyed reflects firms' actual financial performance. However, this relationship becomes less pronounced as CEO power increases, suggesting that CEO power influences CEOs' language choices. Therefore, as CEO power increases, the tone of CEO letters is more likely to be subject to impression management. Overall, this study contributes to the understanding of whether the credibility of CEO letter tones change when CEOs are powerful. An improved understanding of whether CEO power influences CEOs' choice of tone provides valuable insights to various stakeholders, including investors and regulators, helping them better interpret CEO letters, and CEOs' tones in particular, by accounting for CEO power to avoid misjudging firms.

The remainder of the paper is structured as follows: Section 2 reviews the relevant literature and formulates the research hypotheses. Section 3 outlines the research methodology. Section 4 presents the empirical results. Section 5 discusses the research findings and their implications. Section 6 concludes the paper.

2.2 Theoretical background and hypotheses development

Several theories have emerged in accounting and finance research to explain why managers voluntarily disclose information and how they do so. These include economic-based, sociological/organisational, and behavioural and communication theories. While these theories differ in terms of managerial motivations, incentives, behaviours, and audiences, they have been widely used to explain why and how managers use narrative disclosures.

2.2.1 Theoretical literature

The agency theory, developed by Jensen and Meckling (1976), is one of the dominant theories used to explain the relationship between managers and shareholders, as well as managerial motivations for voluntarily disclosing information. The agency theory focuses on the relationship between agents (managers) and principals (shareholders), characterised by contractual obligations and utility maximisation, in which both agents and principals are perceived as rational individuals who make decisions based on their own best interests (Merkl-Davies and Brennan, 2011). The theory assumes that conflicts of interest exist between managers and shareholders, as managers are driven to make self-serving decisions that may not necessarily be in shareholders' best interests (Schleicher and Walker, 2010).

Impression management and incremental information, two competing schools of thought, draw on assumptions of the agency model to explain why managers provide discretionary disclosures. The incremental information schools posit that managers are incentivised to provide voluntary disclosures to mitigate information asymmetry, lower the cost of capital and increase their compensation (Baiman, 1990; Healy and Palepu, 2001). Whereas the impression management school explains that managers are motivated to exploit information asymmetries and behave opportunistically, engaging in biased reporting to influence stakeholders' perceptions of firms' performance and future direction (Merkl-Davies and

Brennan, 2011). Hence, conflicts of interest between managers and shareholders arising from negative organisational outcomes make managers more prone to engage in impression management, thereby influencing stakeholders' perceptions of firms' performance (Aerts, 2005).

While impression management theory was initially introduced in the psychology literature and research on human behaviour, it has been predominantly applied in accounting research to explain corporate narrative disclosures (Clatworthy and Jones, 2003). Hooghiemstra (2000) defines impression management as an area of social psychology associated with individuals' attempts to make a good impression on others. In the context of corporate disclosures, impression management is driven by managerial desire to portray a self-serving view of firm performance, reflecting opportunistic managerial behaviour aimed at manipulating stakeholders' perceptions rather than providing them with useful information (Moreno et al., 2019). Managers are thus assumed to manipulate the content and presentation of information in annual report narratives, leading to biased reporting that misleads and influences investors' interpretation of organisational outcomes (Merkl-Davies and Brennan, 2011).

A substantial body of research documents various forms of impression management used by managers in narrative disclosures to shape stakeholders' perceptions of a firm's performance and future prospects, either by concealing negative outcomes or by performance attribution. These include tone manipulation (e.g., Huang et al., 2018), textual complexity (e.g., Asay et al., 2018a), minimal disclosure of poor performance (e.g., Leung et al., 2015), emphasis on positive outcomes (e.g., Yang and Liu, 2017), rhetorical framing (e.g., Berkin et al., 2025), selective presentation of information (e.g., Edgar et al., 2018), and self-serving bias (e.g., Moreno et al., 2019). These strategic and opportunistic behaviours of managers have given rise to the obfuscation hypothesis (Courtis, 1998), which suggests that managers are not neutral in their narratives as they tend to obfuscate negative information and emphasise positive information (Courtis, 2004; Taurangana and Chong, 2004).

The stakeholder theory has also been widely used to explain narrative disclosures (Asay et al., 2018b; Rutherford, 2003). Due to the challenges posed by information asymmetry, firms signal certain information to shareholders to convey their superiority over competitors, thereby establishing a positive corporate image and reputation (Stern and James, 2016). Smith and Taffler (1992) have employed signalling theory within the context of the obfuscation hypothesis and find that firms tend to actively signal positive information while intentionally obscuring messages conveying negative information. Hence, while the agency theory focuses on firms with poor financial performance, the signalling theory focuses on the managerial behaviour of well-performing firms that intend to signal that they are better than other firms through enhanced transparency and presentation of information (Merkl-Davies and Brennan, 2008).

The legitimacy theory has also been used in accounting and finance research to explain corporate reporting practices (e.g., Berkin et al., 2025; Elsayed, 2025; Li et al., 2023). Within the legitimacy theory, narrative disclosures are assumed to be used to influence perceptions of organisational legitimacy and to preserve a firm's reputation as a legitimate organisation (Villiers and Van Staden, 2006). The legitimacy theory holds that firms enter into a social contract with society and therefore, organisational survival depends on operating within societal norms and boundaries (Dowling and Pfeffer, 1975). Hence, organisational legitimacy is critical for firms' survival and success, and when firms are subject to legitimacy pressures from the surrounding environment, they may adopt several strategies to gain, maintain, and/or repair legitimacy (Suchman, 1995). Impression management strategies, in particular, have been extensively employed in the literature to examine how firms facing legitimacy threats use narrative disclosures to manage organisational legitimacy. A substantial body of research documents different forms of impression management strategies in annual report narratives to legitimise during periods of uncertainty, including the use of assertive and defensive strategies, rhetorical association and disassociation strategies, attribution of

positive and negative outcomes, and selectivity in disclosing information regarding firm performance (e.g., Berkin et al., 2025; Edgar et al., 2018; Li et al., 2023).

Stakeholder theory is another widely used theoretical framework in corporate communication studies (e.g., Chiu and Wang, 2015; Lu and Abeysekera, 2017). Stakeholder theory and legitimacy are similar in positioning firms within a broader social structure in which other communities in society exert influence and are affected by the firm, and in recognising that corporate disclosures respond to the requirements and expectations of diverse stakeholder groups (Zamil et al., 2023). However, stakeholder theory focuses on a wide range of stakeholders, whereas legitimacy theory focuses on societal expectations. Impression management has also been employed as a theoretical framework to examine how firms manage the perceptions of various stakeholder groups. For instance, Bozzolan et al. (2015) examine the tone employed in a firm's press releases, news articles, and analyst reports and find that tones differ across disclosures depending on the salience of the stakeholders, suggesting that the firm's management strategically manipulates tone to manage perceptions of different stakeholder groups.

In addition, the institutional theory assumes that firms tend to comply with institutional norms and rules to meet institutional expectations (DiMaggio and Powell, 1983). Meeting institutional expectations allows firms to gain legitimacy (Suchman, 1995), which in turn allows them to minimise inspections and evaluations that can reveal events and deviations that compromise organisational legitimacy (Meyer and Rowan, 1977). Organisations, therefore, can respond to institutional pressures through corporate disclosures (Haque and Jones, 2020). From the perspective of institutional theory, managers are motivated to engage in impression management and thus influence stakeholders' perceptions to ensure organisational legitimacy (Bansal and Clelland, 2004).

Although the annual report narratives and managerial communication practices have been extensively explained using impression management theory in the accounting and finance literature, other studies challenge the premise that annual report narratives are strategic

actions aimed at exerting influence. Yuthas et al. (2002), in particular, challenge the obfuscation hypothesis of impression management theory and argue that narrative discourses may serve as communicative actions intended to foster mutual understanding rather than as strategic actions aimed at influencing stakeholders' decisions. Grounded in Habermas' principles of communicative action (1984, 1987), their findings suggest that firm communications are more communicative than strategic. Dhanani and Connolly (2015) also examine non-governmental organisations and report that organisations provide truthful disclosures characterised by communicative actions aimed at enhancing understandability. Similarly, Patelli and Pedrini (2013) examine the relationship between firm performance and optimistic tones employed in CEO letters and find that firms tend to engage in communicative actions intended to dialogue with shareholders through sincere disclosures. The authors argue that CEO letters are "routine" disclosure vehicles and that false portrayals of performance are not economically viable, as they would jeopardise organisational legitimacy. However, several studies provide evidence of different impression management strategies employed by CEOs in routine disclosure vehicles, such as CEO letters, to shape stakeholders' perceptions, respond to organisational pressures and restore legitimacy (e.g., Beelitz and Merkl-Davies, 2012; Berkin et al., 2025; Boudt and Thewissen, 2019).

2.2.2 Hypotheses development

According to the preceding discussion, the literature appears to be divided on whether narrative disclosures serve as strategic actions intended to influence stakeholders' perceptions or as communicative actions aimed at dialoguing with stakeholders to reach mutual understanding.

Several studies have examined the relationship between managerial tone and firms' financial performance, yielding inconsistent findings regarding whether narrative disclosure tone reflects actual firm performance or is used for manipulative purposes (e.g., Caserio et al., 2019; Huang et al., 2018; Li et al., 2019). For example, Huang et al. (2020) examine the narrative content of earnings press releases and find that managers strategically inflate the tone to portray a more positive picture of the firm's underlying financial performance. Huang et al. (2018) also investigate earnings press releases and demonstrate that managers strategically employ tone as a complement to earnings management to influence investors' opinions. In addition, Caserio et al. (2019) examine the tone employed in MD&As, comparing that of healthy firms to that of unhealthy firms, and report that managers of unhealthy firms strategically manipulate tone by using overly positive and forward-looking words more than those of healthy firms. Bozzolan et al. (2015) analyse different forms of corporate communications for a particular firm, including press releases, news articles, and analyst reports, and find that the optimism of disclosures varies across stakeholder groups depending on their salience, indicating that managerial tone is primarily used to shape stakeholders' perceptions.

Meanwhile, in their study on the information content of MD&As, Li et al. (2019) find that positive tone is associated with future performance, suggesting that the tone employed is used for informative rather than manipulative purposes. Similarly, Patelli and Pedrini (2013) examine CEO letters and find that a positive tone is associated with both past and future

performance, supporting the view of CEO letters as communicative rather than strategic actions.

Overall, the literature on managerial tone in narrative disclosures appears mixed, with some studies finding that tone communicates actual firm performance (e.g., Dhanani and Connolly, 2015; Li et al., 2019), and others demonstrating that it is used for strategic impression management (e.g., Arslan-Ayaydin et al., 2016; Huang et al., 2014). Although this body of work has significantly improved the understanding of managerial tone in annual report narratives, it has predominantly focused on specific corporate communications, such as earnings press releases (e.g., Arslan-Ayaydin et al., 2020; Huang et al., 2018; Marquez-Illescas et al., 2019) and MD&As (e.g., Caserio et al., 2019; Liu and Nguyen, 2020). In addition, while several studies investigate the tone of CEO letters, they mainly focus on U.S. firms (e.g., Boudt and Thewissen, 2019; Liu and Nguyen, 2020; Patelli and Pedrini, 2013). However, the UK and the US operate under different corporate governance systems. Hence, although the UK's corporate governance system is more flexible than the US's, it encompasses distinct regulations, rules and principles that shape corporate communication practices. This gap, therefore, motivates examining the relationship between the tone of CEO letters and the financial performance of UK firms. As a result, a hypothesis of the relationship between the tone of CEO letters and the financial performance of UK-listed firms is supported. A closer examination of the association between the tone of CEO letters and the financial performance of UK-listed firms, along with additional analysis, can yield a more precise determination of whether CEO letter tone pursues strategic or communicative objectives. Because of the conflicting position in the literature, I state the first hypothesis in the null form:

H_1 : There is a significant relationship between the tone of CEO letters to shareholders and the financial performance of UK-listed firms.

Given that CEO letters are unaudited, the message communicated in such narratives and the choice of language employed by CEOs can, to a significant degree, be shaped as they see fit (Boudt and Thewissen, 2019). Research on the determinants of managerial tone has consistently emphasised the role of managerial characteristics in shaping how managers interpret firm performance and express tone in their communications (e.g., Bassyouny et al., 2020; Buchholz et al., 2018; Liu and Nguyen, 2020). CEO power, in particular, has attracted increased interest in the corporate communication literature, especially regarding disclosure tone (e.g., Cao et al., 2022; DeBoskey et al., 2019; González et al., 2021). The rationale for the relationship between CEO power and narrative disclosure tones is grounded in the observation that CEO power is positively associated with excessive risk-taking (Lewellyn and Muller-Kahle, 2012).

According to DeBoskey et al. (2019), more powerful CEOs are more likely to take advantage of the benefits associated with more optimistic tones due to their increased tolerance for risks related to litigation, regulatory actions, and reputational costs. The authors argue that while most managers tend to use positive language in their disclosures, powerful CEOs not only have stronger incentives to inflate the tone but also have greater opportunities to present an excessively optimistic picture of firms' performance. Cao et al. (2022) investigate the impact of managerial power on annual report tone and document that greater managerial power yields a more positive tone. González et al. (2021) also provide empirical evidence demonstrating that powerful CEOs can influence tone in initial public offerings (IPOs) by avoiding using negative tones and instead using more optimistic language in the information they disclose to the market.

Overall, the literature appears to support the view that CEO power influences managerial tone and their interpretation of firm performance, leading to overly positive tones expressed in their communications. While prior research investigates the relationship between CEO power and narrative disclosure tone (e.g., DeBoskey et al., 2019), as well as the relationship between CEO power and financial performance (e.g., Han et al., 2016), only few examine

the moderating effect of CEO power on the relationship between disclosure tone and financial performance, especially for CEO letters in the context of UK firms. According to Amernic et al. (2010, p. 33), CEO letters and CEO discourse more generally are underexplored in terms of how CEOs use their power to set the tone at the top and attempt to reframe organisational reality for participants. This gap, therefore, motivates examining whether CEO power moderates the association between the tone of CEO letters and financial performance, as it can provide valuable insights into whether CEO power affects the extent to which the tone conveyed in CEO letters serves communicative or manipulative purposes. This leads to the second testable hypothesis:

H_2 : CEO power moderates the significant relationship between the tone of CEO letters and the financial performance of UK-listed firms.

2.3. Methodology

2.3.1 Sample selection and data collection

The sample frame used in this study comprises UK FTSE-350 non-financial firms (see Appendix A2.1 for the list of sampled firms). I exclude financial, investment trust, insurance, and real estate firms, as they have different regulations, accounting practices and financial statement structures than non-financial firms (Schleicher and Walker, 2010). Additionally, only companies listed before January 2013 are included in the sample, as the investigation covers the period 2013 to 2021 inclusive. This selection technique gives an initial total of 182 firms. However, 62 firms are excluded: 34 because their listing dates fall outside the specified criteria above, and 28 due to data availability issues. This sampling criterion yields a final balanced panel dataset covering 9 years from 2013 to 2021 (inclusive), comprising 120 firms and 1080 observations. Additionally, all 120 firms are classified into nine primary

sectors based on the Industrial Classification Benchmark (ICB) structure and code index¹. These sectors include (1) Health Care, (2) Basic Materials, (3) Industrials, (4) Consumer Goods, (5) Consumer Services, (6) Technology, (7) Telecommunications, (8) Energy and (9) Utilities (see Appendix A2.2 for industrial classifications).

All collected data exhibit a panel structure. One of the key advantages of panel data is the possibility to account for unobserved heterogeneity by incorporating individual and time-unobserved effects in the model (Fernández-Val and Weidner, 2018). In addition, panel datasets are more efficient in econometric estimates than cross-sectional or time-series datasets (Hsiao, 2022, p. 3). Panel data, thus, allow researchers to use more observations and to control for unobserved heterogeneity, which is also a potential source of endogeneity (Ullah et al., 2018).

I chose 2013 as the starting point for the sample for several reasons. Initially, the decision recognises the significant time and effort required to manually collect CEO letters from annual reports. Looking into earlier historical periods, therefore, could further lengthen the already time-consuming collection task, especially if the possibility of missing CEO letters in particular firms' archives arises. Additionally, this time frame provides a substantial, consistent dataset of textual data, aligning with the research's objective of employing ML regressors to validate the primary results. Furthermore, this time frame allows me to cover the issuance of the 2014 strategic reporting guidance and the 2018 revised strategic reporting guidance by the Financial Reporting Council (FRC).

This study draws data from four primary sources. CEO letters to shareholders are obtained from each company's official website. However, if some annual reports are unavailable, I use Annualreports.com to obtain them. Furthermore, data for CEO power and corporate

¹ Industry Classification Benchmark (ICB) is a system administered and operated by FTSE Russell, a wholly owned subsidiary of the London Stock Exchange Group (LSEG), that enables comparisons and categorisations of firms by industry and sector. In this study, the structure and code were accessed on June 3, 2023, and downloaded from <https://www.lseg.com/en/ftse-russell/industry-classification-benchmark-icb>.

governance variables are derived from BoardEx, while financial and accounting data are obtained from Thomson Reuters.

2.3.2 Model

In this study, regression analysis is employed to examine the association between the tone of CEO letters and the financial performance of UK firms, and to assess the potential moderating impact of CEO power on this association. Therefore, I develop two models to address the research hypotheses. The first model includes the tone of CEO letters, financial performance, firm-specific characteristics, CEO personality traits, and corporate governance variables as controls. The purpose of this model is to investigate whether the tone of CEO letters and the financial performance of UK firms are significantly associated, as proposed in Hypothesis 1 (*H1*). Meanwhile, the second model introduces and includes CEO power and the interaction term between ROA and CEO power. This model examines the potential moderating effect of CEO power on the relationship between the tone of CEO letters and the financial performance of UK firms, as proposed in Hypothesis 2 (*H2*). Therefore, to explore the associations mentioned above and test the research hypotheses, I develop the following random effects (RE) regression model:

$$\begin{aligned}
 Tone_{it} = & \alpha + \beta_1 ROA_{it} + \beta_2 CEOP_{it} + \beta_3 ROA_CEOP_{it} + \beta_4 Lev_{it} + \beta_5 Fsize_{it} + \beta_6 Openness_{it} \\
 & + \beta_7 Beta_{it} + \beta_8 Bmeetings_{it} + \beta_9 Neds_{it} + \beta_{10} PropFem_{it} + \beta_{11} Acind_{it} \\
 & + \beta_{12} Industry_Dummies_i + \beta_{13} Year_Dummies_t + \mu_i + e_{it}
 \end{aligned}
 \tag{1}$$

Where Tone is the sentiment expressed in the textual content of CEO letters; α is the intercept; ROA is a proxy for firms' current financial performance; CEOP is the composite

measure obtained through the principal component analysis (PCA) of five CEO power-related variables (CEOT, CEOO, CEOR, CEONS, and CEONQ), each characterised by varying loadings and weightings; ROA_CEOP is the interaction between ROA and the CEOP index to capture the moderating impact of CEO power; Lev is the ratio of a firm's debt to its total capital; Fsize is measured as the natural logarithm of total assets; Openness is a personality trait drawn from the Big Five personality framework, reflecting the CEO's openness to experience, measured using machine learning-based linguistic analysis of CEO letters; Beta is a measure of market risk; Bmeetings is the number of board meetings in a given year; Neds is the percentage of non-executive directors on the board; PropFem is the proportion of females on the board; Acind is the proportion of independent board members on the audit committee, industry and year dummies control for differences between industries and over time; μ_i is the firm-specific random effect and e_{it} is the error term.

The analysis employs the random effects (RE) panel regression method following the execution of the Hausman and the Breusch-Pagan Lagrange multiplier (LM) tests, which yield results in favour of the random effects (RE) model over the fixed effects (FE) and pooled OLS models (see Appendix A2.3 for the results of the Hausman and LM tests), suggesting that the random effects (RE) model is valid and efficient for conducting this analysis. The Hausman test statistically assesses the correlation between the unobserved individual effect and the conditioning regressors in the model; failing to reject the exogeneity of the unobserved individual effect provides evidence in support of the random effects (RE) model, while rejecting the exogeneity assumption offers support for the fixed effects (FE) model (Amini et al., 2012, p. 2). In addition, the Breusch-Pagan Lagrange multiplier (LM) test, proposed by Breusch and Pagan (1980), is commonly used to help decide between a random effects (RE) regression model and a simple OLS regression model (Moulton, 1987). Moreover, I conduct additional tests to identify possible issues and verify the robustness and suitability of the presented model for examining the anticipated relationships. Serial correlation, also known as autocorrelation, can bias standard errors and reduce the

efficiency of outcomes in panel data models (Drukker, 2003). Therefore, I use the Wooldridge test to check for autocorrelation in the model, and the results indicate the presence of serial correlation (see Appendix A2.3 for the Wooldridge test results). According to Stock and Watson (2011, pp. 364 - 370), variables in panel data are typically autocorrelated, and standard errors need to allow for both autocorrelation and potential heteroskedasticity; one approach to do so is to use clustered standard errors. Consequently, I use clustered standard errors at the industry level to account for autocorrelation and potential heteroskedasticity.

2.3.3 Variables

2.3.3.1 Dependent variable

I measure the tone of CEO letters (Tone), the model's dependent variable, using TextBlob, a popular Python natural language processing library that performs tasks such as sentiment analysis. The primary objective of TextBlob is to facilitate access to commonly used word-processing operations through a user-friendly interface (Daniel et al., 2017). TextBlob's sentiment analysis uses machine learning (ML) to classify each sentence as positive, negative, or neutral. This classification is determined by matching the text's words to a predefined list of positive and negative terms. Hence, the sentiment score is a floating-point number in the range -1.0 to 1.0, where -1.0 indicates very negative sentiment and 1.0 indicates very positive sentiment. TextBlob has been widely used in accounting and finance research, particularly in studies that extract polarity and sentiment scores from narrative disclosures (e.g., Bochkay et al., 2023; Lagasio, 2024; Siano and Wysocki, 2021). To prepare CEO letters for tone measurement, all letters are preprocessed and cleaned by converting them and minimising their complexity. The cleaning of CEO letters involves removing punctuation, stop words, lowercasing, lemmatising, and stemming (see Appendix A2.4 for a demonstration of the data cleaning process). The overall objective of the data

preprocessing and cleaning stage is to transform raw text into a format that TextBlob can use for sentiment analysis (Akstinaite et al., 2021).

2.3.3.2 Performance variable

I measure firms' financial performance using Return on Assets (ROA), which serves as the model's independent variable. ROA is a widely recognised accounting metric used to assess a company's current financial performance by measuring profitability relative to total assets, which has been used extensively in the literature studying the relation between disclosure tone and firms' financial performance (e.g., Boudt and Thewissen, 2019; Fisher et al., 2019; Martikainen et al., 2023). However, the existing research on the association between ROA and disclosure tone has shown inconsistent findings. While some studies report insignificant associations (e.g., Albitar et al., 2023), others find a significant and positive relationship between disclosure tone and ROA (e.g., Aly et al., 2018; Jain et al., 2023; Lu et al., 2019). Therefore, in the present study, I control for ROA as a proxy for current financial performance. However, due to conflicting positions in the literature, my expectations regarding the relationship between the tone of CEO letters and ROA remain unclear.

2.3.3.3 Moderator variable

To capture CEO power (CEOP), I use principal component analysis (PCA) to construct a CEO power index (CEOP) from five CEO power-related variables, integrating and capturing different aspects of CEO power. The PCA approach to capture CEO power has been extensively employed in accounting and finance research (e.g., Adams et al., 2024, 2025; Islam et al., 2024). In his seminal work, Finkelstein (1992) categorises four distinct types of CEO power: structural power, ownership power, expert power, and prestige power. Although devising a comprehensive measure of CEO power is challenging, the existing literature extensively examines and uses various indicators to proxy for CEO power, including the CEO's tenure to measure expert power (e.g., DeBoskey et al., 2019; Park et al., 2018), CEO's share ownership to measure ownership power (e.g., Jia et al., 2022; Muttakin et al., 2018), CEO's remuneration to measure structural power (e.g., Li et al., 2018; Song and Wan, 2019), CEO's social network size to measure prestige power (e.g., Altunbaş et al., 2020; Fralich and Papadopoulos, 2018), and the CEO's educational background and number of qualifications to measure prestige power (e.g., Haynes et al., 2019; Lisic et al., 2016). Based on previous research findings on the impact of CEO power on managerial tone (e.g., DeBoskey et al., 2019; González et al., 2021), I expect CEO power to significantly moderate the relationship between the CEO letter tones and financial performance.

CEO tenure is an important observable CEO characteristic which has received considerable attention in the management, accounting, and finance literature and has been used as a proxy for various aspects, including expert power and, thus, CEO power (Darouichi et al., 2021). The years the CEO had served on the board reflect the CEO's ability to influence others within the firm and mobilise resources (Walls and Berrone, 2017). According to Veprauskaitė and Adams (2013), the length of a CEO's tenure plays an important role in the development of his/her decision-making autonomy and power within the firm. Hermalin and Weisbach (1998) suggest that as CEOs' tenure increases, their power grows, as they can influence the board's selection process. Hence, when directors are appointed during a

CEO's tenure, they tend to feel a sense of obligation to the CEO, as CEOs are heavily involved in the recruitment, nomination, and appointment of directors, which may reduce the likelihood of those directors disciplining the CEO and, thus, lead to less effective board monitoring (Fan et al., 2019). In addition, Lewis et al. (2014) find that CEO tenure impacts firms' likelihood of disclosing information, and firms led by recently appointed CEOs exhibit a significantly higher tendency to acquiesce than those led by long-tenured ones. Moreover, DeBoskey et al. (2019) document that disclosure tone and CEO tenure are significantly and positively related. In general, two different perspectives exist concerning CEOs' tenure. On the one hand, several scholars provide evidence demonstrating that long-tenured CEOs negatively impact organisational performance (e.g., Nguyen et al., 2024) and tend to prioritise their self-interest and exercise entrenchment (e.g., Chikunda et al., 2025). On the other hand, others, such as Oh et al. (2018), argue that long-tenured CEOs can lead to better organisational outcomes because, during their tenure, they gain firm-specific knowledge and social capital, which helps them better understand stakeholders' interests and how to satisfy them. Saito (2019) also provides evidence suggesting that long-tenured CEOs are not rent seekers, but instead appear to make better exit decisions that improve future organisational performance. Accordingly, in the present study, I include CEO tenure (CEOT) as an indicator of CEO power, measured as the number of years a CEO has held their position.

Furthermore, CEO ownership is another popular and widely used indicator of CEO power in the extant literature (Brahma and Economou, 2024). According to Finkelstein (1992), ownership power relates to the CEO's ownership position in the firm, with those holding larger shareholdings being more powerful and having greater control over boards than those who do not. CEO ownership, thus, allows CEOs to influence the firm's decisions, given their voting rights and equity holdings (Fattoum-Guedri et al., 2018). In addition, CEOs with significant shareholdings, which tend to be more powerful, may prevent involuntary dismissal (Daily and Johnson, 1997). With reference to the "convergence of interest" hypothesis, CEO

share ownership binds CEOs' economic interests and incentives to be aligned with those of shareholders, which may help reduce agency costs (Fama and Jensen, 1983; Rashid, 2016). However, it is also argued that increased CEO share ownership could result in increased expected tenure for self-seeking reasons beyond their point of effectiveness (Brookman and Thistle, 2009; Combs et al., 2007). Consequently, in the current study, CEO share ownership (CEO_{SO}), measured as the value of shares held by the CEO divided by the total number of ordinary shares, is used as an indicator of CEO power.

Moreover, CEO remuneration or compensation is a CEO power indicator that many studies have employed as a proxy for structural power (Brahma and Economou, 2024). According to the agency theory, powerful CEOs may effectively influence remuneration committee members to agree and approve generous pay packages that may not align with the firm's actual financial performance (Veprauskaitė and Adams, 2013). CEO compensation thus reflects CEOs' relative power within firms, as more powerful CEOs tend to secure higher pay packages than their peers (Sheikh, 2022). In addition, Bachmann et al. (2020) provide evidence that more powerful CEOs receive larger cash bonuses tied to non-financial targets, relative to their compensation. Therefore, a higher relative CEO compensation compared to other executives may indicate that the CEO is less likely to rely on other executives for decision-making and execution, suggesting increased CEO power (Jaroenjitkam et al., 2020). CEO compensation is, therefore, one way to capture CEO power more objectively (Finkelstein, 1992; Jiraporn et al., 2012). Consequently, in this study, CEO remuneration (CEOR) is used as a proxy for CEO power, measured as the total annual compensation the CEO received during the period, including salary, bonus, other compensation, and pension, divided by market capitalisation.

Personal prestige or status is also an important source of informal power arising from the institutional environment and the CEOs' engagement with other individuals who occupy authority within their firms (Finkelstein, 1992). CEOs' social network size has been widely used as a proxy for prestige power (e.g., Chikh and Filbien, 2011; Fralich and

Papadopoulos, 2018). Hence, CEOs gain power through their social networks and connections outside the firm (Sheikh, 2022). Powerful CEOs have superior access to information about targets through their personal social networks, allowing them to reduce information asymmetry and better assess target quality (Fralich and Papadopoulos, 2018). In addition, well-connected CEOs can obtain, transmit and validate business-related information through their social connections, which may allow them to respond more effectively to future disruptions in their operational environments, including changes in market conditions, the entry of new competitors, macroeconomic shocks, changes in legislation and cost structures (Griffin et al., 2021). However, Mundi (2024) suggests that CEOs' social networks motivate them to make more complex capital structure decisions that may deteriorate organisational performance. Based on the premise that CEOs' social networks and connections increase CEO power within the firm, I include CEO network size (CEONS), measured as the number of social networks held by CEOs throughout their employment, as an indicator of CEO power.

Additionally, apart from having powerful friends, prestige provides power through managerial educational background and qualifications (Finkelstein, 1992). CEOs' educational background is among the most important characteristics believed to impact corporate strategies (Nawaz, 2022). According to Gounopoulos et al. (2021), CEOs with elite education and advanced degrees are more likely to be associated with higher levels of knowledge, skills, cognitive complexity, intellectual ability and access to social networks. Hence, CEOs' education, which reflects their skills and establishes their prestige, may increase their confidence and sense of self-efficacy (Miller and Xu, 2020). In their study on the association between CEO educational attainment and bank performance, King et al. (2016) find that the quality of CEO education matters, as CEOs who have graduated from top twenty universities and those with higher MBA education appear to achieve superior bank performance outcomes and higher profitability than banks led by CEOs who graduated from non-top twenty universities and those without MBA education. CEOs' social class and elite education

thus reflect their elevated power and control over resources, which increase their approach-related behaviours, including strategic risk-taking, as such individuals tend to focus more on rewards and exhibit a higher level of optimism regarding their chances of success (Kish-Gephart and Campbell, 2015). Therefore, CEOs' educational backgrounds may confer prominence and, therefore, increase their power within firms (Lisic et al., 2016). Accordingly, in this study, I include the CEOs' educational background and number of qualifications (CEONQ), measured as the number of academic qualifications held by the CEO, as an indicator of CEO power.

2.3.3.4 Control variables

I control for additional variables identified in prior studies as having significant effects on disclosure tone, including firm-specific characteristics, CEO personality traits and corporate governance mechanisms (e.g., Arslan-Ayaydin et al., 2021; Bassyouny et al., 2020; DeBoskey et al., 2019; Hennig et al., 2025; Veprauskaitė and Adams, 2013).

Regarding firm-specific characteristics, this study controls for three variables: leverage (Lev), firm size (Fsize), and beta (Beta), which have been widely used in the existing literature (e.g., Arslan-Ayaydin et al., 2021; Ataullah et al., 2018; Buchholz et al., 2018). The debt-to-equity ratio, commonly referred to as leverage, is a significant factor in assessing the risk associated with a firm and its securities (Bowman, 1980). Based on the agency theory framework, highly leveraged firms voluntarily disclose information to mitigate agency costs and, thereby, lower their cost of capital (Jensen and Meckling, 1976). However, previous research reports divergent findings regarding the association between the tone of corporate disclosures and leverage. For instance, Wang and Hussainey (2013) find a positive relationship between disclosure tone and leverage. Meanwhile, several other studies document a negative relationship between disclosure tone and leverage (e.g., Aly et al., 2018; Ataullah et al., 2018; Bakarich et al., 2019). In the present study, I control for leverage

(Lev), measured as a firm's debt ratio to its total capital, and anticipate a negative relationship between the tone of CEO letters and firm leverage. This anticipation stems from the notion that increased debt could prompt CEOs to adopt a more cautious communication approach to alleviate perceived financial threats, leading to a less optimistic tone in their letters.

In addition, firm size is commonly employed in research on corporate disclosures, particularly disclosure tone (e.g., Buchholz et al., 2018; Campbell et al., 2020; Schreck and Raithel, 2018). According to Branco and Rodrigues (2008), larger firms tend to attract greater attention than smaller ones, subjecting them to more public scrutiny from stakeholders since they are more visible and more vulnerable to adverse reactions. Therefore, they are expected to increase the extent of their corporate disclosures to address concerns and communicate information regarding corporate activities (Brammer and Pavelin, 2008). Hence, smaller firms have limited resources and higher costs to cover information production, compared to large firms (Schreck and Raithel, 2018). However, existing research yields inconsistent findings regarding the association between disclosure tone and firm size. For example, several studies find that larger firms are negatively and significantly associated with positive tone (e.g., Bassyouny et al., 2020; Marquez-Illescas et al., 2019), whereas others document that larger firms are positively and significantly related to positive tone (e.g., Arslan-Ayaydin et al., 2021; Buchholz et al., 2018). In the present study, I control for firm size (Fsize) and expect a positive relationship between the tone of CEO letters and firm size, given that larger firms face greater pressures and could, therefore, adopt a positive tone to reassure and maintain stakeholders' confidence.

Furthermore, beta is a commonly used metric for assessing firms' systematic risk, also known as market risk (e.g., Arslan-Ayaydin et al., 2021; Benlemlih et al., 2018). External economic and market conditions mainly cause market risk, which cannot be dispersed and eliminated (Zeng et al., 2020). Beta can be highly informative in explaining corporate disclosures, as firms are incentivised to communicate more information, especially during

periods of heightened political visibility and high systemic risk (Khlif and Hussainey, 2016). In addition, Bakarich et al. (2019) find that firms with more unstable beta are more likely to use a more negative tone in their disclosures. Meanwhile, in their study on narrative disclosure tone and bank risk, Albert et al. (2025) report a significant positive association between disclosure tone and market risks, suggesting that negative tones increase both perceived and actual risks. Consequently, in the current study, I control for beta (Beta), a measure of market risk, and anticipate a negative association between CEO letters' tones and beta, as firms with high beta may adopt a less positive tone to avoid any potential institutional pressures or legitimacy threats.

For CEO personality traits, I control for CEO openness and receptiveness to experience (Openness), a CEO personality trait drawn from the Big Five personality traits framework (McCrae and Costa, 1987), measured via machine-learning-based linguistic analysis of CEO letters. Following Harrison et al. (2019), CEO openness is measured using a personality prediction model in which vectorised textual features derived from preprocessed CEO letters are input into pretrained trait-specific algorithms. The model then provides continuous scores ranging from 1 to 7, with higher values reflecting greater CEO openness to experience, new ideas and risk-taking. Various studies in the literature show that CEO openness influences the implementation of strategic actions, the use of language and organisational performance (e.g., Herrmann and Nadkarni, 2014; Nadkarni and Herrmann, 2017).

Strong and effective corporate governance mechanisms have consistently been linked to improved financial reporting quality and considered important drivers of narrative tone (Albitar et al., 2023; Bassyouny et al., 2020). Board oversight thus plays a crucial role in constraining CEOs' optimistic tone in their disclosures (DeBoskey et al., 2019). Consistent with prior research (e.g., Albitar et al., 2023; Bassyouny et al., 2020; Martikainen et al., 2023), I control for corporate governance variables identified as significant in driving disclosure tones, such as board size (Bsize), board meetings (Bmeetings), the proportion of

females on boards (PropFem), audit committee independence (ACind) and the percentage of non-executive directors on the board (Neds).

Board size is one of the key factors that exerts significant pressure on the board's effective monitoring function, as a larger board may present a broader range of experiences, knowledge and diverse opinions (Liao et al., 2018). According to agency theory, boards with more directors are more likely to effectively monitor firms' management and mitigate the agency problem, as more board members increase the board's vigilance (Singh et al., 2018). Consequently, a larger board may play a key role in overseeing management, promoting higher disclosure quality and reducing agency costs (Vitolla et al., 2020). However, it has also been argued that larger boards might substantially limit their capacity to undertake strategic action because of the obstacles they face in reaching consensus on critical topics (Goodstein et al., 1994). Prior research has yielded inconsistent findings regarding the influence of board size on the tone employed in corporate disclosures. For instance, El-Deeb et al. (2022) find that board size and disclosure tone are significantly and positively associated. Meanwhile, Albitar et al. (2023) report that board size does not appear to affect the net tone regardless of how it is measured. In the present study, I control for board size (Bsize), measured as the total number of board members, and, given the conflicting position in the literature, I expect an association between the tone of CEO letters and board size.

In addition, board meetings are considered a crucial communication channel for outside directors to gain insights into firms' operations and managerial decision-making, facilitating an effective involvement in firms' governance (Liao et al., 2018). Prior research often uses board meetings to assess board diligence and efficiency, suggesting that board meetings influence the quality of voluntary disclosures (e.g., Jizi, 2017; Kent and Stewart, 2008). Regular board meetings and attendance at such meetings are important, as they allow directors to contribute to organisational goals and decision-making, and, especially for outside directors, to maintain membership in the firm by interacting with other organisational actors (Min and Chizema, 2018). According to Adams and Ferreira (2009), board meetings

are crucial for directors to acquire firm-specific information and effectively carry out their oversight responsibilities. In addition, DeBoskey et al. (2019) find that higher board meeting intensity and improved board meeting attendance, which together indicate greater board oversight, are related to a less aggressive disclosure tone when firms have CEOs with longer tenures. Consequently, in the present study, I control for board meetings (Bmeetings), measured as the total number of board meetings in a given year, and anticipate a negative relationship between the CEO letter's tone and the number of board meetings.

Furthermore, given the boards' key role in monitoring firms' executive management, having independent directors is essential to ensure that managerial interests are aligned with those of shareholders rather than with those of directors associated with the manager (Di Vito and Trottier, 2022). Independent directors are commonly believed to be better monitors of CEOs, as they strive to uphold their reputations in the directorship market; therefore, firms with a higher proportion of independent directors may benefit more from enhanced board monitoring (Fan et al., 2019). Consequently, the presence of non-executive independent directors may enhance the board's efficiency and performance, thereby mitigating agency conflicts (Singh et al., 2018). Hence, from the agency perspective, independent directors are more likely to provide executive directors with unbiased guidance free of potential conflicts of interest (Fama and Jensen, 1983). Several studies provide evidence suggesting that board independence is negatively associated with positive disclosure tone (e.g., Albitar et al., 2023; Bassyouny et al., 2020). Therefore, in the current study, I control for board independence (Neds), measured by the proportion of non-executive directors on the board, and expect a negative relationship between the tone of CEO letters and board independence.

Furthermore, gender-diverse boards have been suggested to improve monitoring of firms' management, decision-making processes, and accountability (Chen et al., 2019). Female directors on the board have been associated with enhanced quality and transparency in corporate disclosures (Ben-Amar et al., 2024). Similarly, Albitar et al. (2023) suggest that the presence of women on the board could improve disclosure practices and may influence

disclosure tone, as more women on the board may result in reduced impression management (Albitar et al., 2023). According to Adams and Ferreira (2009), female directors appear to have the same impact as independent directors, and thus significantly improve board inputs and organisational outcomes. However, prior research yields mixed findings regarding the association between the percentage of female directors on the board and disclosure tone. For example, Bassyouny et al. (2020) find that a higher percentage of women on the board is positively and significantly associated with a positive tone. On the other hand, Albitar et al. (2023) report that the proportion of women on boards does not appear to affect disclosure tone. Consequently, in the current study, I control for the presence of female directors on the board (PropFem), measured as the proportion of female directors, and expect that a higher percentage of female directors is associated with a more negative tone.

The audit committee plays a crucial role in the firm's internal corporate governance mechanism by overseeing management activities, especially in financial reporting (Yekini et al., 2015). Consequently, it is essential to consider audit committee characteristics when analysing financial reporting narratives in general and narrative textual features, particularly disclosure tone (Bassyouny et al., 2020). In the UK context, the Smith Committee Report (2003) states that the audit committee should act independently from executives, review firms' internal control systems and risk management, and thus ensure that shareholders' interests are protected in relation to financial reporting and internal control. In addition, the UK Corporate Governance Code (2024) advises boards to establish audit committees comprising at least three independent directors, and two in the case of smaller firms. However, Alderman and Jollineau (2020) argue that retired executives may serve on audit committees, potentially undermining the committee's perceived independence. Makhoul (2024) provides evidence suggesting that the independence of audit committees is significantly and negatively related to impression management in narrative disclosures, thereby mitigating managerial opportunistic behaviours. Moreover, Bassyouny et al. (2020)

document that more independent audit committees are associated with a less positive tone in narrative reporting. Therefore, in the present study, I control for audit committee independence (ACind), measured as the proportion of independent members on the audit committee, and expect a negative relationship between the tone of CEO letters and audit committee independence.

Table 2.1
Variables definitions

Type	Variable	Definition
Dependent variable	CEO letters tone (Tone)	A measure of the overall tone expressed within CEO letters using the TextBlob sentiment analysis tool, calculated as a compound score that ranges between -1.0 to 1.0, where -1.0 is very negative, and 1.0 is very positive.
Independent variable	Return on assets (ROA)	A measure of a firm's profitability in relation to its total assets.
Moderator variable	CEO power (CEOP)	A composite measure derived through Principal Component Analysis (PCA) of 5 dimensions: CEO tenure, CEO ownership, CEO remuneration, CEO network size and the number of CEO qualifications. CEOT – is the number of years the CEO has been in position. CEOO – is the value of shares the CEO holds divided by total ordinary shares. CEOR – is the total annual compensation the CEO received (including salary plus bonus plus other compensation plus employer retirement/pension contribution for the period) divided by market capitalisation. CEONS- is the number of social networks the CEOs hold throughout their employment. CEONQ – is the number of academic qualifications held by the CEO.

Control variables	Leverage (Lev)	A measure of a firm's debt ratio to its total capital.
	Firm size (Fsize)	Measured as the natural logarithm of total assets.
	CEO Openness (Openness)	Measured using machine learning-based linguistic analysis of CEO letters, producing a continuous score ranging from 1 to 7.
	Beta (Beta)	A measure of market risk.
	Board size (Bsize)	Measured as the total number of individuals who hold positions on the board of directors.
	Board meetings (Bmeetings)	Measured as the total number of board meetings in a given year.
	Non-executive directors (Neds)	Measured as the proportion of non-executive directors on the board.
	Proportion of females (PropFem)	Measured as the proportion of female directors on the board.
	Audit committee independence (ACind)	Measured as the proportion of independent members on the audit committee.

This table presents and defines all the variables used in this study.

2.4. Empirical results

2.4.1 Descriptive statistics and correlation results

Table 2.2 presents the descriptive statistics for all variables used in this study. The mean tone is 0.129, with a minimum of -0.064 and a maximum of 0.413, suggesting that CEO letters of UK sampled firms have a moderately positive tone, but not an overly positive one. The performance variable, ROA, has a mean of 0.078, a standard deviation of 0.096, a minimum of -0.532, and a maximum of 0.626, which are similar and in the same range as those reported in UK-based studies (e.g. Al-Shaer and Zaman, 2021; Yau, 2025). As presented in Table 2.2, CEO power variables, namely, CEOT, CEOO, CEOR, CEONS and CEONQ, have different dimensionalities and are measured on different scales with varying variances. These outcomes suggest that it may be necessary to transform the variables to address skewness and stabilise the variance of variables, and then subsequently standardise them to a zero mean and unit variance to ensure they are on the same scale prior to conducting principal component analysis (PCA), which is a common practice (Baxter, 1995; Maadooliat et al., 2015). Consequently, I employ a logarithmic transformation, which has been shown to yield more satisfactory results (Baxter, 1995). Following that, I standardise the transformed variables to ensure they are on the same scale before conducting PCA to generate the CEO power (CEOP) index. Histograms are created for each component, before and after transformation (see Appendix A2.5 for the histogram results). The transformed and standardised CEO power-related variables are then used as the principal components in the PCA to create the CEO power (CEOP) index.

Regarding the control variables, and in particular firm-specific characteristics, the sampled firms have a mean leverage (Lev) of 0.261, a standard deviation of 0.182, a minimum of 0.000, and a maximum of 0.971. Meanwhile, the average firm size (Fsize) is 6.549, with a standard deviation of 0.732, a minimum of 4.695 and a maximum of 8.508. The average

beta for sampled firms is 1.306, a standard deviation of 0.632, a minimum of 0.190 and a maximum of 3.370. In addition, CEOs in the sample have an average openness to experiences score of 5.732, ranging from 4.895 to 6.211. CEO openness is a score of 7, suggesting that CEOs within the sample are open to experiences.

In terms of corporate governance mechanisms, the sampled firms hold an average of 9 board meetings (Bmeetings) per year, ranging from 1 to 34. The results also show that sampled firms have a mean board size (Bsize) of 9 members, ranging from 4 to 17. Moreover, the board members of the sampled firms comprise an average of 61% non-executive directors (Neds), ranging from 0% to 87.5% across firms. Meanwhile, the proportion of females (PropFem) ranges from 0% to 16%, with an average of 9.2% of female directors on boards. Lastly, the average audit committee independence is 94.3%, ranging from 28.6% to 100%.

Table 2.2
Summary descriptive statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Tone variable					
Tone	1080	0.129	0.061	-0.064	0.413
Performance variable					
ROA	1080	0.078	0.096	-0.532	0.626
CEO power variables					
CEOT	1080	5.275	4.481	0.000	31.000
CEOO	1080	0.052	0.043	0.010	0.369
CEOR	1080	0.085	0.075	0.010	0.472
CEONS	1080	1446.069	1578.185	15.000	9904
CEONQ	1080	2.094	0.813	1.000	5.000
Control variables					
Lev	1080	0.261	0.182	0.000	0.971
Fsize	1080	6.549	0.732	4.695	8.508
Beta	1080	1.306	0.632	0.190	3.370
Openness	1080	5.732	0.219	4.895	6.211
Bsize	1080	8.946	1.908	4.000	17.000
Bmeetings	1080	8.679	2.637	1.000	34.000
PropFem	1080	0.092	0.021	0.000	0.160
ACind	1080	0.943	0.117	0.286	1.000
Neds	1080	0.611	0.131	0.000	0.875

This table presents the descriptive statistics for all variables used in this study.

Table 2.3 presents the Pearson correlation matrix for all variables used in this study. This correlation test shows the pairwise linear correlations among all variables, along with their directions and strengths, which can be particularly useful for identifying potential multicollinearity (Bassyouny et al., 2020). As displayed in Table 2.3, the tone of CEO letters is positively and significantly associated with ROA. The tone of CEO letters is also positively and significantly associated with CEOP, indicating that as CEO power increases, tone increases as well. In addition, the tone of CEO letters is significantly and positively related to Fsize, Openness, Neds, PropFem and Bsize, but significantly and negatively related to Bmeetings. Meanwhile, Lev, Beta and ACind appear not to be significantly associated with

tone. Table 2.3 also shows a mix of weak and moderate correlations among the explanatory variables, except for one variable, Bsize, which shows a strong association with the other variables, especially PropFem. This implies the potential for multicollinearity. Therefore, I exclude Bsize from the regression model to enhance the reliability of my findings. Otherwise, issues such as inflated coefficients and unstable standard errors could arise from multicollinearity if left unaddressed.

Table 2.3
Pearson correlation matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) Tone	1.000											
(2) ROA	0.115***	1.000										
(3) CEOP	0.072**	-0.041	1.000									
(4) Lev	0.019	-0.001	0.124***	1.000								
(5) Fsize	0.118***	-0.235***	0.028	0.102***	1.000							
(6) Beta	-0.039	-0.109***	-0.044	-0.121***	-0.114***	1.000						
(7) Openness	0.221***	0.046	0.052*	0.089***	-0.085***	-0.143***	1.000					
(8) Bmeetings	-0.098***	-0.080***	0.051*	0.054*	-0.098***	0.094***	-0.027	1.000				
(9) Neds	0.081***	0.053*	-0.014	0.014	0.237***	-0.075**	0.073**	-0.055*	1.000			
(10) PropFem	0.154***	-0.136***	0.019	0.092***	0.636***	-0.116***	-0.067**	-0.046	0.131***	1.000		
(11) Acind	0.027	-0.080***	0.089***	0.003	0.159***	-0.031	0.041	-0.103***	0.197***	0.118***	1.000	
(12) Bsize	0.138***	-0.105***	0.002	0.124***	0.501***	-0.033	-0.034	-0.017	0.074**	0.801***	0.089***	1.000

This table presents the correlation matrix for all the variables used in this study. Statistical significance at the 1%, 5%, and 10% levels in two-tailed tests is indicated by ***, **, and *, respectively.

2.4.2 Principal component analysis (PCA)

As previously stated, this study uses principal component analysis (PCA) to measure CEO power, which serves as the study's moderator variable. The primary objective of employing principal component analysis (PCA) is to reduce the dimensionality of the dataset comprising several interconnected variables, namely those related to CEO power, while preserving a significant portion of the variability inherent in the dataset (Veprauskaitė and Adams, 2013). This can be accomplished by converting the data into a new set of uncorrelated variables, known as principal components. These principal components are arranged in a specific order, with the initial components capturing most of the variation observed in the original variables that present significant components of CEO power. In the present study, I employ PCA to reduce five variables representing different sources of CEO power: CEOT, CEOO, CEOR, CEONS and CEONQ to one dimension, labelled as the CEOP index.

Table 2.4 presents the results obtained from the principal component analysis (PCA).

Section A of Table 2.4 presents the correlation matrix for the five CEO power variables that comprise the CEOP index. The observed correlation coefficients among the five measures of CEO power are not strong, suggesting that these variables capture distinct aspects of CEO power. The loads for the CEOP index are reported in Section B of Table 2.4. As demonstrated in Table 2.4, Section B, the CEOP index is primarily characterised by CEONS and CEONQ, as shown by their loadings exceeding 0.5 in absolute value. In addition, CEOR contributes moderately, while CEOT and CEOO contribute less compared to others.

Additionally, Section C of Table 2.4 further presents the summary descriptive statistics for the components of the CEOP index. The presence of a mean value of zero arises from the standardisation process that the variables underwent before performing the PCA, to ensure they are on a similar scale. Hence, PCA, a technique for reducing dimensionality, involves centring the data by subtracting each variable's mean from its respective values (Beattie and

Esmonde-White, 2021). Therefore, when these centred-around-the-mean variables are aggregated into the CEOP index, the resulting index mean is zero.

Moreover, the recurring finding of a mean value of zero in the CEO power index across several UK-based studies, such as those of Adams et al. (2024) and Brahma et al. (2025), indicates that this outcome is not exclusive to my research. Instead, it is a common feature of the principal component analysis (PCA) approach employed to analyse these CEO power-related variables. Further, it is essential to acknowledge that the PCA produces a mean of zero only as a mathematical result. However, the meaningful interpretation of the analysis resides in identifying the variation and associations shown by the derived principal components. In the following section, I analyse how the CEOP index relates to the tone of CEO letters and whether CEO power moderates the relationship between CEO letter tones and financial performance.

Table 2.4
Principal component analysis (PCA)

Variables	(1)	(2)	(3)	(4)	(5)
A. Correlation matrix					
(1) CEOT	1.000				
(2) CEOO	0.023	1.000			
(3) CEOR	0.047	0.003	1.000		
(4) CEONS	-0.090***	0.066**	-0.188***	1.000	
(5) CEONQ	0.014	0.098***	0.032	0.262***	1.000
B. Index weights					
CEOP	0.1473	0.0168	0.4092	0.6931	-0.5746
C. Principal component descriptive statistics					
	Obs	Mean	Std. Dev.	Min	Max
CEOP	1080	0	0.809	-2.458	2.161

This table presents: A) the correlation matrix for CEO power-related variables as components of the CEOP index; B) principal components index weights and loadings; C) principal components summary descriptive statistics. Statistical significance at the 1%, 5%, and 10% levels in two-tailed tests is indicated by ***, **, and *, respectively.

2.4.3 Multivariate results

Table 2.5 presents the random effects (RE) panel regression results. As shown in Table 2.5, both models account for approximately 14% of the overall variation in the tone of CEO letters of the sampled companies. Furthermore, Model 1 accounts for 9.5% of the variation within an entity from one year to the next and approximately 19.5% of the variation between entities. Meanwhile, Model 2 demonstrates a similar capability, accounting for 10% of the variation within an entity from one year to the next and approximately 19% of the variation between entities. Although the overall R^2 value is relatively low, it aligns with previous studies in this field (e.g., Albitar et al., 2023; Rahman, 2019). The model's low explanatory capacity could be attributed to its exploratory nature. Nevertheless, before relying on these findings to test the hypotheses, additional post-estimation tests were conducted to ensure the credibility of the results. This involved evaluating the appropriateness of the regression model employed in this study, namely the random effects (RE) panel regression. Additionally, I account for autocorrelation and heteroskedasticity by using clustered standard errors for both models (see the discussion in Section 2.3.2 above).

2.4.3.1 Tone of CEO letters and firms' financial performance

As displayed in Table 2.5, the tone of CEO letters and financial performance (ROA) are positively and significantly associated at the 1% level in both models, thus supporting *H1*. This observation suggests that the tone of CEO letters is congruent with firms' financial performance, as tone becomes more positive when ROA increases. This provides greater support for communicative action theory than for impression management theory, as tone appears to be used in CEO letters to communicate and reflect actual performance, and may therefore be considered communicative rather than manipulative. This finding is consistent with those of Patelli and Pedrini (2013) and Yuthas et al. (2002), who document that disclosure tone is consistent with firms' performance, suggesting that annual report

narratives and, especially, CEO letters' tones are used to convey firms' true financial position, rather than being strategically manipulated to manipulate stakeholders.

2.4.3.2 The moderating impact of CEO power

As demonstrated in Table 2.5, the tone of CEO letters and CEO power are significantly and positively related in Model 2 at the 10% level. This outcome reveals that more powerful CEOs tend to use a more positive tone in their communications, which is consistent with prior research (e.g., Cao et al., 2022; DeBoskey et al., 2019). This is consistent with the principles of impression management theory, which emphasise the deliberate actions and communication strategies CEOs undertake, such as tone management and overly optimistic tones, to strategically influence how shareholders interpret financial performance (e.g., Huang et al., 2014; Merkl-Davies and Brennan, 2011). Thus, when CEOs hold substantial power within the organisational, they have the necessary resources and authority to strategically use their CEO letters to project a favourable perception of the firm (DeBoskey et al., 2019). This may involve highlighting the firm's accomplishments and prospective opportunities to foster a more optimistic outlook in their letters to shareholders.

While powerful CEOs may emphasise impression management techniques, such as adopting positive tones when communicating with shareholders, less powerful CEOs may employ alternative communication strategies. Thus, the results indicate that when CEOs possess less power, there is more support for the communicative action theory. In other words, less powerful CEOs may be more inclined to use less positive and more neutral language. Therefore, less powerful CEOs may tend to employ communication strategies that align with the principles of the communication active theory, due to their limited or insignificant influence within the firm.

Interestingly, as shown in Table 2.5, the interaction term between ROA and the CEO power index (ROA_CEOP), in Model 2, is significantly and negatively related to the tone of CEO

letters at the 5% level. This finding reveals that as CEO power increases, the relationship between CEO letter tones and firms' financial performance becomes less pronounced. Hence, CEO power appears to significantly moderate the association between the tone of CEO letters and ROA. Therefore, *H2* is supported. This outcome suggests that as CEO power increases, the tone of CEO letters is more likely to be subject to impression management. In addition, this finding implies that as CEO power increases, CEOs may be more inclined to shift their communication focus away from financial performance and towards other factors. For instance, they may communicate with shareholders to highlight their strategic leadership, future plans, vision, and advertise the value they add to the firm (Li et al., 2019; Zajac and Westphal, 1995). Hence, the strategic shift in powerful CEOs' communication and the focus on these factors collectively help to develop an optimistic view of the firm's prospects.

2.4.3.3 Control variables

As illustrated in Table 2.5, the tone of CEO letters and leverage (Lev) are negatively associated in both models. However, this relationship is not statistically significant, and therefore, I cannot confidently conclude that CEO letter tones and leverage are associated in this sample. Meanwhile, the tone of CEO letters and firm size (Fsize) are positively and significantly related at the 1% level in both models. These results imply that larger firms tend to use a more positive tone in their communications, consistent with prior research (e.g., Arslan-Ayaydin et al., 2021; Buchholz et al., 2018). In addition, Beta is positively and significantly associated with CEO letter tones in both models at the 10% level, indicating that CEOs of firms with higher betas appear to use more positive tones, possibly to reassure stakeholders and manage their perceptions. I also find that CEO openness to experience (openness) is significantly and positively associated with the tone of CEO letters in both models at the 1% level. These observations imply that CEO personality traits influence their

communication style, as CEOs who are more open to experience, new ideas, and risk-taking tend to use a more optimistic tone in their letters to shareholders.

In terms of corporate governance mechanisms, CEO letter tones and board meetings (Bmeetings) appear to be significantly and negatively associated in both models at the 1% level. This observation indicates that more frequent board meetings, and thus greater board oversight, are associated with less positive tones in CEO letters to shareholders. Therefore, as board meetings increase, CEOs may experience increased pressure to adopt a more cautious or pessimistic tone in their letters to shareholders. This finding is consistent with that of DeBoskey et al. (2019). Surprisingly, I find a statistically significant positive relationship between the proportion of females on boards (PropFem) and the tone of CEO letters in both models at 1% level, consistent with Bassyouny et al. (2020). This can be explained by the relatively low representation of females on UK boards, in contrast to previous studies conducted within the US. As shown in Table 2.2 (descriptive statistics), the average proportion of females on the boards of my sample of FTSE-350 firms is 0.092. Moreover, the proportion of non-executive directors (Neds) and the tone of CEO letters are negatively related in both models, suggesting that as the proportion of Neds on the board increases, the tone of CEO letters becomes less positive, aligning with previous research (e.g., Albitar et al., 2023; Bassyouny et al., 2020). However, this association is insignificant in Model 1, but significant at the 10% level in Model 2, implying that the relationship becomes significant when CEO power is included. This observation suggests that the impact of non-executive directors on CEO letter tones is more pronounced under certain conditions, such as when CEOs are powerful. Lastly, I find a negative but insignificant relationship between the CEO's letter tone and audit committee independence in both models, suggesting that although a more independent audit committee may exert pressure for a more cautious disclosure tone, the effect is not statistically significant in this context.

Table 2.5
Regression results

	(Model 1) Tone			(Model 2) Tone		
	Coef.	Robust std. error	p-value	Coef.	Robust std. error	p-value
ROA	0.073***	0.020	0.000	0.071***	0.017	0.000
CEOP	-	-	-	0.006*	0.003	0.052
ROA_CEOP	-	-	-	-0.036**	0.016	0.021
Lev	-0.004	0.011	0.726	-0.007	0.011	0.522
Fsize	0.015***	0.006	0.008	0.015***	0.006	0.008
Beta	0.008*	0.004	0.071	0.008*	0.004	0.079
Openness	0.048***	0.012	0.000	0.048***	0.012	0.000
Bmeetings	-0.002***	0.000	0.000	-0.002***	0.000	0.000
Neds	-0.025	0.015	0.107	-0.026*	0.014	0.059
PropFem	0.281***	0.098	0.004	0.288***	0.100	0.004
Acind	-0.019	0.023	0.405	-0.020	0.024	0.391
Industry dummies	Yes			Yes		
Year dummies	Yes			Yes		
Constant	-0.227	0.111	0.040	-0.222	0.108	0.039
R^2 : Within=		0.095		R^2 : Within=		0.100
Between		0.195		Between=		0.190
Overall=		0.137		Overall=		0.138
Rho=		0.342		Rho=		0.338
N=		1080		N=		1080

This table reports the results of my examination of the relationship between the tone of CEO letters and the financial performance of UK firms, investigating whether CEO power moderates this relationship. Model 1 tests the relationship between the tone of CEO letters and firms' financial performance. Meanwhile, Model 2 includes the CEOP index. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels with a two-tailed test. Tone is the tone of CEO letters. ROA is a measure of a firm's profitability. CEOP is a composite measure derived through PCA of five CEO power-related variables. Lev is a measure of a firm's debt ratio to its total capital. Fsize is the natural logarithm of total assets. Openness is the CEO's openness to experience. Beta is a measure of market risk. Bmeetings is the total number of board meetings. Neds is the proportion of non-executive directors on the board. PropFem is the proportion of female board members. Acind is the proportion of independent members of the audit committee.

2.4.4 Robustness tests

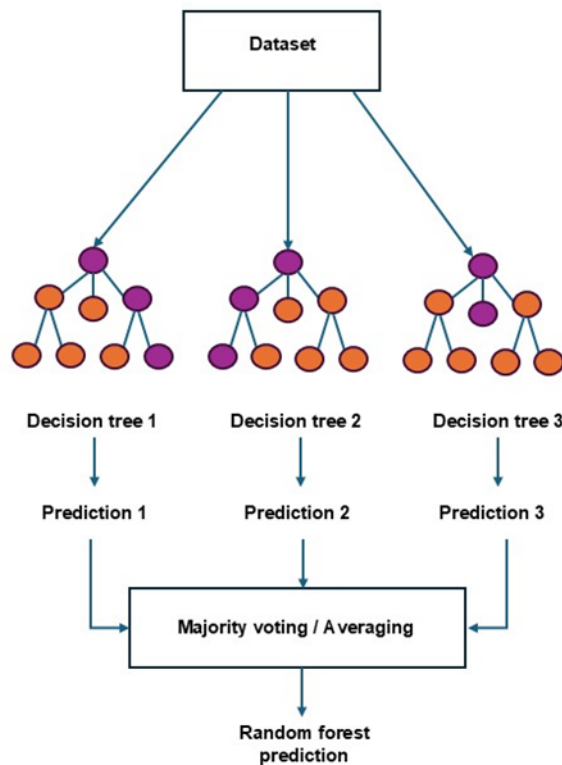
I perform additional robustness tests to validate the associations identified in the primary regression model by adopting an alternative methodology, particularly employing machine learning (ML) techniques. In machine learning, a regression task is a method for predicting a continuous dependent variable, to understand the relationship between the dependent variable and a set of independent variables (Venkatesh et al., 2023). Regression is a supervised learning technique that facilitates the discovery of relationships among variables and the prediction of the target variable, the dependent variable, based on the predictor variables (Hoang and Wiegratz, 2023). The primary objective is, therefore, to validate the associations established in my regression model, which investigates the relationship between the tone of CEO letters and the financial performance of UK-listed firms, and whether CEO power moderates this association. Consequently, by adopting a machine learning approach, the goal is to assess whether the relationships identified in the primary regression model are meaningful and whether the interaction term between ROA and CEO power adds explanatory power, which collectively helps validate the primary regression results.

A wide range of supervised machine learning methods and algorithms has been widely used for regression tasks, such as random forests, support vector machines (SVM), and regularised linear methods, given their ability to uncover complex relationships between variables (D'Amato et al., 2022; Hoang and Wiegratz, 2023). However, SVM-based methods require continuous variables to be scaled and transformed, whereas decision tree-based methods do not (Krupa and Minutti-Meza, 2022). In addition, for numerical data, decision tree-based machine learning methods are commonly used more than linear methods, as they can automatically capture relevant nonlinearities and interaction effects (Hoang and Wiegratz, 2023). Random forest techniques, in particular, have consistently been shown to outperform other machine learning techniques, especially for prediction tasks (e.g., Barboza

et al., 2017; Barboza and Altman, 2024). Therefore, in the present study, I employ the random forest (RF) regressor, a supervised learning algorithm based on decision trees (D'Amato et al., 2022), to predict the tone of CEO letters using variables from the primary regression model. Figure 2.1 illustrates the structure of the random forest. As shown in Figure 2.1, the random forest regressor (RF) learns patterns in the training dataset using decision trees. Each decision tree in a random forest is developed using random subsets of the data, with each tree trained on a portion of the complete dataset, and then the results from each individual tree are aggregated to give a prediction for every observation (Speiser et al., 2019).

Figure 2.1

Structure of a random forest



Consequently, I develop two machine learning models to predict the target variable, tone of CEO letters, using the same variables used in the primary regression model. The first model (Model 1) includes only ROA, Lev, Fsize, Beta, Openess, Bmeetings, NEDs, PropFem, and ACind. Meanwhile, the second model (Model 2) adds CEO power and the interaction term between ROA and CEO power (ROA_CEOP) to assess whether predictive performance improves with their inclusion. The primary objective of supervised learning techniques is to make out-of-sample predictions using previously unseen observations to accurately assess prediction performance, which requires dividing the data into a training and a testing set (Hoang and Wiegatz, 2023). Therefore, I systematically split the dataset into 80% for training and 20% for testing to evaluate the predictive accuracy of the random forest regressor. This step ensures the training set is used to learn relationships among the variables, and the testing set is used to evaluate the model on new, unseen data. Hence, the purpose of the data split is to train the random forest regressor (RF) on the training set with categorised examples to predict the tone of new, unseen observations in the testing set.

Consistent with prior research (e.g., Elamir, 2020; D'Amato et al., 2022), I evaluate the models' predictive performance using four commonly employed regression metrics: mean squared error (MSE), root mean square error (RMSE), R-squared, and mean absolute error (MAE). These metrics play a crucial role in evaluating the random forest regressor's predictive performance. For instance, R-squared denotes the proportion of the variance in the dependent variable explained by the predictor variables, while MSE represents the average squared difference between the predicted values and the actual target values (Coenen et al., 2022). In addition, RMSE is the square root of the mean squared error in a model's predictions and is considered a robust error metric for numerical predictions (Moreno et al., 2022). In contrast, MAE represents the mean absolute prediction error, a commonly used metric for measuring out-of-sample predictive performance (Kim and Won, 2018). Therefore, the goal is to assess whether the tone of CEO letters can be effectively predicted using the same variables and interaction term used in the primary regression

model. Hence, if the random forest regressor (RF) can significantly predict the tone of CEO letters using these features, then the validity and robustness of the associations identified within the primary regression model can be supported.

Table 2.6 presents the evaluation metrics results. As shown in Table 2.6, in Model 1, where no CEO power or the interaction term between financial performance and CEO power are included, the R-squared of 0.275, and, more importantly, the low MSE of 0.003, MAE of 0.042, and RMSE of 0.055 suggest that the variables used in the primary regression provide meaningful explanatory power for the tone of CEO letters, indicating that they matter.

Therefore, one can conclude that the associations identified in the primary regression model are valid as they appear to contribute to explaining tone. Meanwhile, in Model 2, which includes CEO power and the interaction term between ROA and CEO power, while MSE and MAE remain the same, a slight drop in RMSE and an increase in R-squared are observed. Hence, despite the changes being small, this reveals that CEO power and its moderating impact add explanatory value by accounting for additional variation in tone. This further validates that the association between the tone of CEO letters and firms' financial performance is influenced by CEO power, albeit modestly. Together, these results confirm that the relationships identified in the primary regression model are meaningful rather than statistical noise, and that, to a certain extent, CEO power influences the relationship between CEO letter tones and financial performance.

Table 2.6
Evaluation metrics for robustness checks

Evaluation metrics	(Model 1) Tone	(Model 2) Tone
	Random forest regressor (RF)	Random forest regressor (RF)
Mean squared error	0.003	0.003
R-squared	0.275	0.280
Mean absolute error	0.042	0.042
Root mean squared error	0.055	0.054

This table presents the evaluation metrics used to assess the random forest regressor's predictive performance. Model (1) does not include CEO power or the interaction term between firm performance and CEO power. Meanwhile, Model (2) includes CEO power and the interaction term between firm performance and CEO power.

2.5. Discussion and implications

CEO letters are important corporate communications that CEOs use to communicate with shareholders. They provide an open forum for CEOs to convey their firms' financial performance, position, strategies, and visions (Yan et al., 2019). However, because these letters are unaudited, CEOs may have complete freedom to determine their content and structure, thereby increasing the likelihood of strategic and opportunistic disclosures (Boudt and Thewissen, 2019). In this case, one could expect CEOs to exploit information asymmetry by engaging in biased reporting, such as impression management. This is particularly important to understand, as such narratives can influence investors' investment decisions.

Consequently, in this study, I examine the association between the tone of CEO letters and the financial performance of UK FTSE-350 firms and investigate whether CEO power moderates this relationship. This is necessitated by the mixed evidence in the literature on

whether such narratives are used to communicate true financial performance (e.g., Patelli and Pedrini, 2013) or to strategically manipulate stakeholders' interests (e.g., Boudt and Thewissen, 2019).

I find that the tone of CEO letters is positively and significantly related to financial performance, which aligns with the principles of the communicative action theory. This observation indicates that the tone expressed in CEO letters of UK firms is consistent with financial performance. I also find that the tone of CEO letters is positively and significantly associated with CEO power, suggesting that tone becomes more positive as CEO power increases. Interestingly, I find that the association between CEO letter tone and financial performance becomes less pronounced as CEO power increases. This finding reveals that CEO power influences CEOs' choice of tones conveyed in their letters. Therefore, as CEO power increases, the tone of CEO letters may be more subject to impression management.

In addition, I find that certain firm-specific characteristics, CEO personality traits and corporate governance mechanisms are significantly related to the tone of CEO letters. For instance, larger firms and those with higher market risks employ a more positive tone. In addition, CEOs who are open to new experiences also appear to use more positive tones. Moreover, effective corporate governance mechanisms, such as increased board meetings and a greater number of non-executive directors, especially in the presence of CEO power, lead to a less positive tone in the CEO letters of UK firms. Meanwhile, an increased number of female directors on the board appears to be significantly and positively related to tone, but this may be due to low female representation on UK firms' boards.

Furthermore, to validate the reliability of my results and the identified associations, I use the random forest regressor (RF) to predict tone using the same variables as in the primary regression model. The objective is to determine whether the independent, moderator, and control variables can explain tone. If these variables achieve good predictive performance when predicting tone, it will show that they matter and that the associations identified in the regression models are valid. Similarly, if performance improves when CEO power and the

ROA_CEOP interaction term are introduced, this suggests that CEO power influences the relationship between tone and firm performance. The results confirm that the relationships identified in the primary regression model are meaningful rather than statistical noise, and that, to a certain extent, CEO power influences the relationship between CEO letter tones and financial performance.

The primary contribution of this study is to expand upon the existing body of empirical research, specifically regarding the conflicting position in the literature on whether CEOs employ optimistic tones in their letters as a strategic action to cultivate positive perceptions or as a communicative action. The present study provides empirical evidence suggesting that the CEO letters of UK firms appear to reflect actual performance. However, when CEO power increases, the likelihood of impression management, such as the employment of overly optimistic tones, increases, suggesting that CEO power influences CEOs' language choices and that more powerful CEOs are more likely to express a positive tone, which may potentially influence stakeholders' perceptions. This finding has important implications for various stakeholders. For instance, investors should be careful when interpreting CEO letters, especially when CEOs are powerful, to avoid misjudging a firm. That is because increased CEO power may increase the likelihood that the information communicated is strategically framed to influence perceptions rather than communicate actual firm performance. Therefore, investors should not rely solely on CEO letters when making investment decisions, although firms' CEO letter tones and firm performance appear to be positively related. That is because powerful CEOs are more likely to adopt an overly positive tone that does not necessarily reflect true firm performance, which may influence investors' perceptions. In addition, regulatory bodies and board directors may benefit from these findings and consider adopting regulations or oversight systems to ensure CEO letters remain transparent, reliable, and consistent with financial performance, even in contexts of increased CEO power.

2.6 Conclusion

This study employs a random effects (RE) panel regression to investigate the association between the tone of CEO letters and the financial performance of UK FTSE 350 non-financial firms from 2013 to 2021, controlling for firm-specific characteristics, CEO personality traits and corporate governance mechanisms. I also examine the moderating impact of CEO power on this relationship. The findings reveal that the tone of CEO letters and financial performance are positively related. However, more importantly, this relationship becomes less pronounced as CEO power increases, suggesting that the tone of CEO letters is influenced by CEO power. This finding indicates that the likelihood that the tone of CEO letters may be subject to impression management increases with CEO power.

The findings of this study also reveal that firm-specific characteristics, CEO personality traits and corporate governance mechanisms are significantly associated with the tone of CEO letters. For instance, larger firms and those with higher market risk appear to be associated with a more positive tone. The tone of CEO letters and that of leveraged firms seems negatively related; however, the association is not significant. Considering CEO personality traits, I find a significant and positive association between CEO openness to experience and the tone of CEO letters. Regarding corporate governance mechanisms, I find that the increased number of board meetings results in a less positive tone. In addition, although I find that a greater number of non-executive directors on the board reduces positive tone, this effect is significant only when CEO power is present. Notably, I find a significant positive relationship between the proportion of female directors on boards and the tone of CEO letters, which may be due to the relatively low representation of females on UK boards. Lastly, audit committees' independence and the tone of CEO letters appear to be negatively related. However, this relationship is not statistically significant, suggesting that although a more independent audit committee may exert pressure for less positive tones, the effect is not statistically significant in this context.

Together, these findings expand the existing empirical literature, especially regarding the conflicting position in the literature on whether tones expressed in CEO letters pursue strategic objectives, such as persuading stakeholders and influencing their perceptions of the firms' performance, or communicating the firms' underlying performance. This study's findings, thus, demonstrate that CEO power influences how CEOs express tone in their letters, suggesting that the tone of CEO letters is more likely to be subject to impression management when CEOs possess greater power.

Despite the importance and relevance of this study's findings, some limitations prompt future research. For instance, this study uses ROA as an accounting performance measure, which is among the most widely used measures in the extant literature when investigating the relationship between disclosure tone and current firm performance (e.g., Boudt and Thewissen, 2019; Martikainen et al., 2023; Mousa et al., 2022). Future research could therefore use other performance measures, such as market-based measures like Tobin's Q or extend this analysis by including future ROA to examine whether the tone of CEO letters and future performance are consistent and whether CEO power moderates this relationship. In addition, the scope of this study is limited to UK firms. However, given that countries have different corporate governance systems and regulatory environments, which may also affect levels of CEO power and managerial incentives to engage in impression management, future research could examine whether CEO power moderates the relationship between disclosure tone and firm performance in other countries with different governance structures. Future studies could also conduct cross-country comparisons to determine whether the moderating effect of CEO power on disclosure tone and firm performance is specific to certain countries and their corporate governance structures, or a common phenomenon across countries and corporate governance frameworks.

Appendix

Appendix A2.1

List of sampled firms

1	Astrazeneca plc	41	Croda International plc	81	Inchcape plc
2	Unilever plc	42	Dechra Pharmaceuticals plc	82	Mediclinic International plc
3	BP plc	43	Drax Group plc	83	Mitie Group plc
4	Diageo plc	44	Easyjet plc	84	Oxford Instruments plc
5	Rio Tinto plc	45	Greggs plc	85	Pagegroup plc
6	Glencore plc	46	Halma plc	86	Pennon Group plc
7	British American Tobacco plc	47	Howden Joinery Group plc	87	Playtech plc
			International Distributions		
8	GSK plc	48	Services plc	88	Qinetiq Group plc
9	Anglo American plc	49	Intertek Group plc	89	RedRow plc
10	Reckitt Benckiser Group plc	50	ITV plc	90	Rotork plc
11	RELX plc	51	Johnson Matthey plc	91	Serco Group plc
12	National Grid plc	52	Kingfisher plc	92	Smiths Group plc
13	Compass Group plc	53	Mitchells and Butlers plc	93	Spectris plc
14	Vodafone Group plc	54	Moneysupermarket.com plc	94	Spirent Communications plc
15	BAE Systems plc	55	Next plc	95	Tate and Lyle plc
16	Experian plc	56	Ocado Group plc	96	Telecom Plus plc
17	SSE plc	57	Pearson plc	97	Travis Perkins plc
18	Tesco plc	58	RS Group plc	98	Tullow Oil plc
19	Antofagasta plc	59	Sainsbury plc	99	Vesuvius plc
20	Imperial Brands plc	60	Severn Trent plc	100	Victrex plc
21	Associated British Foods plc	61	DS Smith plc	101	Videndum plc
22	BT Group plc	62	Taylor Wimpey plc	102	Weir Group plc
23	Rentokil Initial plc	63	John Wood Group plc	103	WH Smith plc
24	Smith and Nephew plc	64	Balfour Beatty plc	104	Whitbread plc
25	Babcock International Group plc	65	Bellway plc	105	AG Barr plc
26	Bunzl plc	66	Bodycote plc	106	C and C Group plc
27	Fresnillo plc	67	Centrica plc	107	Chemring Group plc
28	Harbour Energy plc	68	Computacenter plc	108	Clarkson plc
	InterContinental Hotels Group				
29	plc	69	Cranswick plc	109	DiscoverIE Group plc
30	Marks and Spencer Group plc	70	DCC plc	110	Elementis plc
31	Mondi plc	71	Diploma plc	111	Essentra plc
32	Rolls-Royce Holdings plc	72	Domino's Pizza Group plc	112	Hilton Food Group plc
33	United Utilities plc	73	Dunelm Group plc	113	Keller Group plc
34	Berkeley Group Holdings plc	74	FirstGroup plc	114	Marshalls plc
35	Barrat Developments plc	75	Frasers Group plc	115	Morgan Advanced Materials plc
36	Britvic plc	76	Future plc	116	Morgan Sindall Group plc
37	Burberry Group plc	77	Games Workshop plc	117	Mobico Group plc
38	Capricorn Energy plc	78	Genus plc	118	Premier Foods plc
39	Carnival plc	79	Grafton Group plc	119	Synthomer plc
40	Coats Group plc	80	Hays plc	120	Shell plc

This table lists the sampled firms.

Appendix A2.2
Industrial classifications

Panel A

Industrial Classification Benchmark (ICB)	Number of companies	Number of CEO letters examined
Basic materials	12	108
Consumer goods	17	153
Consumer services	28	252
Energy	6	54
Health care	5	45
Industrials	39	351
Technology	2	18
Telecommunications	4	36
Utilities	7	63
Total	120	1080

This table provides the industrial classifications of the sampled FTSE-350 non-financial firms.

Panel B

Industry classification	Sector
Basic materials	Industrial materials, Industrial metals and mining, Precious metals and mining, and Chemicals.
Consumer goods	Beverages, Food producers, Tobacco, Personal care, drug and grocery stores.
Consumer services	Automobiles and parts, Consumer services, Household goods and home construction, Leisure goods, Personal goods, Media, Retailers, and Travel and leisure.
Energy	Oil, gas and coal, and Alternative energy.
Health care	Health care providers, Medical equipment and services, and Pharmaceuticals and biotechnology.
Industrials	Construction and materials, Aerospace and defence, Electronic and electrical equipment, General industrials, Industrial engineering, Industrial support services, and Industrial transportation.
Technology	Software and computer services, and Technology hardware and equipment.
Telecommunications	Telecommunications equipment and Telecommunications service providers.
Utilities	Electricity, Gas, water and multi-utilities, and Waste and disposal services.

This table lists the sectors included under each industry classification.

Appendix A2.3

Hausman, Breusch-Pagan Lagrange multiplier (LM) and Wooldridge tests

Hausman test

Test of H0: Difference in coefficients not systematic

$$\begin{aligned}\chi^2(01) &= (b-B)'[(V_b - V_B)^{-1}](b-B) \\ &= 12.31\end{aligned}$$

$$\text{Prob} > \chi^2 = 0.7807$$

($V_b - V_B$ is not positive definite)

Based on the Hausman test result ($p > 0.05$), I fail to reject the exogeneity of the unobserved individual effect, providing statistical evidence in favour of the random effects (RE) model (see Amini et al., 2012).

Breusch-Pagan Lagrange multiplier (LM) test for random effects

$$\text{tone}[id,t] = Xb + u[id] + e[id,t]$$

Estimated results:

	Var	SD = sqrt(Var)
-----+-----		
tone	.0037333	.0611006
e	.0022213	.0471304
u	.0011526	.0339497

Test: $\text{Var}(u) = 0$

$$\chi^2(01) = 397.84$$

$$\text{Prob} > \chi^2 = 0.0000$$

Based on the results of the Breusch-Pagan LM test ($p < 0.05$), I reject the null hypothesis that random effects are insignificant (variance across firms is zero, and thus there are no significant differences across units), and conclude that the random effects (RE) model is more appropriate than the simple OLS regression model (see Breusch and Pagan, 1980).

Wooldridge test for autocorrelation

Serial Autocorrelation

Model (1)	Model (2)
Wooldridge test for autocorrelation in panel data H0: no first-order autocorrelation F(1, 119) = 22.305 Prob > F = 0.0000	Wooldridge test for autocorrelation in panel data H0: no first-order autocorrelation F(1, 119) = 22.911 Prob > F = 0.0000

Based on the results of the Wooldridge test for autocorrelation for both models ($p < 0.05$), I reject the null hypothesis of no first-order autocorrelation.

Appendix A2.4

Demonstration of data cleaning for tone measurement

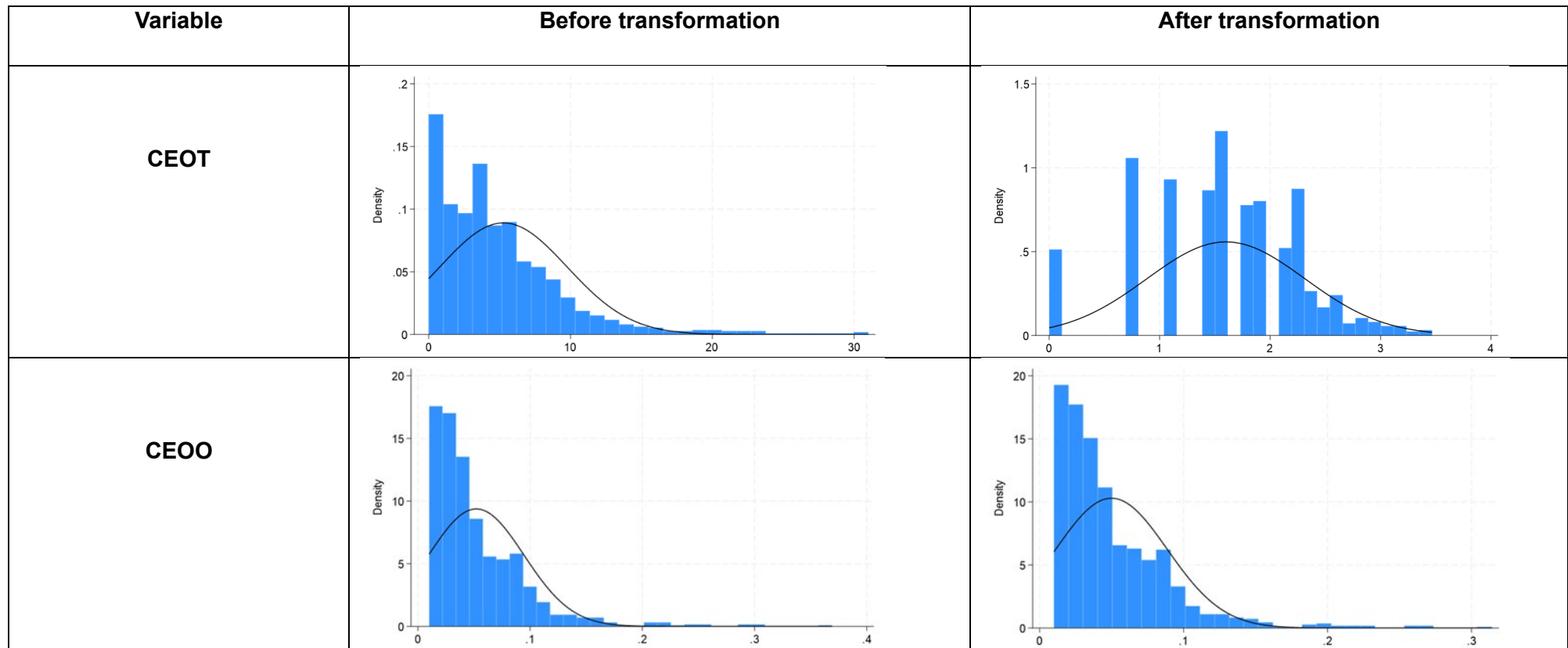
A portion of Antofagasta's 2018 CEO letter

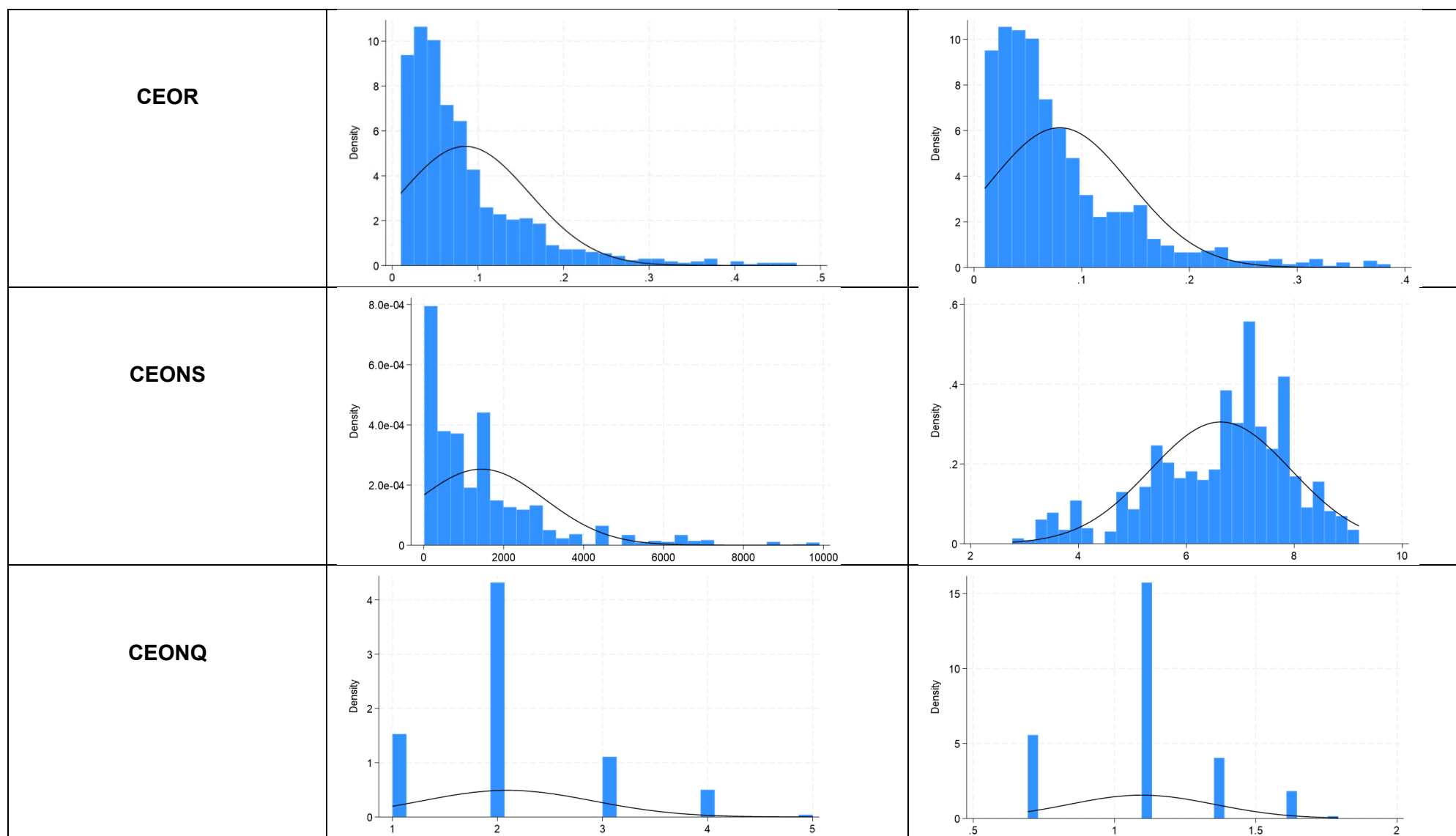
Before preprocessing	DEAR SHAREHOLDERS, I am pleased to share with you the significant progress we have made during the year in many important areas. This progress demonstrates the transformational changes under way in the Group that underpin a set of solid results in 2018. As part of our annual strategy review with the Board, during the year we devoted time to discussing how best to describe the ultimate purpose of what we do as an organisation. This is what motivates all of us who work at Antofagasta, beyond the immediate tasks that everyone is expected to accomplish in her or his specific role.
After preprocessing	dear shareholder please share significant progress made year many important area progress demonstrate transform change way group underpin set solid result part annual strategy review board year devote time discuss best describe ultimate purpose organise motivate work antofagasta beyond immediate task everyone expect accomplish specific role
The overall tone of the letter	0.186 (Positive)

This table shows a portion of text extracted from Antofagasta's CEO letter for 2018, before and after preprocessing (data cleaning). TextBlob then uses the preprocessed text for sentiment analysis.

Appendix A2.5

Histograms for the distribution of the CEO power index components





This table compares the distributions of the CEO power-related variables before and after the components transformation.

Chapter 3

Multi-horizon bankruptcy prediction using Long Short-Term Memory (LSTM) networks

3.1 Introduction

Over the past several decades, corporate financial distress and bankruptcy have become familiar economic realities to many firms, which have posed significant challenges in the financial markets and severely affected firms' stakeholders and society at large (Altman et al., 2019). Consequently, bankruptcy prediction models have emerged as a significant research domain in corporate finance, primarily due to their crucial roles in risk management, regulatory oversight, and investment decisions.

Since the late 1960s, many academic scholars have developed traditional bankruptcy forecasting models using diverse methodologies, such as the pioneering studies of Beaver (1966) and Altman (1968). During the following two decades, several other contributions were made to the research on bankruptcy prediction models, including those proposed by Taffler (1984) and Zmijewski (1984). Despite the different statistical analysis methods employed in these studies, one shared factor is the predominant reliance on accounting, market, and macroeconomic indicators to forecast the likelihood of firms' failure. However, following the high-profile bankruptcies and the collapse of major organisations, there has been an increased interest in corporate failure prediction models, as shareholders have become more cautious about the risks associated with such collapses (Jones, 2023). Therefore, improving the methodologies and the capacity to explain and predict corporate failure has become a key focus in the literature.

Over the years, there has been a growing recognition and interest in the potential and predictive value of qualitative data in forecasting models. Thus, a large body of research incorporates textual analysis into corporate failure prediction models, demonstrating the

explanatory power of language in annual report narratives for predicting corporate failures (e.g., Huang et al., 2023; Tang et al., 2020). For instance, Elsayed and Elshandidy (2020) examine narrative-related disclosures and find that they offer incremental predictive ability relative to financial ratios in traditional models, reporting that they can predict corporate failures up to two years before the actual failure. Similarly, Lopatta et al. (2017) find that firms at risk of bankruptcy tend to use more negative language in their 10-K filings up to three years prior to the actual bankruptcy filings, suggesting that narrative tone has explanatory power in predicting corporate bankruptcies. Moreover, Mai et al. (2019) document that using textual data from 10-K annual filings, in conjunction with accounting-based and market-based data, significantly improves the model's predictive accuracy, revealing the predictive power of textual disclosures.

Although prior studies report that the information content of narrative disclosures enhances the accuracy of bankruptcy prediction models, an important question remains about the extent to which these text-based features derived from corporate narrative disclosures improve predictive performance, particularly across multiple prediction horizons. This leaves significant gaps in understanding the role of textual factors in bankruptcy prediction models, particularly regarding whether all textual features extracted from accounting narratives contribute equally to predictive performance, or whether certain linguistic features are more informative than others as the prediction horizon changes. The present study addresses this gap by examining whether the inclusion of text-based features derived from CEO letters can enhance the understanding of firms' bankruptcy risks across various prediction horizons. Specifically, it investigates the extent to which textual features derived from CEO letters complement financial data in assessing multi-horizon bankruptcy risks. Thus, this study extends the existing literature in two key aspects.

First, CEO letters to shareholders, an essential component of annual reports, serve as a critical source of qualitative data that has attracted significant attention for their potential to convey forward-looking information (Amernic et al., 2010). Such narratives have consistently

been found to be used by investors when making investment decisions, indicating their prominence and importance (Kiattikulwattana, 2019; Patelli and Pedrini, 2013). In addition, various studies have established that linguistic cues and signals embedded in CEO letters supplement the information provided in quantitative disclosures to forecast firms' future performance (e.g., Abrahamson and Amir, 2006; Boudt and Thewissen, 2019). However, despite the promising information value of CEO letters, extracting textual features from such rich qualitative data sources and incorporating them into multi-horizon bankruptcy prediction models has received limited attention. Accordingly, this study expands on the existing literature by incorporating multiple textual dimensions of CEO letters into a multiclass bankruptcy prediction model that accounts for varying levels of financial distress over various prediction horizons.

Second, although integrating qualitative and quantitative data has been established to significantly enhance the accuracy of bankruptcy prediction models, less is known about whether the predictive contributions of textual elements embedded in narrative disclosures differ across varying stages of financial distress. Thus, there is no clear evidence in the literature on whether all linguistic features derived from annual report narratives provide equally strong predictive signals for identifying imminent and longer-term financial distress, or whether their predictive value varies across different stages of distress and prediction horizons. The present study, therefore, contributes to the bankruptcy prediction literature by investigating the role and informativeness of each textual factor derived from CEO letters in predicting corporate failures across firms with varying levels of distress over different prediction horizons.

These distinctive aspects collectively motivate me to investigate the incremental explanatory power of the textual data derived from CEO letters for multi-horizon bankruptcy risk prediction and to assess the individual contribution of each textual feature to predictive performance. The present study draws on a sample of UK FTSE 350-listed firms from 2002 to 2022, encompassing both firms that failed and are legally declared insolvent and those

that remained operational and active throughout that period. Using a two-layer stacked Long Short-Term Memory (LSTM) model, I develop three configurations that share the same architecture but differ in their input features: the first includes only financial data, the second includes only textual data, and the third integrates both. In addition, a three-year rolling window is employed, allowing the model to capture and learn temporal dependencies in sequential data and thereby predict whether a firm is likely to remain operational or fail within one, two, or three years. Financial indicators include accounting-based, market-based, and macroeconomic variables. Whereas, textual indicators include the readability of CEO letters, as measured by the BOG index; the sentiment conveyed in CEO letters; CEO turnover; and three representative topics in CEO letters, which are: Customer relations, Operations, and Production activities and environment that are derived through Latent Dirichlet Allocation (LDA). My findings demonstrate that integrating textual data with financial data significantly enhances predictive performance, underscoring the incremental explanatory power of the textual information contained in CEO letters. Interestingly, although all textual variables appear to enhance predictive performance, I find that the informativeness and individual contributions of text-based features extracted from CEO letters vary with the prediction horizon. In particular, the readability of CEO letters, compared to other text-based features, appears to provide the most informative predictive signals of imminent bankruptcy risk, enhancing the model's prediction of short-term bankruptcy risks. Meanwhile, the representative topics of CEO letters appear to be more informative than other textual factors for improving the detection of medium-term bankruptcy risks. Lastly, CEO turnover appears to be the most informative text-based feature for enhancing the identification of longer-term bankruptcy risks.

Taken together, the findings of this study empirically indicate that the information content of CEO letters is an important factor in assessing multi-horizon bankruptcy risks. These findings thus enrich the ongoing discussion of the usefulness of textual information conveyed in accounting narratives by demonstrating that the predictive value and the individual

contribution of textual features extracted from CEO letters differ across financial distress stages and prediction horizons. In practice, an improved understanding of the role and informativeness of the textual information provided in CEO letters in enhancing the accuracy and reliability of bankruptcy prediction models facilitates more informed decision-making for various stakeholders. Thus, by prioritising the textual features most relevant to the prediction horizon of their interest, investors, banks, and regulators could use such textual dimensions of CEO letters in their evaluations to make more informed investment decisions, risk assessments and timely regulatory interventions.

The remainder of this paper proceeds as follows. Section 2 discusses the theoretical considerations and relevant prior literature. Section 3 outlines the research methodology, discusses the design and formulation of the two-layer stacked Long Short-Term Memory (LSTM) model, and describes the evaluation metrics used to assess predictive performance. Section 4 presents the empirical results. Section 5 discusses the research findings and their implications. Section 6 concludes the paper.

3.2 Theoretical background

Corporate failure prediction has emerged as a significant research domain in accounting and finance, with conventional models heavily relying on financial-based measures (e.g., Altman, 1968; Beaver, 1966; Taffler, 1984). Although these traditional bankruptcy models have undergone thorough validation and remain widely recognised as valuable tools for bankruptcy prediction, they have faced significant criticism over the years due to their assumptions and limitations (Hillegeist et al., 2004). For example, Jayasekera (2018) critiques these models for relying solely on the backward nature of financial statements as their input and for their rigid assumptions, such as linearity and normality. Moreover, Balcaen and Ooghe (2006) argue that corporate failure prediction models limited to financial ratios exhibit serious drawbacks, as these models implicitly assume that all critical failure/success

factors are reflected in balance sheets and financial statements. However, financial statements and ratios derived from them do not fully capture all relevant information about firms' performance (Lev, 2018; Tang et al., 2020), thereby limiting the effectiveness of financial ratios as sole predictors of future firm performance. Consequently, over the last decade, studies have adopted a combined approach that integrates financial and non-financial variables, including qualitative data, to enhance the predictive power and accuracy of bankruptcy prediction models (e.g., Huang et al., 2023; Mai et al., 2019; Tang et al., 2020).

Qualitative information provides critical insights into predicting corporate failure risk while improving traditional corporate failure models' accuracy and predictive power (Ahmed et al., 2024). Several studies support this argument by demonstrating that the information value of accounting narratives enhances the explanatory power of bankruptcy prediction models. For instance, within the UK context, Elsayed and Elshandidy (2020) employ logit regression models to predict corporate failure using narrative-related disclosures, providing evidence that such disclosures provide valuable information that can predict corporate failure up to two years before actual failure. Within the Chinese context, Huang et al. (2023) demonstrate that the sentiment reflected in MD&A sections and audit reports can significantly enhance the performance of financial distress prediction models. Within the US context, Mai et al. (2019) provide evidence suggesting that deep learning models achieve improved predictive performance when textual disclosures are included in the model to predict a firm's bankruptcy risks.

Several theoretical perspectives underpin the incorporation of narrative data into bankruptcy prediction models, providing a robust foundation for understanding how and why firms communicate and disclose information through annual report narratives, particularly during periods of financial distress. For instance, according to the agency theory, one of the dominant theories in this field of research, the interests of managers and stakeholders do not always align, leading to potential conflicts of interest, as managers are assumed to be utility-

maximising individuals who act in their self-interest (Schleicher and Walker, 2010). The theory thus asserts that these conflicts of interest create information asymmetries, in which managers possess more information about the firm's operations and actual financial position (Jensen and Meckling, 1976).

Although increased corporate disclosures are intended to reduce information asymmetries and agency costs (Healy and Palepu, 2001), managers retain discretion over the characteristics of these disclosures, such as quality (Brown and Hillegeist, 2007), frequency (Buskirk, 2012) and tone (Bassyouny et al., 2020). This discretion is critical, especially during periods of financial distress, as empirical evidence suggests that managerial incentives and agency-driven motives, including the disclosure and withholding of specific information, shape narrative disclosures (Edgar et al., 2018). For instance, career concerns may incentivise managers to withhold negative information to address these issues before they become evident, or to release it to shareholders to avoid career-related costs (Baginski et al., 2018). Additionally, various studies provide evidence demonstrating that managers voluntarily disclose information to lower agency costs, improve their reputation and increase their compensation, which aligns with the assumption of agency theory that managers are self-interested (e.g., Chung et al., 2015; Kumar, 2024; Obeng et al., 2021).

The impression management theory uses the assumptions rooted in and builds on the agency theory by explaining managerial disclosure strategies as opportunistic behaviour, where managers strategically craft accounting narratives to influence stakeholders' perceptions (Merkl-Davies and Brennan, 2011). The theory elucidates that managers strategically manipulate the content and presentation of information in annual report narratives to strategically persuade stakeholders and influence their opinions of the firm's performance and future outlook (Boudt and Thewissen, 2019; Moreno et al., 2019).

Impression management comprises various strategies that managers use to construct a positive image of the firm, even when the firm is performing poorly. Merkl-Davies et al. (2011) discuss two impression management strategies: concealment and attribution.

Concealment can be achieved by obfuscating poor performance through tone and readability manipulation (e.g., Asay et al., 2018a; Huang et al., 2018), rhetorical manipulation (e.g., Aerts and Yan, 2017) and minimal disclosures on negative outcomes, and emphasising positive outcomes (e.g., Yang and Liu, 2017). Meanwhile, attribution is a defensive framing technique managers use to claim responsibility for success while shifting the blame for bad news away from themselves and onto external factors (e.g., Edgar et al., 2018; Talbot and Boiral, 2018).

Leung et al. (2015) provide empirical evidence suggesting that managers of firms with a higher risk of financial distress deliberately conceal information in narrative disclosures about poor organisational performance and future outlooks to divert stakeholders' attention from adverse outcomes. In addition, Keusch et al. (2012) find that during crises, managers of financially distressed firms tend to use more defensive attributions in annual report narratives to present themselves in the best possible light and dissociate themselves from any negative outcomes. Meanwhile, Lopatta et al. (2017) provide evidence demonstrating that firms at risk of bankruptcy use more cautious language and negative words in their corporate disclosures.

Moreover, signalling theory offers an additional perspective on how managers use accounting narratives to signal information and communicate with external stakeholders. According to signalling theory, managers of firms facing elevated risk levels are incentivised to disclose additional risk-related information, demonstrating their ability to identify and manage risks effectively (Elshandidy and Shrivies, 2016). These disclosures serve as a signalling mechanism, distinguishing well-managed firms from those with weaker risk management practices. Therefore, narrative disclosures, particularly during periods of financial distress, signal firms' stability and strategic direction, thereby mitigating the information asymmetry problem and the costs of distress, such as the cost of capital, reputation and contracting (Cheynel, 2013; Healy and Palepu, 2001). For instance, Hersel et

al. (2023) find that during periods of distress, managers employ language and tone strategically to signal their identification of the risks and commitment to managing them.

According to Dobler (2008), incentives significantly drive managerial decisions to voluntarily disclose risk-related information. Thus, beyond signalling financial stability and risk management, firms must also maintain their organisational legitimacy. The legitimacy theory asserts that a "social contract" exists between organisations and the broader society, which can be jeopardised by legal, economic or social sanctions, resulting in its organisational legitimacy and, thereby, its continued existence being threatened (Dowling and Pfeffer, 1975). Hence, managers are incentivised to voluntarily reveal events or instances that could threaten their organisational legitimacy, in an attempt to gain, maintain, and/or repair it (O'Donovan, 2002).

Given that the legitimacy theory is rooted in stakeholders' perceptions, Cormier and Gordon (2001) explain that any managerial responses and strategies implemented to influence how external parties view their firms must be accompanied by voluntary disclosures. In line with this argument, Bodnaruk et al. (2015) demonstrate that managers use a cautious tone in their narrative disclosures during a crisis to manipulate stakeholders and mitigate shareholder litigation. Several studies suggest that when the organisational legitimacy and firms' access to resources are threatened, managers of poorly performing firms are incentivised to use narrative disclosures to employ legitimisation techniques to influence stakeholders' perceptions of their actual performance while also maintaining their legitimacy (e.g., Deegan, 2002; Hummel and Schlick, 2016). Other studies contend that ethical responsibilities, litigation, and reputation risks incentivise managers to disclose risk-related information voluntarily (e.g., Cormier and Gordon, 2001; Holder-Webb and Cohen, 2007). In this context, CEO letters to shareholders may serve as a strategic tool for managing stakeholders' perceptions and maintaining organisational legitimacy.

Moreover, processing fluency provides additional insights into how corporate communications' linguistic aspects influence stakeholders' opinions and their ability to

process fluency. Within narrative disclosures, processing fluency denotes an individual's experience of how easy it is to process and understand the information provided in corporate communications (Rennekamp, 2012). According to the principles of processing fluency, textual disclosures with high fluency and, thereby, greater readability enhance stakeholders' processing and comprehension of the provided information, which in turn boosts their trust and confidence (Asay et al., 2017). Consequently, this fluency helps shrink the information gap among investors, maintaining organisational legitimacy and increasing transparency (Guo et al., 2021).

Previous research highlights the communicative relevance of linguistic features embedded within narrative disclosures, particularly in understanding corporate failure risks. The readability of corporate communications reflects the clarity and ease of comprehending the information provided in such disclosures, with empirical evidence demonstrating that distressed firms tend to employ less fluent language to obfuscate poor performance (e.g., Bonsall et al., 2017; Li, 2008). In contrast, more fluent and readable disclosures facilitate stakeholders' comprehension of the information provided, positively influencing their views on managerial transparency and credibility even during periods of uncertainty (Asay et al., 2017). In addition, sentiment analysis and the tone conveyed in accounting narratives have been shown to be effective in predicting bankruptcy risk (e.g., Lopatta et al., 2017; Mayew et al., 2015). For instance, Ahmed et al. (2024) and Del Gaudio et al. (2020) find that disclosures with a net positive tone are associated with lower bankruptcy risk. Huang et al. (2023) documents that the sentiment extracted from annual report narratives significantly enhances predictive performance, recommending that narrative disclosure sentiment be included in financial distress prediction models. Similarly, it has been established that topic modelling and, thus, identifying and extracting themes and patterns within corporate communications have the potential to predict bankruptcy (e.g., Yang et al., 2018). Therefore, the accounting narratives of firms approaching bankruptcy may contain specific words, patterns, and themes that signal corporate failure.

Overall, the interplay between these theoretical perspectives highlights the strategic nature of narrative disclosures, emphasising the importance of incorporating narrative data into bankruptcy prediction models to better understand how firms communicate financial performance during periods of distress. Agency theory explains managerial incentives for voluntary disclosures, whereas impression management elucidates the strategic and opportunistic behaviour that managers use to manipulate narrative data and shape stakeholders' views. Signalling theory provides insight into how firms signal information to stakeholders to distinguish themselves from others, while legitimacy theory underscores the role of narrative information in maintaining organisational legitimacy. Lastly, how stakeholders process and perceive the information provided in narrative disclosures is shaped by processing fluency, which links managerial disclosure strategic choices to stakeholders' cognitive biases in their assessments of these disclosures. Together, these frameworks provide a more comprehensive understanding of why managers disclose information, especially during periods of instability, and how they do so.

3.3 Methodology

3.3.1 Sample selection and data collection

The present study examines the contribution of quantitative and qualitative data for a sample of UK FTSE 350-listed firms from 2002 to 2022. Three criteria are used to collect the data samples. First, I exclude all financial firms (financial, real estate, and insurance firms) due to their different accounting and financial reporting practices (Schleicher and Walker, 2010). Second, I require that sampled firms have at least three years of financial data and CEO letters. Third, following prior UK-based studies (e.g., Elsayed and Elshandidy, 2020; Tinoco and Wilson, 2013), I define corporate failure due to insolvency and bankruptcy by drawing on both legal and financial distress perspectives. It is important to note that delisting from the

stock market does not necessarily imply bankruptcy. Hence, firms may be delisted for non-bankruptcy-related reasons, such as privatisation, mergers, acquisitions, or non-compliance with listing requirements, without formally filing for insolvency. Therefore, a firm is declared legally insolvent (i.e., bankrupt) if it has filed for insolvency and is officially classified as "in administrative receivership, in administration, company voluntary arrangement, creditor's voluntary liquidation, in liquidation or when it is dissolved and assumed valueless" on the GOV.UK Companies House Register (Christidis and Gregory, 2010).² This definition, thus, enhances the scope and predictive performance of empirical models, facilitating more accurate classification of firms at risk of failure (Tinoco and Wilson, 2013).

Alongside the legal approach, this study draws on the financial approach in the literature, which classifies firms as distressed when they concurrently endure a decline in market value and their operational profitability is insufficient to cover their financial obligations (Tinoco and Wilson, 2013). Importantly, in this study, this financial criterion is used as a predictor of bankruptcy rather than a direct classification criterion for bankrupt firms. Hence, this study captures financial distress by incorporating a combination of accounting, market and macroeconomic indicators that serve as predictive factors within the model rather than direct criteria for classifying firms as financially distressed. Therefore, in this study, a firm is considered failed and bankrupt if it has formally filed for insolvency proceedings, has been delisted for bankruptcy-related reasons, and consequently, appears on GOV.UK Companies House Register, with status of "in administrative receivership, in administration, company voluntary arrangement, creditors' voluntary liquidation or in liquidation". This methodology provides a more accurate classification of financially distressed firms, enabling a more robust assessment of their risk of bankruptcy and corporate failure. Furthermore, the sample's solvent firms include those that have not undergone legal proceedings, such as insolvency filings, are active and expected to remain operational throughout the study's

² See <https://www.pwc.co.uk/assets/pdf/insolvency-in-brief.pdf> for the technical and legal insolvency (bankruptcy) terminology and procedures used to define corporate failure in the UK.

sampling period, thus maintaining a clear distinction from bankrupt firms. This sampling criterion yields a final sample of 174 firms and 522 firm-year observations for the period 2002 to 2022 (see Appendix A3.1 for the list of sampled firms).

The timeframe from 2002 to 2022 is chosen to ensure the availability of consistent financial data and annual reports, facilitating the development of a three-year rolling window for each firm. In particular, the timeframe concludes in 2022 to ensure data completeness, as several firms that failed after this year did not have annual reports released in the year preceding their failure. Such firms, therefore, could not be included in the analysis, as the model relies on quantitative and qualitative data from one, two and three years before firms' actual failure. In addition, manually collecting CEO letters and extracting their text requires substantial time and effort, especially for those of failed firms, which are formatted to prevent text extraction, necessitating OCR software.³ Therefore, exploring earlier historical periods may further lengthen the already time-consuming collection task. Moreover, this twenty-year period encompasses various economic conditions, including political instability (e.g., Brexit), economic downturns (e.g., the 2008 Global Financial Crisis and the COVID-19 Pandemic), and recovery, allowing the model to account for diverse macroeconomic conditions and, therefore, improving its practical relevance. At the same time, this timeframe provides a substantial, consistent dataset of financial and textual data, enabling the model to learn and capture the patterns of financial distress that gradually emerge and evolve in the years preceding failure.

All accounting and market-based data are obtained from Thomson Reuters Eikon, while macroeconomic data are obtained from the Office for National Statistics (2025) and the Economic Policy Uncertainty Index (EPU) developed by Baker et al. (2016).⁴ In addition, all

³ I use ABBY FineReader OCR software to extract the textual content of CEO letters from the annual reports of delisted firms.

⁴ Official website for UK Economy Policy Uncertainty index (UPU): https://www.policyuncertainty.com/uk_monthly.html

annual reports of active firms are downloaded from their official websites, whereas those of failed firms are retrieved from the UK Companies House register – GOV.UK (Companies House, 2015).⁵ Finally, I use the StyleWriter 4 software to scan each CEO letter and obtain the BOG measure.

3.3.2 Predictive factors

3.3.2.1 Corporate failure indicator

To develop a robust multiclass corporate failure prediction model, I construct a categorical indicator of corporate failure that classifies firms into four categories. Active firms that are operational throughout the sampling period and are expected to remain active are labelled 0. While distressed firms expected to fail and cease operations within a year are labelled 1, those within two years are labelled 2, and those within three years are labelled 3. Thus, unlike conventional binary classification models, which predict whether a firm will remain active or fail, this multiclass approach classifies firms by the projected timing of their failure.

The motivation for using a multiclass approach with a time-varying panel dataset and incorporating multiple prediction horizons is to capture the evolving nature of financial distress. Consequently, each firm in the dataset has data spanning three years, and the model learns from previous years' information when making predictions. This approach stands in contrast to single-period classification (static) models, which have been criticised for yielding biased and inconsistent estimates of the probabilities they seek to approximate (Shumway, 2001), thus necessitating more dynamic predictive models that account for

⁵ For active firms, if there are any missing annual reports from corporate official websites, annual reports are obtained from <https://www.annualreports.com/>

temporal dependencies and changes in firms' risk of failure and health over time (Lukason and Laitinen, 2019).

The selection of a three-year window is based on empirical and practical grounds. Empirical studies on corporate failure prediction suggest that financial distress and failure are not instantaneous; rather, they develop gradually, with firms often experiencing progressive deterioration in their financial health and stability before filing for bankruptcy (e.g., Zhou et al., 2022). In addition, studies, such as those by Zhou et al. (2016) and Sun et al. (2021) demonstrate that prediction models that account for multiple forecasting horizons perform better and more efficiently in capturing firms' potential risk of financial distress and failure than binary models with a one-year prediction horizon. At the same time, extending the prediction horizon beyond three years, such as four or five years, may introduce uncertainty into the model and reduce its predictive performance and accuracy. According to Quaedvlieg (2021), the variance of loss differences often increases with the prediction horizon; therefore, adding horizons to the model may decrease its power. In addition, du Jardin (2015) analyses the relationship between model accuracy and prediction horizon in prior research, finding a general pattern in which accuracy tends to decrease as the prediction horizon increases. Moreover, data availability issues may arise when extending the timeframe beyond three years, especially for failed firms, as some may lack annual reports and/or financial data. Consequently, this study employs a three-year window to account for the temporal dimensions that lead to corporate failure, while ensuring that the model's predictive power is not compromised by noise introduced by adding additional horizons, which may be uninformative and thus affect the model's practical relevance and reliability.

3.3.2.2 Financial predictors

Consistent with prior research (e.g., Elsayed and Elshandidy, 2020; Tinoco and Wilson, 2013), this study incorporates a comprehensive set of financial indicators to assess firms' financial health and stability. Table 3.1 presents the list of financial predictors used in this study. As demonstrated in Table 3.1, these indicators include accounting-based, market-based, and macroeconomic variables, given their prominence in the literature and relevance to the study.

For the accounting-based variables, this study employs the five financial ratios introduced by Altman (1968) in his Z-score model: Working Capital/Total Assets (WC/TA), Retained Earnings/Total Assets (RE/TA), Earnings Before Interest and Taxes/Total Assets (EBIT/TA), Market Value of Equity/Total Liabilities (MVQ/TL) and Sales/Total Assets (S/TA). These accounting ratios are appealing on several grounds. Theoretically, they have been extensively documented in the bankruptcy prediction literature as significant predictors of bankruptcy across various firms and industries (Almamy et al., 2016; Barboza et al., 2017). In addition, these ratios provide essential information on various aspects of a firm's overall financial health, including liquidity, profitability, leverage, solvency, and activity (Altman et al., 2017). This emphasises the predictive power and reliability of these accounting ratios in predicting corporate failure. Therefore, by incorporating these five accounting ratios, the model is designed to capture the key factors of a firm's financial health that signal potential bankruptcies.

In addition to accounting ratios, I include two market-based variables: the log of a firm's equity price (PRICE) and the volatility of market returns (Volatility). Market-based variables have been widely recognised in bankruptcy literature. Academics such as Balcaen and Ooghe (2006) argue that relying solely on accounting ratios derived from financial statements does not include all the information relevant to predicting bankruptcy, and incorporating market-based variables is very likely to help address this deficiency. Similarly,

Hillegeist et al. (2004) suggest that the stock market is an alternative information source, drawing on financial statement data and other sources. More recently, Alam et al. (2021) include both financial and market-based variables in their corporate failure prediction model and document that it can accurately predict corporate failure. Therefore, incorporating market-based variables into bankruptcy prediction models is appealing on several grounds. Firstly, market prices reflect information in financial statements, as well as additional information not included in these statements (Agarwal and Taffler, 2008). Secondly, market prices complement historical accounting data in predicting bankruptcy, reflecting anticipated future cash flows, whereas accounting ratios alone reflect firms' historical performance (Tinoco and Wilson, 2013). Thirdly, according to Beaver et al. (2005), market variables provide a direct evaluation of volatility, a valuable predictor of corporate bankruptcy, as increased volatility is associated with a higher likelihood of bankruptcy and is not reflected in financial statements.

Furthermore, I include two macroeconomic variables: the annual Retail Price Index (RPI) as a proxy for inflation, and the annual Economic Policy Uncertainty Index (EPU) as a measure of economic policy uncertainty in the UK market, both used at the level form.⁶ According to Jones (2017), external economic conditions and the overall economic state are expected to influence the frequency of corporate bankruptcies. Khoja et al. (2019) also suggest that external factors, such as macroeconomic conditions, that can impact earnings, debt, equity or cash flow, are very likely to influence the risk of bankruptcy. Hence, during periods of rising inflation, financially distressed firms may be significantly affected, increasing their risk of bankruptcy (Tinoco and Wilson, 2013). Similarly, a decline in market performance may signal broader economic stress or external shocks, potentially leading to greater losses and costs, and therefore, increasing the likelihood of bankruptcy (Branch, 2002). The inclusion of

⁶ Both RPI and EPU were also measured as annual percentage changes and included in the model; however, an enhanced predictive performance was observed when they were included at the absolute level, suggesting that annual fluctuations may be introducing noise into the model.

macroeconomic indicators is further supported by recent studies, such as that of Tinoco et al. (2018), which demonstrate that integrating macroeconomic indicators with accounting and market data improves the performance and power of bankruptcy prediction models.

Table 3.1
List of financial predictors

Type	Factor	Definition
Accounting	Working Capital/Total Assets	A measure of the firm's net liquid assets relative to its total capitalisation.
	Retained Earnings/Total Assets	A measure of the firm's cumulative profitability over time.
	Earnings Before Interest and Taxes/Total Assets	A measure of the firm's assets' true productivity, excluding any tax or leverage factors.
	Market Value of Equity/Total Liabilities	A measure of the extent to which the firm's assets can decline in value before liabilities surpass assets, leading to the firm's insolvency.
	Sales/Total Assets	A measure of the firm's management capacity to navigate competitive conditions, reflecting the ability of the firm's assets to generate sales.
Market	PRICE	A measure of the logarithm of firms' equity price.
	Volatility	A measure of total risk, calculated as the standard deviation of historical market returns.
Macroeconomic	RPI	A measure of inflation in the economy.
	EPU	A measure of economic policy uncertainty.

This table presents and defines the financial indicators used in this study.

3.3.2.3 Textual predictors

A key innovation of this study lies in the use of the information content of CEO letters to predict firms' multi-horizon bankruptcy risk. I select CEO letters as the primary source of textual data in this study for several reasons. First, across all communication outlets, CEO letters in annual reports serve as an important source of information for investors, as CEOs use such narratives to provide a qualitative description and summary of firms' current performance and to convey their views on future prospects and objectives (Kiattikulwattana, 2019). In addition, studying the information content of CEO letters can reveal how CEO mindsets operate during periods of global boom and expansion, thereby improving the assessment of matters such as risk management policies and practices (Amernic et al., 2010, p. 6). Hence, an improved understanding of such matters can be relevant to various stakeholders seeking early insights into potential firms' financial issues. Moreover, several studies in the literature demonstrate that the information contained in CEO letters is valuable for predicting firms' future performance (e.g., Boudt and Thewissen, 2019; Patelli and Pedrini, 2013). Thus, numerous studies in the literature emphasise the explanatory and predictive power of textual disclosures, particularly in predicting corporate failure.⁷ Therefore, by extracting text-based features from CEO letters, this study highlights the communicative and predictive relevance of the linguistic features in such disclosures as textual predictors of corporate failure. Table 3.2 presents the list of text-based features extracted from CEO letters and used as textual predictors in this study.

⁷ Mai et al. (2019) incorporate textual disclosures in their machine learning bankruptcy prediction model, demonstrating that they significantly improve the model's predictive power. Similarly, Elsayed and Elshandidy (2020) and Zhao et al. (2023) investigate whether narrative-related disclosures can predict corporate failure and find that they significantly improve the model's predictive performance.

To prepare CEO letters for feature extraction, I clean all letters using widely adopted data preprocessing techniques, including lowercasing, removal of punctuation and stop-words, and tokenisation (Uysal and Gunal, 2014). Such methods are particularly important as they enhance the quality of information extracted from textual data, ensure consistency and eliminate noise. Nonetheless, it is important to note that in this study, I do not apply word normalisation techniques (e.g., stemming and lemmatisation) to preserve hidden semantic structures and contextual meanings, which is especially important for topic modelling. This methodological choice aligns with the findings of Schofield and Mimno (2016), who investigate the influence of stemming and lemmatisation on topic modelling performance, demonstrating that they degrade topic stability and provide no meaningful improvement in fit or topic coherence. Similarly, Banks et al. (2018) recommend not performing stemming or lemmatising when generating topic models or extracting emerging topics from textual data.

After the CEO letters have been prepared and cleaned, I first conduct sentiment analysis to assess whether the sentiment conveyed in the letters provides valuable information beyond the disclosed financial data. A substantial body of literature examines the predictive power of sentiment across different narrative disclosures. For instance, Huang et al. (2023) investigate the sentiment reflected in the audit reports and management discussion and analysis (MD&A) sections of listed firms in the Chinese market, revealing that textual sentiment scores significantly improve the performance of their financial distress prediction model. Meanwhile, Ahmed et al. (2024) investigate the tone of several narrative sections of annual reports of UK firms, including the chairman's statement, performance commentary and the business review section and find that they provide significant incremental predictive power. Similarly, Lopatta et al. (2017) examine whether the language employed in 10-K filings of US firms can reflect firms' risk of failure, documenting that firms with a higher likelihood of bankruptcy employ more negative words than financially stable firms.

In this study, I measure the overall sentiment of each preprocessed letter using the VADER sentiment analysis technique, "Valence Aware Dictionary and sEntiment Reasoner".⁸ VADER is a lexicon and rule-based approach that assigns sentiment scores based on the valence of words and their contextual aspects (Hutto and Gilbert, 2014). In particular, it provides a compound sentiment score for a given text that reflects its overall sentiment intensity and valence. I then incorporate the resulting sentiment compound scores for each CEO letter, ranging from -1 (stronger overall presence of negative sentiment) to 1 (stronger overall presence of positive sentiment), as a textual predictor of bankruptcy risk in the model, named "Sentiment". Consistent with prior research, I expect that the sentiment conveyed in CEO letters to provide incremental predictive power, thereby enhancing the model's performance.

In addition, I assess the readability of CEO letters to investigate whether textual complexity reflects a firm's risk of bankruptcy. Prior research examining the relationship between corporate performance and the readability of annual report narratives has consistently demonstrated that high-performing firms' narratives are easier to read than those of low-performing firms (e.g., Asay et al., 2018a; Bakarich et al., 2019). According to Fabrizio and Kim (2019), firms with poor performance deliberately maintain a certain level of opacity in their textual disclosures, making them more complex and less readable to obscure negative outcomes. Such increased textual complexity and its variability can therefore signal deteriorating financial performance, making the readability of CEO letters a credible textual predictor of corporate failure. Consequently, in the present study, I measure the readability of CEO letters using the BOG index, proposed by Bonsall et al. (2017), which assesses the

⁸ I also measure the net sentiment score for each CEO letter using the Loughran-McDonald (LM) wordlist by subtracting the total count of negative words from the total count of positive words. However, this measure did not significantly affect the model's predictive performance, leading me to exclude it to avoid multicollinearity issues introduced by including both sentiment measures.

readability of financial texts using a graded wordlist rather than relying solely on syllable count or word length.

Although the BOG index is cited less often than other readability metrics, its inclusion is expected to enhance the model's predictive power, as its assessment of textual complexity is designed to address concerns raised about other readability metrics.⁹ In line with Bonsall et al. (2017), I use StyleWriter 4, a commercial software, to obtain the BOG index for each CEO letter in the sample. StyleWriter 4 software analyses each document's sentence Bog, word Bog, and pep up writing style (Pep), providing a comprehensive BOG index that reflects the document's overall readability and complexity based on these three distinct linguistic aspects. BOG index values are classified as: 0–39 ("Good"), 40–69 ("Fair"), 70–100 ("Poor"), and >100 ("Bad"). In the present study, I create a binary categorical variable named "BOG", where a value of 1 denotes firms with poor and bad readability (BOG index values ≥ 70), and a value of 0 indicates firms with good and fair readability (BOG index values ≤ 69). Consistent with prior research, I expect firms with harder-to-read CEO letters to be associated with higher bankruptcy risk, while firms with easier-to-read CEO letters to be associated with lower bankruptcy risk.

Furthermore, I perform topic modelling to examine whether latent representative topics and underlying themes embedded in CEO letters provide incremental information indicative of firms' future performance and likelihood of failure. Identifying topics and themes in narrative disclosures may provide valuable cues to differentiate between bankrupt and non-bankrupt firms, as the topics and issues discussed in such disclosures may differ significantly (Yang et al., 2018). Among the seminal studies in this area are those of Tennyson et al. (1990) and

⁹ There is an ongoing debate concerning the most effective and suitable readability measure for financial texts in the literature. The FOG index, for example, is widely applied to measure the readability of text disclosures; however, it is criticised for the poor specification of its "complex words" component, as it tends to label frequent financial terms as complex when investors can easily understand them (Loughran and McDonald, 2014). Similarly, the file size readability metric is also argued to be an unreliable readability proxy, as documents might be considered large due to their textual content and the inclusion of other elements (Bonsall et al., 2017).

Smith and Taffler (2000), who investigate the relationship between the thematic content of annual report narratives and corporate failure, demonstrating that underlying themes in narrative disclosures are particularly useful for distinguishing between bankrupt and healthy firms. Conaway and Wardrope (2010) also identify key topics reflected in CEO letters and find that CEOs strategically select and omit certain themes to shape readers' minds, indicating that the information content of narrative disclosures serves a purpose beyond only communicating factual information to stakeholders. Therefore, prior research suggests that latent topics and themes in annual report narratives may contain useful information that could improve the ability to discriminate between bankrupt and non-bankrupt firms.

In this study, I employ the Latent Dirichlet Allocation (LDA) topic modelling technique to extract representative topics from CEO letters, assigning each letter to the topic with the highest probability based on its word distribution. LDA is one of the most popular statistical and topic models that assumes documents exhibit multiple topics, where each word in each document is drawn from one of the topics, and the selected topics are determined by each document's topic distribution (Blei, 2012). Therefore, it can be a particularly effective technique for investigating the "hidden" thematic structure and latent topics within collections of text documents, making it attractive for communication research (Maier et al., 2018). In the context of narrative disclosures, Dyer et al. (2017) suggest that LDA is well-suited for analysing textual disclosures, as it facilitates the objective and replicable examination of the thematic content of lengthy documents and relies on a very limited set of assumptions likely to be met in narrative disclosures. Thus, one of the key advantages of LDA is that it is an unsupervised probabilistic model that uses the likelihood approach to learn and uncover topics that often appear in a corpus of documents, without the need for predefined word lists or topic classifications and instead depends on the observation that words commonly appearing together are often semantically related (Brown et al., 2020; Lewis and Young, 2019).

Consequently, I extract representative topics embedded in CEO letters of failed and active firms and use them as textual predictors in my model. In addition, I employ coherence and perplexity measures to determine the optimal number of latent topics to extract from CEO letters by evaluating LDA models with varying numbers of topics (Röder et al., 2015; Zhao et al., 2015). Accordingly, I select the three-topic model for its highest coherence score and lowest perplexity among the other topic counts, reflecting an optimal balance between semantic clarity and statistical fit (see Appendix A3.2 for coherence and perplexity scores). In addition, to interpret the identified topics, which is a common challenge in topic modelling (Hoberg and Lewis, 2017) I extract and analyse the top twenty-five most frequent words within each topic and limit the interpretation to those words with statistically significant Bonferroni-adjusted p-values (see Appendix A3.2 for topic interpretations). The three identified representative topics, named "Topic 1", "Topic 2", and "Topic 3", are represented as binary variables in my model, with each CEO letter's representative topic variable set to 1 and the remaining topic variables set to 0. Consistent with prior evidence, I expect that representative topics extracted from CEO letters provide valuable information, enabling improved differentiation between bankrupt and active firms.

Lastly, I incorporate CEO turnover as an indicator of corporate failure in the model to capture early warning signals of deteriorating financial performance. Previous studies have consistently linked CEO turnover to poor performance (e.g., Bushman et al., 2010; Jenter and Kanaan, 2015; Kato and Long, 2006). Thus, boards often respond early by dismissing CEOs as a corrective measure before performance issues become widely known, especially to shareholders (Ertugrul and Krishnan, 2011; Martin and Combs, 2011). Consequently, changes in the CEO may serve as an early indicator of corporate failure, reflecting underlying financial distress. In the present study, I create a binary categorical variable named "CEO turnover", in which a value of 0 indicates that the same CEO remained in position throughout the three-year window, and a value of 1 denotes that a CEO change occurred during the three-year window. Consistent with previous findings, I expect CEO

turnover to provide useful information that improves the model's ability to differentiate between bankrupt and healthy firms, with distressed firms exhibiting a higher likelihood of turnover than non-distressed firms.

Table 3.2
List of textual factors

Type	Factor	Definition
Textual	Sentiment	A measure of the overall sentiment expressed in CEO letters using the VADER sentiment analysis tool, calculated as a compound score between -1 (stronger presence of overall negative sentiment) and 1 (stronger presence of overall positive sentiment).
	Representative topics	A measure of the underlying representative topics within CEO letters, presented as three binary variables: Topic 1: Customer relations Topic 2: Operations Topic 3: Production activities and environment Where each CEO letter's representative topic variable is set to 1, and the remaining topic variables are set to 0.
	BOG	A measure of text readability presented as a binary variable, where a value of 0 represents good/fair readability, with CEO letters of BOG indexes below 70. In contrast, a value of 1 indicates poor/bad readability, with CEO letters of BOG indexes above 69.
	CEO turnover	A measure of CEO turnover presented as a binary variable, where a value of 0 indicates that the same CEO remained in position throughout the three-year window, and a value of 1 denotes that a CEO change occurred.

This table presents and defines the textual features derived from CEO letters and used in this study.

3.3.3. Model

This study utilises Long Short-Term Memory (LSTM) networks, a special type of recurrent neural network (RNN), to predict firms' risk of corporate failure within a multiclass classification framework. I develop a two-layer stacked LSTM model and create three configurations that share the same architecture but differ in their input variables. One includes only financial data (Model A), one includes only textual data (Model B), and one integrates both (Model C). LSTM networks have been widely recognised for their suitability for sequential data analysis, particularly in forecasting models (Fischer and Krauss, 2018; Graves, 2012). Thus, they are designed to capture temporal dependencies and longer-term patterns, rather than overlooking the gradual progression of financial distress, by exploiting the time-series structure of datasets, making them particularly suitable for my study.

I use a three-year rolling window to generate input sequences for the LSTM model. The dataset comprises firm-year observations, with each firm represented by a sequence of three years of data. Hence, each firm in Model A is represented by three years of financial data only; each firm in Model B is represented by three years of textual data only; each firm in Model C is represented by three years of financial and textual data. In addition, the dataset is then split into training and testing sets. In particular, 80% of the data is assigned to the training set, comprising firms that failed and remained operational from 2002 to 2018, while the remaining 20% is assigned to the testing set, comprising firms that failed and remained operational from 2019 to 2022.

3.3.3.1 Data preparation

Data preparation, a mandatory step, involves techniques such as data cleaning, transformation, and normalisation that properly prepare the data to serve as the model's input for machine learning algorithms (García et al., 2015, pp. 11-16). Hence, machine learning (ML) algorithms are highly dependent on the output of this stage. Therefore, if the raw data is inadequate or contains extreme or redundant information, the model cannot produce accurate results or uncover valuable information.

Numerous well-established data preparation techniques and data mining tasks have been proven essential for addressing various critical issues that could affect the performance of machine learning algorithms.¹⁰ The application of machine learning (ML) classifiers, such as neural networks, requires researchers' intervention and judgment, especially in variable transformation and outlier handling, given the data issues commonly encountered in bankruptcy datasets (Jones et al., 2017). Such issues can, therefore, significantly impact model estimation and predictive performance. Consistent with most studies on corporate failure prediction, I winsorise extreme values within the dataset at the 5th and 95th percentiles to minimise outliers (e.g., Altman et al., 2017; du Jardin, 2021; Elsayed and Elshandidy, 2020).¹¹ In addition, issues related to variance stability pose significant challenges for most business datasets, especially in bankruptcy research. Therefore, it is common practice to transform the data (Jones et al., 2017). Consequently, I employ the Yeo-Johnson

¹⁰ There is consensus in the literature on the importance of data preprocessing before modelling bankruptcy, and handling outliers is one of the most essential tasks to reduce bias arising from outliers (Nyitrai and Virág, 2019). In addition, financial data and features in bankruptcy prediction models often exhibit highly skewed distributions, which can be addressed through data transformation techniques rather than by reducing the number of variables, thereby improving the model's predictive performance (Son et al., 2019). Scaling and normalising input variables can also generally improve the predictive performance of classifiers (Jones et al., 2017). Lastly, handling data imbalance is another important issue to prevent an inaccurate prediction model due to data bias towards the majority class, which is the non-bankrupt firms (Smiti and Soui, 2020).

¹¹ Handling outliers is particularly important for enhancing the predictive power of bankruptcy prediction models, including neural networks and logistic regression (Nyitrai and Virág, 2019).

transformation, developed by Yeo and Johnson (2000), which addresses both positive and negative data, to transform input variables that require transformation to reduce skewness, approximate normality, and stabilise variance.

Moreover, scaling features so that all values fall within the same range and are consistent is particularly important for classification algorithms (Shi et al., 2024). Thus, it is recommended to scale datasets when constructing neural networks for financial and economic forecasting (Ticknor, 2013). Therefore, consistent with prior research on corporate bankruptcy prediction (e.g., Kim et al., 2022; Thor and Postek, 2024), I apply min-max scaling, a widely used technique in machine learning (ML), to ensure that all input features are consistent and on the same scale, which is crucial for the model's predictive power and performance. Lastly, given the challenging problem of data imbalance, particularly in multiclass classification models, handling it during training is essential to avoid bias towards the dominant class, which can lead to inaccurate predictions (Fernández et al., 2018, pp. 19-36). In the context of corporate failure prediction models, healthy firms (the majority class) outnumber distressed firms (the minority class), an empirical fact that raises concerns for classification tasks. This is because machine learning (ML) models are often biased towards the majority class, overlooking examples from the remaining minority classes (Thabtah et al., 2020). Several solutions, including undersampling and oversampling techniques, have been previously proposed to address the class imbalance at the data level. The undersampling technique addresses class imbalance by removing examples from the majority class, while the oversampling technique increases the size of the minority classes until all classes are balanced (Veganzones and Séverin, 2018). However, several academics have criticised undersampling for ignoring many examples from the majority class that could contain valuable information (e.g., Douzas et al., 2018). Given the drawbacks of undersampling, I apply SMOTE (synthetic minority over-sampling technique), proposed by Chawla et al. (2002), to the training dataset to address class imbalance. This effective oversampling

technique creates synthetic examples for the minority classes along the line segment by connecting each example to its K-nearest neighbours (Elreedy and Atiya, 2019).

3.3.3.2 LSTM model architecture

Over the years, recurrent neural networks (RNNs) have been widely used for sequential and time-series prediction (Hewamalage et al., 2021). Such neural networks can remember previous observations by maintaining a cell state that is updated at each time step. However, the main issue with the simple or vanilla RNN is that it poses challenges for learning and modelling long-term dependencies, such as the vanishing and exploding gradient problems (Barthélémy et al., 2024). Long Short-Term Memory (LSTM) networks, a special type of RNN, have been proposed to solve these issues by establishing linear connectivity between cell states and using gated mechanisms (Hochreiter and Schmidhuber, 1997).

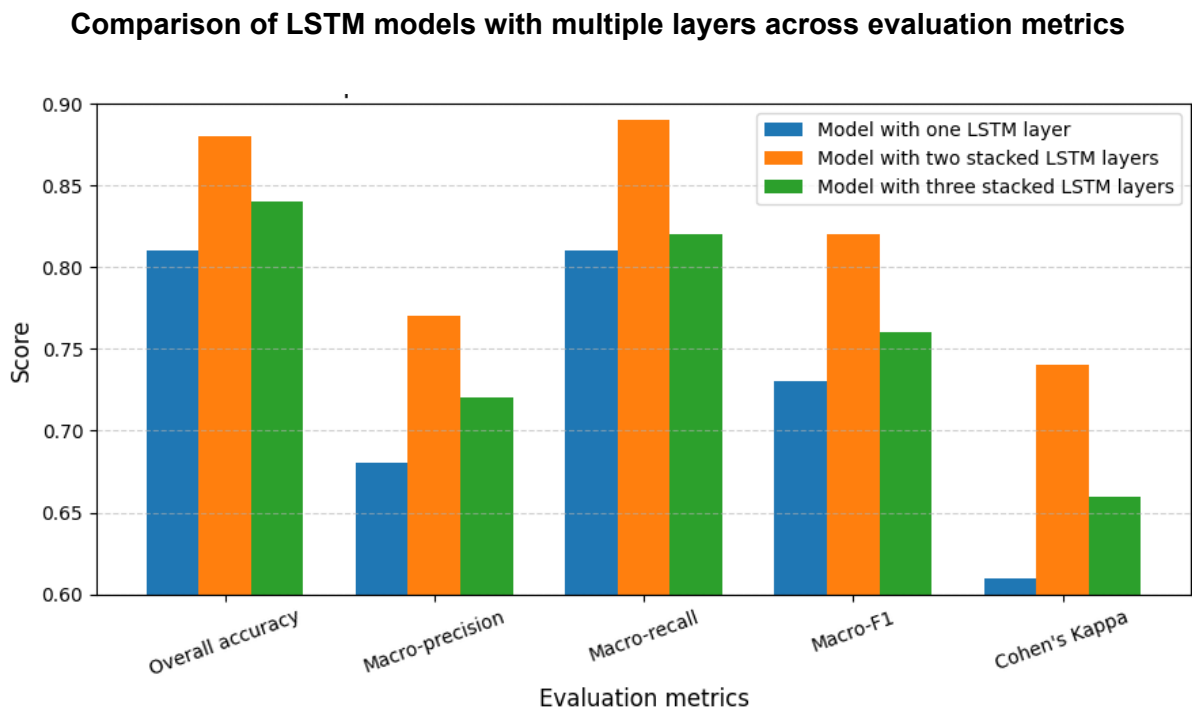
The LSTM architecture typically consists of a set of memory blocks, each containing one or more self-connected memory cells and three gates, which are the input, output, and forget gates that serve to regulate information flow through the model (Graves, 2012). These multiplicative gates facilitate LSTM memory cells' storage and retrieval of information over extended periods, effectively mitigating the vanishing gradient problem in vanilla RNNs.

Thus, based on the output of the sigmoid functions that range between 0 and 1, which depends on the relevance of information, the input gate determines which information to add to the cell state, the forget gate determines which information to retain and which to forget, and the output gate determines which information from the memory cell gets used as output.

A stacked LSTM model with multiple LSTM layers, each containing multiple hidden memory units, has been consistently used in sequential data tasks and for time-series prediction to improve abstraction and the model's ability to understand long-term data dependencies across time steps (e.g., Borovkova and Tsiamas, 2019; Singh et al., 2025). In this study, I first preserve the chronological order of the data within each firm's three-year sequence to

ensure the model learns from evolving patterns and predicts future performance. Then, following the approach adopted by Gülmez (2023), I conduct three experiments with one, two, and three LSTM layers to determine the optimal number of layers. Figure 3.1 illustrates the predictive performance of these models across several evaluation metrics. As shown in Figure 3.1, the model with two stacked LSTM layers achieves the strongest and most balanced overall predictive performance, indicating that it is best suited for the model.

Figure 3.1



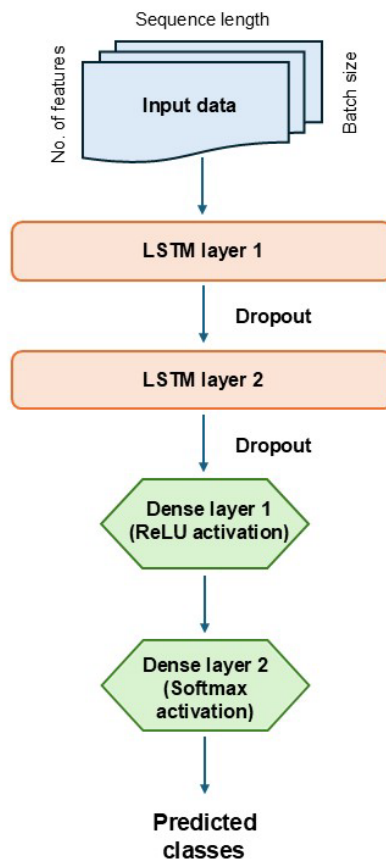
Consequently, in the present study, I develop a two-layer stacked LSTM model and create three configurations that share the same architecture but differ in their input features: the first includes only financial data (Model A), the second includes only textual data (Model B), and the third integrates both (Model C). All three configurations are trained for 134 epochs, determined by the lowest observed validation loss, ensuring fairness among configurations. Figure 3.2 illustrates the architecture of the Long Short-Term Memory (LSTM) network

implemented in this study. As shown in Figure 3.2, the model comprises two LSTM layers and two dense layers (LSTM 1 = 256, LSTM 2 = 32, dense 1 = 64, and dense 2 = 4). It is common for the two LSTM layers to differ in size, with the first layer being larger than the second, as the first LSTM layer captures general, non-specific features that require more capacity, while the second targets more specific, refined ones (Borovkova and Tsiamas, 2019). Then the output from the second LSTM layer is passed through a fully connected dense layer with a rectified linear unit (ReLU), a widely used activation function that enables the model to learn non-linear representations of the data (Borovykh et al., 2018). ReLU has been shown to perform better than other activation functions, enabling the network to discover complex patterns and enhancing learning rates, which, in turn, improve classification and predictive performance (e.g., Orimoloye et al., 2020; Sharma et al., 2020). Following that, the output from the first dense layer is passed to the second dense layer, which is the output layer with softmax activation, commonly used for multiclass classification problems (Chen et al., 2024), that transforms the raw output into class probabilities.

Consequently, as a whole, the output of LSTM layer 1 is fed to LSTM layer 2, and subsequently, the output of LSTM layer 2 is fed into dense layer 1 with ReLU activation, and then, the output of dense layer 1 is passed through dense layer 2, which is the output layer with softmax activation that produces class probabilities. Following that, predicted class labels are assigned by taking the argmax of the model's generated probability vector for each instance, selecting the class with the highest probability. In addition, I apply dropout regularisation rates of 50% between LSTM layer 1 and LSTM layer 2, and 40% between LSTM layer 2 and dense layer 1. Dropout regularisation is a standard technique to prevent neural networks from overfitting, as it randomly drops units during training, preventing excessive "co-adaptation" among neurons (Ibraheem et al., 2025; Koivu et al., 2022). Given that the class labels are integers (0, 1, 2, and 3), I use the sparse categorical cross-entropy loss function, which is efficient for multiclass classification with integer-encoded labels, to quantify the divergence between the predicted class probabilities and the true integer class

labels (Terven et al., 2025). I also use Adam optimiser, a widely used optimisation algorithm for neural networks (Nikou et al., 2019; Vo et al., 2019). Lastly, as recommended by Bengio (2012), I use a batch size of 32, since values above 10 take advantage of the speed-up of matrix-matrix multiplications.

Figure 3.2
Model architecture for the two-layer stacked LSTM model



3.3.3.3 Model evaluation metrics

Consistent with prior studies on multiclass financial distress prediction (e.g., Chen et al., 2024; Jiang and Jones, 2018; Sun et al., 2023), I use a confusion matrix to evaluate the performance of the multiclass bankruptcy prediction model (see Appendix A3.3 for the evaluation metrics' calculation formulas). To evaluate the overall model performance, I calculate overall classification accuracy, macro-precision, macro-recall, macro-F1 score, and Cohen's kappa. Overall classification accuracy is a widely used metric in multiclass classification, directly derived from the confusion matrix and reflects the proportion of correctly predicted instances across all classes (Jeong, 2025). Similarly, macro-average metrics, including macro-precision, macro-recall, and macro-F1, are frequently used to evaluate overall multiclass classification performance by separately assessing each class's performance and then averaging across classes, treating all classes equally, regardless of their sizes (Sun et al., 2023). In addition, to evaluate class-wise performance, I calculate precision, recall, and the F1 score for each class, which is especially relevant in multiclass classification tasks, as it assesses the model's performance in identifying each class (Chen et al., 2024). Cohen's kappa is another statistical test used in multiclass classification problems that measures the degree of agreement beyond chance between the model's predicted and actual classifications (Jeong, 2025).

Following Sokolova and Lapalme (2009), I do not include receiver operating characteristic (ROC) curves within the evaluation metrics, as when expressed in the form of the area under the curve (AUC), the measure is only applicable in the event that there are two classes and lacks a clear generalisation to multiple classes. Hence, while various extensions to the ROC curve have been proposed for multiclass classification, there is no clear consensus on the most suitable approach. Furthermore, as noted by Ben-David (2008), both ROC curves and Cohen's kappa measure classifier accuracy while accounting for random successes, but in distinct ways. Importantly, their research suggests that Cohen's kappa provides a simple,

comprehensible alternative to ROC curves and AUCs when they are deemed unsuitable, such as in multiclass problems.

3.4. Empirical Results

3.4.1 Overall prediction performance

Table 3.3 presents the overall predictive performance of the three LSTM model configurations that share the same architecture but differ in their input features. The first model configuration incorporates only financial ratios (Model A); the second includes only textual data (Model B); and the third integrates both financial and textual data (Model C). The results presented in Table 3.3 show each model's overall predictive performance across all four outcome classes using macro-average evaluation metrics, including classification accuracy, macro-precision, macro-recall, and macro-F1 score. Overall predictive performance is also evaluated using Cohen's kappa to assess the level of agreement between each model's predicted and actual classifications beyond random chance, reflecting each model's overall reliability. Class 0 represents solvent firms expected to remain active, whereas Classes 1, 2, and 3 represent distressed firms expected to fail within one, two, and three years, respectively.

As illustrated in Table 3.3, Model A, which relies solely on financial ratios (accounting, market, and macroeconomic variables), achieves an overall classification accuracy of 73%, indicating that it successfully classifies 73% of the firms across all four classes. The macro-precision of 59% indicates that, on average, 59% of classifications across all four classes are correct. This result suggests that Model A struggles to differentiate between the varying levels of financial distress; therefore, it often misclassifies firms, leading to a high false-positive rate. In practice, such misclassifications may limit the model's reliability as an early warning system and, consequently, raise concerns for stakeholders. In contrast, the macro-

recall of 76% indicates that, on average, the model captures 76% of all true instances across all four classes. The macro-F1 of 64% indicates a moderate yet imperfect balance between precision and recall, suggesting that while the model performs well at capturing true cases across all classes, it is less precise at distinguishing among classes. Lastly, the Cohen's kappa of 0.49 indicates moderate agreement between the predicted and actual classes, suggesting that the model performs beyond chance but is not highly reliable. Together, the results indicate that Model A performs well at detecting most true classes but tends to misclassify certain classes, which, overall, limits the model's reliability. This limitation, therefore, motivates the inclusion of additional bankruptcy indicators, such as qualitative data, to capture patterns and cues that quantitative data alone cannot, thereby enhancing the model's predictive performance and reducing misclassification rates.

In contrast to Model A, Model B uses only text-based features extracted from CEO letters (readability, sentiment, CEO turnover and representative topics). As shown in Table 3.3, Model B achieves an overall accuracy of 66%, correctly classifying 66% of firms across all four classes, compared to Model A (73%). This observation suggests that while qualitative data conveys valuable cues about corporate health, relying solely on it is insufficient to accurately distinguish between classes. In addition, the macro-precision of 53%, compared to 59% in Model A, indicates that slightly more than 50% of predictions for each class are correct. This thus demonstrates a high false-positive rate and, consequently, implies that Model B fails to accurately distinguish all four classes. The macro-recall of 49% further indicates that only 49% of true cases across classes are identified. Similarly, the macro-F1 score of 48% compared with 64% in Model A, suggests a weak balance between precision and recall, revealing the model's imbalance and inconsistency across classes. Lastly, the Cohen's kappa of 0.23, compared to 0.49 in Model A, further demonstrates the model's weak agreement between predicted and actual classifications. Overall, these results indicate that qualitative data alone contain valuable predictive information; however, their discriminative capacity remains limited when used alone. Therefore, one can conclude that textual

variables cannot replace financial ratios; they can complement them to achieve a significantly enhanced overall predictive performance.

Finally, as shown in Table 3.3, Model C, which integrates financial ratios with textual variables extracted from CEO letters, achieves a significantly improved overall predictive performance across all evaluation metrics. The overall classification accuracy of 88% demonstrates Model C's superior ability to classify firms across all four classes correctly. This significant improvement over Model A (73%) shows that the inclusion of qualitative data substantially enhances the model's ability to distinguish between classes. The macro-precision of 77% indicates that, on average, 77% of predictions across all four classes are correct, marking a significant improvement and reduction in false positives compared to Model A (59%). This observed improvement reveals that incorporating both financial and textual data enhances the model's reliability as an early warning system. Similarly, the macro-recall of 89% shows a significant improvement over Model A (76%), indicating that including textual features enhances the model's ability to identify the majority of true instances across all four classes. The macro-F1 score of 82% compared to that of Model A (64%) further demonstrates Model C's strong performance and trade-off between precision and recall, suggesting that predictions display both consistency and balance across all four classes. Lastly, the Cohen's kappa of 0.74, compared to 0.49 in Model A, indicates a substantial agreement between the predicted and actual classifications, demonstrating the model's reliability. Collectively, the observed improvements in overall predictive performance and the improved balance between precision and recall reveal that CEO letters convey valuable linguistic cues, complementing traditional financial ratios and thereby enhancing the model's predictive power and effectiveness as an early warning tool.

Table 3.3**Overall predictive performance of Models A, B and C**

	Overall accuracy	Macro- precision	Macro- recall	Macro- F1 score	Cohen's kappa
Model A	0.73	0.59	0.76	0.64	0.49
Model B	0.66	0.53	0.49	0.48	0.23
Model C	0.88	0.77	0.89	0.82	0.74

This table presents the metrics used to evaluate the overall performance of Models A, B, and C.

3.4.2 Class-wise performance

Table 3.4 presents the performance results across classes for all three models. These results elucidate each model's performance over different forecasting horizons, providing insights into how textual data complements traditional financial ratios in distinguishing among varying levels of financial distress.

As shown in Table 3.4, for the detection of firms expected to remain active (Class 0), Model C achieves the best performance, with 96% precision and 88% recall, yielding an F1 score of 92%. On the contrary, Model A achieves 91% precision and 72% recall, resulting in an F1 score of 80%, indicating that Model C is better at detecting active firms expected to remain active than Model A. Model B, however, exhibits the weakest performance, with a precision of 79% and recall of 74%, leading to an F1 score of 76%, which indicates that text-based features alone are less reliable for detecting firms expected to remain active than when combined with financial ratios. Together, these outcomes demonstrate that integrating financial ratios and textual variables enhances the model's ability to capture more true instances of Class 0, make fewer misclassifications when predicting firms expected to remain active, and provide more reliable Class 0 predictions overall.

A similar pattern is generally observed in the detection of financially distressed firms expected to fail within one, two, and three years, with the integrated model (Model C) consistently outperforming Models A and B. For instance, in identifying firms with short-term bankruptcy risk (Class 1), Model C achieves 73% precision and 89% recall, yielding an F1 score of 80%. Model A, on the other hand, achieves 54% precision and 78% recall, resulting in an F1 score of 64%. This significant decline across all metrics demonstrates Model C's superiority over Model A in identifying more actual instances of firms on the verge of bankruptcy, making fewer mistakes, and performing significantly better overall. Model B, however, achieves the lowest performance despite having the highest precision of 75%, as it appears to miss a significant portion of true Class 1 instances, with a recall of 33%, resulting in an F1 score of 46%. This failure to identify many of the actual instances of Class 1, which in turn leads to increased false alarms, demonstrates that while textual variables alone provide valuable cues of imminent failure risk, relying solely on them in a multiclass bankruptcy prediction is insufficient to accurately and reliably identify firms likely to fail in a year. Taken together, these results indicate that integrating financial and textual data yields the best performance and the trade-off between precision and recall, thereby significantly improving the model's ability to detect firms with imminent bankruptcy risk.

For the detection of firms with medium-term bankruptcy risk (Class 2), Model C again outperforms Models A and B, achieving 73% precision and a 89% recall, resulting in an F1 score of 80%. Model A, on the other hand, achieves 47% precision and 78% recall, yielding an F1 score of 58%. The significant imbalance between precision and recall in Model A suggests that the model captures general trends within this class but fails to differentiate them from other classes accurately. Therefore, while relying solely on financial ratios yields moderate and reasonable performance in predicting firms expected to fail in two years, it is inferior to using both financial and textual variables, as shown in Model C. In addition, again, Model B performs poorly relative to Models A and C, with a precision of 33% and a recall of 44%, leading to an F1 score of 38%, further confirming that using text-based features alone

without financial ratios is insufficient to assess firms' medium-term bankruptcy risks, as it fails to capture many true cases and introduces a high rate of false positives. Collectively, these results demonstrate that including both quantitative and qualitative data yields more consistent, accurate, and reliable predictions of firms likely to fail in two years than financial data alone.

For the detection of firms with longer-term bankruptcy risk (Class 3), Model C also demonstrates stronger, more balanced predictive performance, with 67% precision and 89% recall, yielding an F1 score of 76%. In contrast, Model A achieves 44% precision and 78% recall, resulting in an F1 score of 56%. This observed significant decline in predictive performance across all evaluation metrics in Model A, compared to Model C, demonstrates that integrating financial and textual data enables more effective detection of firms expected to fail within three years, fewer alarms, and stronger overall performance. In contrast, Model B again achieves the weakest performance, with 25% precision and 44% recall, yielding an F1 score of 32%. Put together, these results confirm that integrating financial and textual data, compared to financial data alone, not only improves the prediction of imminent and medium-term bankruptcy risks but also longer-term bankruptcy risks, where the model appears to be learning better by capturing the gradual shifts in the linguistic patterns of CEO letters and the progressive deterioration in financial performance.

Lastly, a similar pattern is observed across all three models, with predictive performance declining as the prediction horizon increases. However, this decline and its steepness differ between models. For instance, in Model C, predictive performance remains consistent and stable when moving from one year before failure to two years before failure, and only slightly drops at three years before failure. This observation indicates that combining financial and textual data enables the model to better capture deeper and earlier signs of distress that financial ratios alone cannot. In contrast, Model A appears to perform better at short horizons than at longer-term horizons, with a moderate decline in predictive performance across prediction horizons. Although Model B also appears to perform significantly better at

short horizons than at longer ones, the substantial decline across prediction horizons, in comparison to Model A and C, indicates that it performs relatively poorly for longer-term predictions. Taken together, these observations demonstrate that including financial and textual data enables the model to maintain robust predictive performance across prediction horizons.

Table 3.4

Class-wise performance of Model A, B and C

	Evaluation metrics	Firms to remain active (Class 0)	Firms to fail in one year (Class 1)	Firms to fail in two years (Class 2)	Firms to fail in three years (Class 3)
Model A	Precision	0.91	0.54	0.47	0.44
	Recall	0.72	0.78	0.78	0.78
	F1 score	0.80	0.64	0.58	0.56
Model B	Precision	0.79	0.75	0.33	0.25
	Recall	0.74	0.33	0.44	0.44
	F1 score	0.76	0.46	0.38	0.32
Model C	Precision	0.96	0.73	0.73	0.67
	Recall	0.88	0.89	0.89	0.89
	F1 score	0.92	0.80	0.80	0.76

This table presents the metrics used to evaluate the class-wise performance of Models A, B, and C.

3.4.3 Additional analysis

While prior research demonstrates that incorporating textual features into prediction models enhances predictive accuracy (e.g., Huang et al., 2023; Mai et al., 2019; Tang et al., 2020), the extent to which these extracted features from accounting narratives contribute to predicting bankruptcy risks across different time horizons remains unclear. To investigate the individual contribution of each text-based feature derived from CEO letters, I conduct a series of feature ablation tests on the baseline Model C. Ablation testing is a well-known technique in machine learning (ML) used to investigate the individual contribution of certain components, including certain input features, on model performance to identify which features are responsible for improved performance and which introduce noise and unnecessary complexity (Bojer, 2022; Neghab et al., 2025). Therefore, to assess the individual contribution of each textual variable extracted from CEO letters to prediction performance, Model C is re-estimated multiple times, each time excluding one text-based feature while keeping all remaining inputs constant. Table 3.5 presents the baseline Model C's overall classification accuracy after excluding one textual variable at a time to evaluate the effect of each exclusion on the model's performance.

As illustrated in Table 3.5, excluding sentiment from the baseline Model C results in the lowest yet notable 8% reduction in overall classification accuracy, compared to other features, suggesting that the sentiment conveyed in CEO letters contributes to improving the model's predictive performance. Meanwhile, omitting readability, as measured by BOG, reduces overall accuracy by 9%, suggesting that CEO letter readability enhances the model's ability to distinguish between non-distressed and distressed firms. More importantly, excluding the three representative topics: Customer relations, Operations, and Production activities and environment extracted from CEO letters through topic modelling results in a significant 10% decline in overall classification accuracy. This observation demonstrates that the underlying themes in CEO letters offer informative and crucial cues for distinguishing

between firms with varying levels of financial distress. Interestingly, excluding CEO turnover reduces overall accuracy by 13%, indicating that CEO replacements and leadership instability provide valuable information, significantly improving the model's predictive performance.

Collectively, these outcomes demonstrate that each text-based feature adds incremental value beyond traditional financial ratios, thus improving the model's overall classification accuracy. These findings are in line with previous research that emphasises the information value of narrative disclosures and highlights the importance of textual features in supplementing traditional financial features in corporate failure prediction models (e.g., Elsayed and Elshandidy, 2020; Mai et al., 2019; Tang et al., 2020).

Table 3.5

Feature ablation testing for Model C's overall accuracy

Feature excluded	Features included	Overall accuracy of the baseline Model C	Overall accuracy after feature exclusion	Accuracy loss
Sentiment	financial ratios + BOG + representative topics + CEO turnover	0.88	0.80	(0.08)
Readability	financial ratios + sentiment + representative topics + CEO turnover	0.88	0.79	(0.09)
Representative topics	financial ratios + BOG + sentiment + CEO turnover	0.88	0.78	(0.10)
CEO turnover	financial ratios + BOG + sentiment + representative topics	0.88	0.75	(0.13)

This table presents Model C's overall performance after removing specific textual features, evaluating each feature's contribution to the model's classification accuracy.

Furthermore, Table 3.6 presents the class-wise performance results for Model C after omitting one textual variable at a time, providing valuable insights into how each textual feature derived from CEO letters contributes to predicting whether firms are likely to remain active or fail within one, two, or three years.

For detecting firms expected to remain active (Class 0), smaller declines in performance are observed across some metrics compared with Classes 1, 2, and 3. As shown in Table 3.6, when sentiment is excluded, precision falls to 93% and recall to 79%, resulting in an F1 score of 85%. The decline in the model's performance and the balance between precision and recall indicate that sentiment helps the model differentiate more effectively between stable firms expected to remain active and those at risk of bankruptcy. Meanwhile, when

readability is not included in the model, precision drops to 90% and recall to 81%, yielding an F1 score of 86%. This observed reduction across all metrics suggests that the readability of CEO letters also enhances the model's ability to identify firms expected to remain active. In addition, excluding the representative topics embedded in CEO letters reduces precision to 93% and recall to 77%, resulting in an F1 score of 84%. This observation implies that the latent thematic content of CEO letters is particularly informative for distinguishing between non-distressed and distressed firms. More importantly, omitting CEO turnover leads to a slight increase in precision to 97% but a significant drop in recall to 72%, yielding an F1 score of 82%. This observed reduction in the model's effectiveness in balancing precision and recall suggests that CEO turnover, among all other textual features, plays the most supportive role in enhancing the model's ability to identify firms expected to remain active.

In addition, in predicting firms with short-term bankruptcy risk (Class 1), excluding sentiment reduces precision, recall, and F1 score to 67%, suggesting that CEO letters' sentiment provides valuable linguistic cues indicative of imminent bankruptcy risk. The exclusion of readability, however, leads to a more notable drop in precision to 60% and recall to 67%, yielding an F1 score of 63%. This significant drop in the model's predictive performance at detecting firms expected to fail in a year demonstrates that the readability of CEO letters significantly enhances the model's ability to identify imminent failure risks. Similarly, a significant decline in precision to 54% and recall to 78%, resulting in an F1 score of 64% is observed when representative topics are not included in the model, suggesting that the thematic content of CEO letters is also crucial for identifying firms approaching bankruptcy. Meanwhile, excluding CEO turnover leads to a less pronounced reduction in precision to 58% and recall to 78%, yielding an F1 score of 67%, suggesting that although it may not be as informative as the readability and thematic content of CEO letters, it still contributes to improving the prediction of firms expected to fail in a year.

Moreover, in identifying firms with medium-term bankruptcy risk (Class 2), excluding sentiment reduces precision to 57%, while recall remains at 89%, resulting in an F1 score of

70%, indicating that the sentiment of CEO letters is informative for detecting firms expected to fail within two years. However, when readability is not included in the model, precision falls to 58% and recall to 78%, yielding an F1 score of 67%, suggesting that the readability of CEO letters provides more valuable information than sentiment for identifying firms with medium-term bankruptcy risk. More importantly, excluding representative topics in CEO letters results in the most pronounced performance decline, with precision dropping to 50% and recall to 78%, leading to an F1 score of 61%. Such a significant reduction in predictive performance indicates that the latent thematic content of CEO letters provides crucial information and cues about firms in transitional distress phases. Omitting CEO turnover also reduces performance, but not as significantly as when representative topics in CEO letters are excluded, with precision dropping to 53% and recall remaining at 89%, resulting in an F1 score of 67%, suggesting that changes in CEO offer meaningful information for detecting firms expected to fail within two years.

Lastly, in detecting firms with longer-term bankruptcy risk (Class 3), excluding the sentiment of CEO letters reduces precision to 50%, while recall remains at 89%, yielding an F1 score of 64%, suggesting that sentiment enhances the detection of firms expected to fail within three years. More notably, when readability is omitted, precision falls to 46% and recall to 67%, leading to an F1 score of 55%. This dramatic decline in performance indicates that the readability of the CEO letters significantly enhances the model's ability to capture early warning signs of emerging distress, making it a critical qualitative variable for predicting firms expected to fail within three years. Excluding representative topics also results in a decline in performance, though not as significant as when readability is excluded, with precision falling to 57% and recall remaining at 89%, yielding an F1 score of 70%. Interestingly, omitting CEO turnover from the model leads to a substantial drop in performance, with precision declining to 38% and recall remaining at 89%, resulting in an F1 score of 53%. This drop in performance indicates that CEO turnover significantly improves the prediction of firms expected to fail within three years.

Table 3.6

Feature ablation testing results for the baseline Model C's class-wise performance.

	Evaluation metrics	Firms to remain active (Class 0)	Firms to fail in one year (Class 1)	Firms to fail in two years (Class 2)	Firms to fail in three years (Class 3)
Baseline Model C	Precision	0.96	0.73	0.73	0.67
	Recall	0.88	0.89	0.89	0.89
	F1 score	0.92	0.80	0.80	0.76
Sentiment excluded from Model C	Precision	0.93	0.67	0.57	0.50
	Recall	0.79	0.67	0.89	0.89
	F1 score	0.85	0.67	0.70	0.64
Readability excluded from Model C	Precision	0.90	0.60	0.58	0.46
	Recall	0.81	0.67	0.78	0.67
	F1 score	0.86	0.63	0.67	0.55
Representative topics excluded from Model C	Precision	0.93	0.54	0.50	0.57
	Recall	0.77	0.78	0.78	0.89
	F1 score	0.84	0.64	0.61	0.70
CEO turnover excluded from Model C	Precision	0.97	0.58	0.53	0.38
	Recall	0.72	0.78	0.89	0.89
	F1 score	0.82	0.67	0.67	0.53

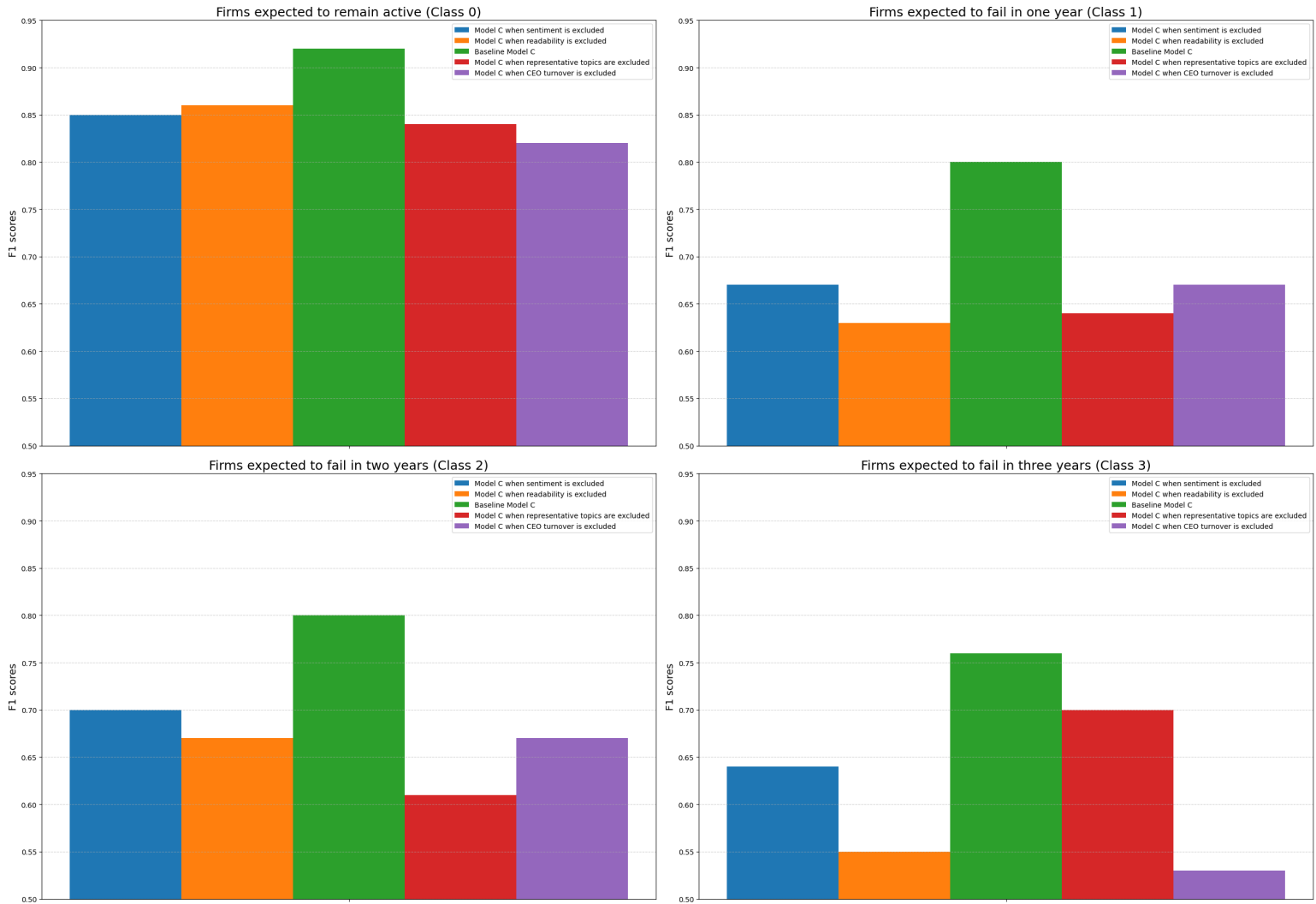
This table presents the results of Model C's class-wise performance after omitting one textual feature at a time.

Moreover, to visually interpret and better evaluate the contribution of each textual feature to the baseline Model C's class-wise predictive performance, Figure 3.3 presents a bar plot showing Model C's F1 score alongside the F1 scores achieved after excluding each text-based feature. As shown in Figure 3.3, although all text-based features derived from CEO letters contribute to improving the detection of active firms expected to remain active, CEO turnover appears to be the most important textual feature in enhancing the model's ability to distinguish between active and failing firms. In predicting firms expected to fail within a year, the most pronounced decline in performance is observed when the readability of CEO letters is excluded, followed by representative topics, indicating that these text-based features provide valuable information for improving the model's ability to detect firms approaching bankruptcy. Meanwhile, in predicting firms expected to fail in two years, the representative topics of CEO letters appear more informative than other textual factors, substantially improving the detection of firms with medium-term bankruptcy risk. Lastly, in identifying firms with longer-term bankruptcy risks, CEO turnover, followed by the readability of CEO letters, emerge as the most useful text-based features in significantly enhancing the model's ability to detect firms expected to fail within three years.

Collectively, these findings reveal that although all textual variables appear to offer incremental predictive value, the impact and individual contribution of each feature on the model's predictive performance vary with the prediction horizon. For instance, the readability of CEO letters, among other textual features, appears to contribute most to improving the model's ability to identify firms at imminent risk of bankruptcy. Meanwhile, the representative topics in CEO letters, as captured by topic modelling, appear to provide the strongest predictive signals of medium-term bankruptcy risk, thereby enhancing the model's ability to detect medium-term failures. Lastly, CEO turnover appears to be the most important textual feature contributing to the model's improved ability to identify firms with longer-term bankruptcy risk.

Figure 3.3

F1 scores after excluding textual features



3.4.4 Robustness check

To assess the validity of my findings, I compare the predictive performance of the LSTM model with that of a multinomial logistic regression (MLR) model trained on the same dataset and input features (see Appendix A3.4 for additional details on how both models are implemented). Several studies in the literature that employ artificial neural networks (e.g., Brockett et al., 2006), deep learning (e.g., Mai et al., 2019), and LSTM networks (e.g., Fischer and Krauss, 2018) for financial distress and market prediction, have used logistic regression as a benchmark model. Consequently, MLR serves as a benchmark methodology in my study to examine whether the LSTM network's capturing of long-term dependencies within sequential information and non-linear relationships among predictors adds value when assessing multi-horizon bankruptcy risks compared to a conventional MLR model.

Logistic regression is one of the widely utilised models in bankruptcy research (e.g., Charitou et al., 2004; Elsayed and Elshandidy, 2020; Ohlson, 1980), which assumes a logit relationship between the predictors and the binary default event (Mai et al., 2019).

Multinomial logistic regression (MLR) is a direct extension of the binary logistic regression model that allows the outcome variable to include more than two categories or additional states of distress (Chen et al., 2024; Johnsen and Melicher, 1994). Due to the three-year sequence design and inclusion of interrelated predictor variables, high multicollinearity is expected. In an MLR model, multicollinearity does not necessarily negatively affect the reliability or predictive power of the model; however, it may not yield accurate results about any given predictor or about the redundancy of predictors with others, making it difficult to derive accurate inferences about the associations between explanatory and dependent variables (Midi et al., 2010; Myers, 2000). The solutions proposed to mitigate multicollinearity in logistic regression models include integrating variables, augmenting sample size, eliminating highly correlated variables, applying regularisation techniques or implementing principal component analysis (Senaviratna and Cooray, 2019).

Given that the present study uses the MLR model only as a predictive performance benchmark, rather than for statistical inference, I apply regularisation rather than omitting or changing predictor variables to maintain fair comparability with the LSTM model and ensure that both models are trained using the same input features. In particular, I apply L2 regularisation, which has been consistently noted to be a superior regularisation technique for handling multicollinearity and preventing overfitting (e.g., Pham, 2025; Si et al., 2025; Yang et al., 2023). The manual and grid search approaches, which are important to decrease the risk of overfitting, are among the most widely used for hyperparameter optimisation; however, manual search often fails to find the optimum (Nystrup et al., 2020). Therefore, consistent with Nagula and Alexakis (2025), I use grid search to tune and optimise the regularisation parameter (C) to identify the best C value for L2 regularisation, which helps avoid model overfitting. Based on the hyperparameter tuning results, the optimal regularisation parameter appears to be $C = 10$, which I use for the final MLR model.

Table 3.7 presents the overall predictive performance of the MLR model relative to the LSTM model (my baseline, Model C). As shown in Table 3.7, the LSTM model outperforms the MLR model across all evaluation metrics. In particular, the MLR model achieves an overall classification accuracy of 79%, compared to 88% for the LSTM model, suggesting that the LSTM model's ability to capture complex nonlinear and sequential patterns improves its capacity to distinguish among all four classes. In addition, the macro-precision, macro-recall, and macro-F1 of the MLR model (66%, 76%, and 69%, respectively) are significantly lower than those of the LSTM model (77%, 89%, and 82%, respectively), indicating that predictions by LSTM across all four outcome classes include fewer false alarms and are more consistent and balanced. Lastly, the Cohen's kappa of 0.55 shows a moderate agreement between the predicted and actual classifications, compared to the strong agreement of 0.74 achieved by the LSTM model. Collectively, the observed improvements in the LSTM model's overall predictive performance demonstrate the LSTM network's

effectiveness in deriving meaningful information from sequential data and capturing temporal dependencies for predicting multi-horizon bankruptcy risks, compared with the MLR model.

Table 3.7

Overall performance of the LSTM and MLR models

	Overall accuracy	Macro-precision	Macro-recall	Macro- F1 score	Cohen's kappa
LSTM model	0.88	0.77	0.89	0.82	0.74
MLR model	0.79	0.66	0.76	0.69	0.55

This table presents the metrics used to evaluate the overall performance of the LSTM and MLR models.

Furthermore, Table 3.8 presents the class-wise performance of both the LSTM and MLR models, showing how each performs at short-term and longer-term horizons. As illustrated in Table 3.8, for the detection of firms expected to remain active (Class 0), the MLR model achieves 90% precision and 80% recall, yielding an F1 score of 85%. However, the LSTM model achieves stronger performance, with 96% precision and 88% recall, resulting in an F1 score of 92%. This observation indicates that the LSTM model has a stronger capacity to identify firms likely to remain active, with a lower misclassification rate than the MLR model.

Interestingly, in predicting firms expected to fail within a year (Class 1), the MLR achieves a precision of 75%, which is higher than that of 73% achieved by the LSTM model; however, the recall of 67% and an F1 score of 71% are lower than those of 89% and 80% respectively, obtained by the LSTM model. These outcomes reveal that the LSTM model, compared to the MLR model, demonstrates a more balanced capacity to identify firms expected to fail within a year and is more effective at capturing most actual instances of firms on the verge of failure, but it yields slightly more false positives.

In detecting firms expected to fail within two years (Class 2), the MLR achieves 54% precision and 78% recall, leading to an F1 score of 64%. In contrast, the LSTM model achieves a precision of 73% and a recall of 89%, resulting in an improved F1 score of 80%. These results demonstrate the LSTM model's effectiveness in capturing long-term dependencies in sequential data over extended periods, as the LSTM's performance in predicting firms with medium-term bankruptcy risk remains consistent and balanced. In contrast, the MLR model's performance drops significantly.

A similar pattern is observed in the identification of firms expected to fail within three years (Class 3), where the MLR achieves 47% precision and 78% recall, yielding an F1 score of 58%, while the LSTM model achieves a precision of 67% and a recall of 89%, resulting in an F1 score of 76%. Again, these outcomes reveal that the LSTM model outperforms the MLR model in detecting firms expected to fail within three years, demonstrating stronger and more balanced performance.

Overall, both models appear to perform best at detecting firms likely to remain active, followed by those likely to fail within a year. However, a significant difference in performance stability and consistency between the models becomes evident when the prediction horizon is extended from one-year-ahead to two-year-ahead and three-year-ahead predictions. As shown in Table 3.8, the predictive performance of the MLR model declines markedly as the prediction horizon increases from one year before failure to three years before failure. The predictive performance of the LSTM model, on the other hand, remains robust and consistent for two-year-ahead predictions, followed by a less sharp decline for three-year-ahead predictions than in the MLR model. Thus, this increased stability in predictive performance as the prediction horizon increases reflects the LSTM model's effectiveness and strong capacity to capture long-term dependencies and non-linear relationships within sequential data, making it better suited for assessing multi-horizon bankruptcy risks than MLR.

Table 3.8

Class-wise performance of the LSTM and MLR models

	Evaluation metrics	Firms to remain active (Class 0)	Firms to fail in one year (Class 1)	Firms to fail in two years (Class 2)	Firms to fail in three years (Class 3)
LSTM model	Precision	0.96	0.73	0.73	0.67
	Recall	0.88	0.89	0.89	0.89
	F1 score	0.92	0.80	0.80	0.76
MLR model	Precision	0.90	0.75	0.54	0.47
	Recall	0.80	0.67	0.78	0.78
	F1 score	0.85	0.71	0.64	0.58

This table presents the metrics used to evaluate the class-wise performance of the LSTM and MLR models.

3.5 Discussion and Implications

This study investigates whether the inclusion of text-based features derived from CEO letters can enhance the understanding of firms' bankruptcy risks across various prediction horizons. Specifically, it examines the extent to which these textual features complement financial data in assessing multi-horizon bankruptcy risks. The results demonstrate that incorporating textual data along with financial data significantly improves the model's predictive performance and reliability. The integrated model, Model C, which combines quantitative features (accounting-based, market-based, and macroeconomic variables) with qualitative features derived from CEO letters (readability, sentiment, representative topics, and CEO turnover), substantially outperforms other models that rely solely on either financial data or textual data. These findings emphasise that the linguistic features embedded in CEO letters convey informative and predictive signals, complementing financial ratios when assessing firms' multi-horizon bankruptcy risks.

The overall and class-wise predictive performance results offer valuable insights into how text-based features derived from CEO letters complement traditional financial ratios in distinguishing between the varying levels of financial distress. Model A, which relies solely on financial data, demonstrates reasonable overall predictive performance across the four classes, with a robust overall accuracy but a notable imbalance between macro-precision and macro-recall. The results indicate that Model A performs overall well, capturing most of the true classes across all four classes. However, it lacks precision in assigning cases to their correct classes, leading to a high misclassification rate. In practice, such an imbalance undermines the model's practical reliability as an early warning tool and, therefore, raises concerns for stakeholders. These false alarms and the lack of precision, for example, may lead investors to misallocate resources, banks to make inefficient credit decisions, or regulators to intervene unnecessarily based on inaccurate warnings. At the class-wise level, predictive performance varies significantly. In particular, the model effectively and accurately detects solvent firms expected to remain active. However, in predicting whether a firm is

likely to fail in one, two, or three years, Model A appears to perform reasonably well at detecting short-term failures but not at longer-term failures. Hence, predictive performance declines moderately as the forecasting horizon increases, indicating that, in a multiclass setting, relying solely on financial data may not reliably and accurately identify early signs of emerging distress.

Moreover, the predictive performance of Model B, which only uses textual features obtained from CEO letters, is relatively weaker than that of Model A. The results demonstrate that while the textual features derived from CEO letters offer valuable information, relying solely on them appears insufficient to distinguish among varying levels of distress and, therefore, to serve as an early warning system. This finding suggests that text-based features may be more effective when combined with financial data than when used alone. Thus, integrating such informative features with financial data would significantly enhance the model's ability to capture distress patterns reflected in both numerical representations of firm performance and linguistic cues in textual disclosures. This is further substantiated by Model C's superior performance, which integrates both financial and textual data. In particular, Model C shows strong overall predictive performance across all four classes, with improved classification accuracy and more balanced macro-average metrics, reflecting improved discriminative capacity.

At the class level, in addition to stronger, more balanced performance in identifying active firms and those expected to fail in a year, Model C also achieves consistent, robust predictive performance even as the prediction horizon increases, unlike both Models A and B. This finding indicates that the text-based features derived from CEO letters play a significant complementary role, providing valuable information that significantly improves the model's ability to detect medium-term and longer-term failures, which financial ratios alone do not seem to capture. Taken together, these results demonstrate that including both financial and textual data yields a more robust and accurate distinction between the classes,

enhancing the model's predictive performance and reliability as an early warning and risk-assessment tool.

Moreover, the results of the feature ablation tests provide valuable insights that can better clarify the role and impact of each textual feature extracted from CEO letters in improving prediction performance. Interestingly, based on evaluating the model's predictive performance and its ability to balance precision and recall (F1 scores), I find that the contribution and informativeness of textual features vary across prediction horizons. For instance, I find that the readability of CEO letters is the most informative textual indicator for identifying firms at imminent risk of bankruptcy, and the second most informative for detecting firms at longer-term risk of bankruptcy. Hence, the readability of CEO letters appears to significantly improve the model's prediction of firms expected to fail in one year and in three years, suggesting that textual complexity offers valuable signals of near-term distress and early warning signals of emerging distress. An extensive body of literature has consistently linked low readability to poor financial performance, suggesting that managers deliberately use more complex and less readable language in their disclosures to obfuscate poor performance (e.g., Asay et al., 2018a; Fabrizio and Kim, 2019; Kim et al., 2019). In addition, Bakarich et al. (2019) investigate whether the qualitative characteristics of 10-Ks vary across corporate life cycles, demonstrating that firms in the decline stages tend to use more complex and ambiguous communications than those in the maturity stages, which have simpler and more readable disclosures. Therefore, the model appears to effectively capture and learn from patterns of gradual shifts in textual complexity over the years, reflecting managerial efforts to either obfuscate poor performance or downplay emerging problems, which may explain the informativeness of the readability of CEO letters in improving the model's ability to detect short-term and longer-term bankruptcy risks.

Interestingly, the representative topics in CEO letters extracted through topic modelling appear to be the most informative among all other textual features for enhancing the model's ability in predicting firms expected to fail within two years, suggesting that the latent thematic

structures of CEO letters provide incremental information for identifying medium-term bankruptcy risks. Representative topics in CEO letters also appear to be the second-most informative textual feature for detecting firms at imminent bankruptcy risk, significantly improving the model's ability to predict firms likely to fail in one year. These findings align with prior research highlighting the potential of the underlying themes of corporate narratives to explain bankruptcy and differentiate between bankrupt and non-bankrupt firms (e.g., Yang et al., 2018). In their seminal study on the relationship between financial distress and management's narrative disclosures, Tennyson et al. (1990) find that the annual report narratives of firms approaching bankruptcy exhibit distinct thematic patterns that set them apart from those of healthy firms. Similarly, in one of the very first studies examining the content of chairman's statements in the UK context, Smith and Taffler (2000) demonstrate that the underlying themes enable high-level discrimination in the categorisation of financially distressed and non-distressed firms. Therefore, representative topics derived from CEO letters appear to capture latent thematic signals and patterns associated with near-term and medium-term distress, enhancing the model's ability to identify and differentiate among varying levels of financial distress.

Lastly, I find that CEO turnover is the most informative textual feature for detecting longer-term failures, suggesting that changes in top management appear to provide meaningful and early warning signals of impending corporate failure that significantly improve the model's ability to identify firms expected to fail in three years. Numerous studies in the literature provide empirical evidence indicating that CEO turnover is negatively related to financial performance (e.g., Fiordelisi and Ricci, 2014; Jenter and Kanaan, 2015). According to Martin and Combs (2011), when signs of poor performance emerge, boards respond by dismissing CEOs early, before these performance issues become widely known, especially to shareholders. Consistent with these views, CEO replacement appears to capture early signs of financial instability and poor performance, which commonly precede corporate

failure, explaining its predictive value and contributing to improving the model's ability to identify firms expected to fail in three years.

Moreover, upon comparing the predictive performance of the LSTM and MLR models, both trained on the same datasets, with the same input features, and prepared in the same way, I find that the LSTM model is significantly better at assessing multi-horizon bankruptcy risks than the MLR model. This finding is consistent with prior studies on bankruptcy prediction, which show that neural networks outperform logistic regression models (e.g., Brockett et al., 2006; Zhang et al., 1999). In this study, the LSTM model clearly demonstrates its superiority over MLR. In terms of overall predictive performance, the LSTM model outperforms the MLR across all evaluation metrics, achieving improved results across all four classes. In terms of class-wise performance, the LSTM model also shows improved, consistent, and more balanced performance for all classes. These results, therefore, underscore the effectiveness of LSTM networks' memory cells in processing sequential information and capturing temporal patterns and changes to predict whether firms are likely to remain active or fail within one, two, or three years.

The findings of this study make several significant empirical and practical contributions to bankruptcy prediction literature. First, this study extends the literature beyond binary classification models by developing a multiclass classification model to assess multi-horizon bankruptcy risk, using LSTM networks to capture the sequential and evolving nature of financial distress. In particular, the LSTM model employed in this study predicts whether a firm is likely to remain operational or fail within one, two, or three years, reflecting varying levels of financial distress. A key empirical contribution of this study lies in integrating text-based features derived from CEO letters with accounting, market, and macroeconomic variables, and in examining the extent to which these textual features enhance the model's predictive performance across multiple prediction horizons. The results demonstrate that combining financial and textual data significantly enhances the model's discriminative power

and reliability as an early warning system, thereby promoting the inclusion of the information content of CEO letters in bankruptcy research.

More importantly, the feature ablation tests provide valuable insights into how and to what extent the textual features derived from CEO letters enhance the detection of solvent firms expected to remain active, as well as those with short-term, medium-term, and longer-term bankruptcy risks. Consequently, the findings of the present study enhance the understanding of how textual data complements financial data by demonstrating that the importance and predictive value of text-based features vary across prediction horizons. Thus, although all textual variables appear to enhance the model's predictive performance, some are more informative than others depending on the prediction window. This study, therefore, highlights the informational value of CEO letters for predicting multi-horizon bankruptcy risks and contributes to the growing body of literature underscoring the predictive power of textual features extracted from narrative disclosures.

In practice, the findings of this study have significant implications for stakeholders interested in the early detection of financial distress. The observed enhancement in predictive performance underscores that the linguistic aspects of CEO letters convey informative and predictive signals, significantly improving the reliability of bankruptcy prediction models as early warning systems. Thus, an improved understanding of how the informativeness of each textual variable extracted from CEO letters varies across prediction horizons may provide stakeholders with a strategic advantage when assessing multi-horizon bankruptcy risks. For instance, investors may significantly benefit from tracking CEO turnover, as it appears to provide valuable cues that signal differences between distressed and non-distressed firms, leading to more informed investment decisions.

In addition, investors, banks, and regulatory bodies could gain valuable insights by evaluating the readability and textual complexity of CEO letters to detect near-term financial distress; identifying and interpreting the dominant topics embedded in CEO letters to assess medium-term bankruptcy risks; and tracking CEO turnover to detect early warning signals of

emerging distress. Therefore, stakeholders may prioritise textual features derived from CEO letters that are most relevant to the prediction horizon of their interest and, accordingly, include them in their evaluations. Hence, an improved understanding of the individual roles and contributions of text-based features derived from CEO letters could greatly assist investors in making more informed investment decisions, banks in reducing credit risks and preventing loan losses, and enhance regulators' capacity to undertake timely regulatory actions. These findings, therefore, offer important insights into how the information content of CEO letters may be used alongside financial data for assessing multi-horizon bankruptcy risks, promoting their inclusion in bankruptcy prediction models for more reliable early warning systems that can contribute to more informed decision-making, strengthened risk assessments, and facilitating earlier and more effective policy interventions.

3.6 Conclusion

This research underscores the essential role of text-based features derived from CEO letters in improving the predictive performance of multiclass bankruptcy prediction models.

Specifically, it sheds light on the importance and informativeness of the textual features extracted from CEO letters across different prediction horizons, enhancing the understanding of how text-based features complement financial ratios in assessing firms' multi-horizon bankruptcy risk. The results indicate that while financial ratios are crucial for evaluating firms' financial health and likelihood of bankruptcy, key corporate communications, such as CEO letters, also contain valuable information that provides explanatory power for predicting multi-horizon bankruptcy risks. Therefore, when combined, quantitative and qualitative data can significantly improve the model's predictive performance and reliability as an early warning tool.

More importantly, I find that the impact and contribution of each text-based feature derived from CEO letters in complementing financial ratios and enhancing predictive performance

vary across prediction horizons. For instance, the readability of CEO letters appears to be the most informative textual indicator for enhancing the model's ability to identify firms expected to fail in a year, and the second most informative for detecting firms expected to fail in three years. The representative topics, on the other hand, seem to be the most informative among all textual features for improving the model's ability to predict firms likely to fail in two years and the second-most informative for identifying firms likely to fail within a year. In addition, CEO turnover appears to be the most informative text-based feature, compared to the other textual predictors, in enhancing the model's ability to detect firms expected to fail in three years and to distinguish between firms expected to remain active and those expected to fail. Lastly, the sentiment conveyed in CEO letters appears to generally enhance the model's predictive performance. However, it does not emerge as the most informative textual feature for any of the imminent, medium-term, or early warning signs of distress.

The present study addresses an underexplored question in the literature regarding the extent to which textual features derived from narrative disclosures improve predictive performance across multiple prediction horizons. The findings demonstrate that text-based features extracted from CEO letters do not all contribute equally to predictive performance enhancement; instead, the predictive value and informativeness of each textual feature appear to vary with the prediction horizon. In practice, an improved understanding of the role and informativeness of different textual dimensions of CEO letters in enhancing the performance and reliability of multi-horizon bankruptcy prediction models facilitates more informed decision-making for various stakeholders, including investors, regulators and banks.

Despite the importance and practical relevance of the findings, this study has several limitations that prompt further investigations. First, this study uses CEO letters as they contain critical information that offers valuable insights into firms' future direction and performance. However, CEO letters are the only textual data source in this study. Future research could, therefore, examine other critical corporate communication outlets that may

also contain valuable information on the likelihood of bankruptcy and compare the information content of CEO letters with that of other corporate communication outlets to determine whether certain narrative disclosures are more informative of a firm's likelihood of failure than others. In addition, I include CEO turnover as a textual predictor of corporate failure by investigating whether the same CEO remained in position throughout the three-year window or whether a CEO change occurred during that window. However, I do not account for whether the CEO's termination was forced or voluntary, which future studies could include. Third, in this study, I extract representative topics from CEO letters using the Latent Dirichlet Allocation (LDA) topic modelling technique, based on the whole sample of firms, without categorising the dominant topics by industry due to the small sample size. Future research with a larger sample could therefore identify industry-specific topics to capture representative topics for each industry. These identified limitations may present promising avenues for future research on corporate failure prediction.

Appendix

Appendix A3.1

List of sampled firms

Active firms				Failed firms	
1	Astrazeneca plc	51	Inchcape plc	101	Qinetiq Group plc
2	Unilever plc	52	Johnson Matthey plc	102	Senior plc
3	RELX Plc	53	Marks and Spencer Group plc	103	Softcat plc
4	Rolls-Royce Holdings plc	54	Mobico Group plc	104	Spirent Communications plc
5	BP plc	55	Moneysupermarket.com plc	105	Travis Perkins plc
6	British American Tobacco plc	56	Ocado Group plc	106	Vesuvius plc
7	Rio Tinto plc	57	Pearson plc	107	Victrex plc
8	BAE Systems plc	58	Persimmon plc	108	Volution Group plc
9	GSK plc	59	Renishaw plc	109	Wizz Air Holdings plc
10	National Grid plc	60	RS Group plc	110	Airtel Africa plc
11	Diageo plc	61	Sainsbury plc	111	Shell plc
12	Anglo American plc	62	Serco Group plc	112	Compass Group plc
13	Experian plc	63	Spectris plc	113	Glencore plc
14	Reckitt Benckiser Group plc	64	Tate and Lyle plc	114	Berkeley Group Holdings plc
15	Tesco plc	65	Taylor Wimpey plc	115	Easyjet plc
16	Vodafone Group plc	66	United Utilities plc	116	ITV plc
17	Imperial Brands plc	67	Whitbread plc	117	Burberry Group plc
18	Antofagasta plc	68	WPP plc	118	Croda International plc
19	SSE plc	69	4Imprint Group plc	119	Fresnillo plc
20	BT Group plc	70	Balfour Beatty plc	120	Intertek Group plc
21	Ashtead Group plc	71	AG Barr plc	121	Mondi plc
22	Associated British Foods plc	72	Bellway plc	122	Auto Trader plc
23	Babcock International Group plc	73	Bodycote plc	123	C and C Group plc
				1	Connaught plc
				2	Hampson Industries plc
				3	Thomas Cook Group plc
				4	Debenhams plc
				5	Studio Retail Group plc
				6	Johnston Press plc
				7	Carillion plc
				8	Mecom Group plc
				9	Morse plc
				10	Mouchel Group plc
				11	Petropavlovsk plc
				12	Hibu plc
				13	Afren plc
				14	Avocet Mining plc
				15	New World Resources plc
				16	Whatman plc
				17	Gyrus Group plc
				18	Wellstream Holdings plc
				19	Imperial Energy Corporation plc
				20	Luminar Group plc
				21	Isoft Group plc
				22	Woolworths Group plc
				23	Jarvis plc

24	InterContinental Hotels Group plc	74	Chemring Group plc	124	DiscoverIE Group plc	24	Baltimore Technologies plc
25	Next plc	75	Clarkson plc	125	Energiean plc	25	Energis plc
26	Sage Group plc	76	Coats Group plc	126	IWG plc	26	First Technology plc
27	Bunzl plc	77	Computacenter plc	127	Telecom Plus plc	27	Heywood Williams Group plc
28	Centrica plc	78	Cranswick plc	128	WH Smith plc	28	Eaga plc
					Aston Martin Lagonda Global		
29	Hikma Pharmaceuticals plc	79	Crest Nicholson Holdings plc	129	Holdings plc	29	Alfred McAlpine plc
30	Severn Trent plc	80	Currys plc	130	Diversified Energy Company plc	30	Carpetright plc
31	Convatec Group plc	81	Domino's Pizza Group plc	131	Dunelm Group plc	31	Xchanging plc
32	Halma plc	82	Elementis plc	132	Future plc	32	Infinis Energy plc
33	IMI plc	83	FirstGroup plc	133	Helios Towers plc	33	Georgia Healthcare Group plc
34	Informa plc	84	Frasers Group plc	134	Kainos Group plc	34	NMC Health plc
35	Melrose plc	85	Genuit Group plc	135	NCC Group plc	35	JJB Sports plc
36	Rentokil Initial plc	86	Genus plc	136	Spire Healthcare Group plc	36	HMV Group plc
37	Smith and Nephew plc	87	Hill and Smith plc			37	Interserve plc
38	Smiths Group plc	88	Hilton Food Group plc			38	Cineworld Group plc
39	Spirax Group plc	89	Ibstock plc				
40	Weir Group plc	90	Keller Group plc				
41	Carnival plc	91	Marshalls plc				
42	DCC plc	92	Mitchells and Butlers plc				
43	Diploma plc	93	Mitie Group plc				
44	Drax Group plc	94	Morgan Advanced Materials plc				
45	Entain plc	95	Morgan Sindall Group plc				
46	Games Workshop Group plc	96	Oxford Instruments plc				
47	Grafton Group plc	97	PageGroup plc				
48	Greggs plc	98	Pets At Home Group plc				
49	Hays plc	99	Playtech plc				
50	Howden Joinery Group plc	100	Premier Foods plc				

This table lists the sampled firms.

Appendix A3.2

Topics interpretation, coherence and perplexity scores

Topic 1

Customer relations

Word	Bonferroni adj. p-value	USAS Semantic Tags
year	0.000	Time: period
customer(s)	0.000	Business: selling People
strong	0.000	Toughness: strong/weak Evaluation: good/bad Power, organising Ability: ability, intelligence Egoism
sales	0.000	Business: selling People Money: price
group	0.000	Groups and affiliation Generally kinds, groups, examples
people	0.000	People

This table presents the most significant words in Topic 1, along with their associated semantic tags derived from USAS. After extracting the most frequent twenty-five words, I assess the statistical significance of these words by calculating the p-value and the Bonferroni-adjusted p-value for each word. Words with a Bonferroni-adjusted p-value greater than 0.002 are considered non-significant in this analysis and, therefore, are excluded. The significance level (Bonferroni adj. p-value ≤ 0.002) is calculated after dividing the original significance level (0.05) by the twenty-five top frequent words extracted.

Topic 2
Operations

Word	Bonferroni adj. p-value	USAS Semantic Tags
per	0.000	Grammatical bin
increased	0.000	Quantities Affect: modify, change
product	0.000	Objects generally General actions, making, etc. Affect: cause/connected
cost(s)	0.000	Money: price
net	0.000	Quantities
stores	0.000	Business: selling Getting and giving; possession
operations	0.000	General actions, making, etc.
end	0.000	Objects generally Linear order Affect: cause/connected Time: beginning and ending

This table presents the most significant words in Topic 2, along with their associated semantic tags derived from USAS. After extracting the most frequent twenty-five words, I assess the statistical significance of these words by calculating the p-value and the Bonferroni-adjusted p-value for each word. Words with a Bonferroni-adjusted p-value greater than 0.002 are considered non-significant in this analysis and, therefore, are excluded. The significance level (Bonferroni adj. p-value ≤ 0.002) is calculated after dividing the original significance level (0.05) by the twenty-five top frequent words extracted.

Topic 3

Production activities and environment

Word	Bonferroni adj. p-Value	USAS Semantic Tags
production	0.000	Objects generally Linear order Affect: cause/connected General actions, making etc.
safety	0.000	Safety/Danger
deliver	0.000	Putting, taking, pulling, pushing, transporting Speech: communicative Ability: success and failure
energy	0.000	Science and technology in general Interest/boredom/excited/energetic
work	0.000	Work and employment: generally General actions, making, etc. Ability: success and failure Affect: modify, change Affect: cause/connected
operational	0.000	General actions, making, etc.
gas	0.000	Substances and materials generally: gas
project	0.000	General actions, making, etc. Education in general

This table presents the most significant words in Topic 3, along with their associated semantic tags derived from USAS. After extracting the most frequent twenty-five words, I assess the statistical significance of these words by calculating the p-value and the Bonferroni-adjusted p-value for each word. Words with a Bonferroni-adjusted p-value greater than 0.002 are considered non-significant in this analysis and, therefore, are excluded. The significance level (Bonferroni adj. p-value ≤ 0.002) is calculated after dividing the original significance level (0.05) by the twenty-five top frequent words extracted.

Chi-Square test and adjusted residuals for topic distribution across classes

		Class 0	Class 1	Class 2	Class 3	Total
Topic 1	Count	194 (165.70)	3 (15.43)	5 (15.43)	10 (15.43)	212
	Adj. Res.	6.10	-4.26	-3.58	-1.86	
Topic 2	Count	92 (110.99)	20 (10.34)	17 (10.34)	13 (10.34)	142
	Adj. Res.	-4.52	3.66	2.52	1.01	
Topic 3	Count	122 (131.31)	15 (12.23)	16 (12.23)	15 (12.23)	168
	Adj. Res.	-2.11	1.00	1.36	1.00	
Chi2 Statistic					44.153	
p-value					0.000	
Cramér's V					0.206	
Bonferroni-corrected α					0.00417	

In this table, count represents the observed frequencies, while expected frequencies are in parentheses. Adj. Res. presents the adjusted residuals for the three topics across all four classes. The Bonferroni-corrected α is 0.00417, calculated by dividing the original significance level (0.05) by the twelve comparisons (3 topics $\times$ 4 classes). To determine whether an adjusted residual is statistically significant, I use the Bonferroni-corrected p-values. Therefore, an adjusted residual is considered statistically significant if its Bonferroni-corrected p-value is below 0.00417. Figures in bold are statistically significant.

Coherence and perplexity results

	Number of topics				
	Two topics	Three topics	Four topics	Five topics	Six topics
Log perplexity	-7.984	-7.959	-7.954	-7.953	-7.950
Coherence score	0.274	0.279	0.278	0.268	0.271

This table shows the coherence and perplexity scores for Latent Dirichlet Allocation (LDA) models with topic counts ranging from two to six. The model with three topics achieves the highest coherence while also maintaining a low perplexity, yielding a good balance between the two and, therefore, supporting the extraction of three topics and their use as textual predictors within the model.

Appendix A3.3

Evaluation metrics' calculation formulas

Multiclass confusion matrix

Actual labels	Prediction labels				Total
	0	1	2	3	
0	A_0P_0	A_0P_1	A_0P_2	A_0P_3	A_0
1	A_1P_0	A_1P_1	A_1P_2	A_1P_3	A_1
2	A_2P_0	A_2P_1	A_2P_2	A_2P_3	A_2
3	A_3P_0	A_3P_1	A_3P_2	A_3P_3	A_3
Total	P_0	P_1	P_2	P_3	AP

This table presents the multiclass confusion matrix. Rows represent actual class labels, also known as ground truth labels. Columns represent predicted class labels. A_iP_j denotes the number of observations whose actual class label (ground truth label) is i and whose predicted label is j and AP is the total number of observations ($AP = P_0 + P_1 + P_2 + P_3 = A_0 + A_1 + A_2 + A_3$).

The evaluation metrics' calculation formulas are as follows:

$$\text{Overall classification accuracy} = \frac{A_0P_0 + A_1P_1 + A_2P_2 + A_3P_3}{AP} \quad (1)$$

Where, A_0P_0 denotes the number of observations whose actual and predicted class is 0 (active firms), A_1P_1 is the number of samples whose actual and predicted class is 1 (firms expected to fail in a year), A_2P_2 is the number of samples whose actual and predicted class is 2 (firms expected to fail in two years), A_3P_3 is the number of samples whose actual and predicted class is 3 (firms expected to fail in three years), and AP is the total number of observations ($AP = P_0 + P_1 + P_2 + P_3 = A_0 + A_1 + A_2 + A_3$).

The formulas for calculating precision, recall, and F1 score for each class are as follows, using Class 0 (active firms) as an example:

$$Precision_0 = \frac{A_0 P_0}{P_0} = \frac{A_0 P_0}{A_0 P_0 + A_1 P_0 + A_2 P_0 + A_3 P_0} \quad (2)$$

$$Recall_0 = \frac{A_0 P_0}{A_0} = \frac{A_0 P_0}{A_0 P_0 + A_0 P_1 + A_0 P_2 + A_0 P_3} \quad (3)$$

$$F1\ score_0 = \frac{2 * Precision_0 * Recall_0}{Precision_0 + Recall_0} \quad (4)$$

Appendix A3.4

Pseudocode for the implementation of the LSTM and MLR models

Algorithm 1 Data preparation for LSTM and MLR models

Require: CEO_letters

Ensure: train_set_{preprocessed}, test_set_{preprocessed} # Training and testing ready sets

Stage 1: New features extraction

Preprocess letters by lowercasing, removing punctuations and stop words, and tokenisation

1: clean_letters ← *text_processing*(CEO_letters)

Vader sentiment extraction

2: sent_scores ← *compute_vader_sentiment*(clean_letters)

Representative topics extraction through LDA topic modelling

3: representative_topics ← *get_dominant_topics*(clean_letters)

Stage 2: Features concatenation

Concatenate the new extracted features (f12: f15) to the already obtained features (f1: f11)

Total number of features now is 15

The 15 features are:

f1 ← WC/TA # Working Capital/ Total Assets

f2 ← RE/TA # Retained Earnings/Total Assets

f3 ← EBIT/TA # Earnings Before Interest and Taxes/ Total Assets

f4 ← MVQ/TL # Market Value of Equity/ Total Liabilities

f5 ← S/TA # Sales/ Total Assets

f6 ← Price # PRICE

f7 ← Volatility # Volatility

f8 ← Annual_RPI # RPI

f9 ← Annual_EPU # EPU

f10 ← BOG # BOG

f11 ← CEO_turnover # CEO turnover

f12 ← Vader_Sentiment # Sentiment

f13 ← Rep_topic1 # Representative topic1: Customer relations

f14 ← Rep_topic2 # Representative topic 2: Operations

f15 ← Rep_topic3 # Representative topic 3: Production activities and environment

Stage 3: Splitting the data into exclusive training and testing sets

train_set: 15 features, data of failed and active firms from 2002–2018

test_set: 15 features, data of failed and active firms from 2019-2022

Data preparation

For training set

$\text{train_set}_{\text{wins}} \leftarrow \text{winsorise}(\text{train_set}_{\text{raw}}[\text{f1, f2, f3, f4 and f12}])$

$\text{train_set}_{\text{yeo}} \leftarrow \text{Yeo-Johnson_transformation}(\text{train_set}_{\text{raw}}[\text{f5, f7 and f8}], \text{train_set}_{\text{wins}}[\text{f4 and f12}])$

$\text{train_set}_{\text{scale}} \leftarrow \text{minmax_scaling}(\text{train_set}_{\text{raw}}[\text{f6 and f9}], \text{train_set}_{\text{wins}}[\text{f1, f2 and f3}], \text{train_set}_{\text{yeo}}[\text{f4, f5, f7, f8 and f12}])$

For testing set

$\text{test_set}_{\text{wins}} \leftarrow \text{winsorise}(\text{test_set}_{\text{raw}}[\text{f1, f2, f3, f4 and f12}])$

$\text{test_set}_{\text{yeo}} \leftarrow \text{Yeo-Johnson_transformation}(\text{test_set}_{\text{raw}}[\text{f5, f7 and f8}], \text{test_set}_{\text{wins}}[\text{f4 and f12}])$

$\text{test_set}_{\text{scale}} \leftarrow \text{minmax_scaling}(\text{test_set}_{\text{raw}}[\text{f6 and f9}], \text{test_set}_{\text{wins}}[\text{f1, f2 and f3}], \text{test_set}_{\text{yeo}}[\text{f4, f5, f7, f8 and f12}])$

Training and testing ready sets

$\text{train_set}_{\text{preprocessed}} \leftarrow \text{train_set}_{\text{scale}}$

$\text{test_set}_{\text{preprocessed}} \leftarrow \text{test_set}_{\text{scale}}$

Algorithm 2: LSTM and MLR prediction models

Require: $\text{train_set}_{\text{preprocessed}}$, and $\text{test_set}_{\text{preprocessed}}$

Ensure: predicted classes

```
#three_years_seq(train_set_preprocessed)
```

```
firms ← train_set_preprocessed["Firm_Name"]  
year ← train_set_preprocessed["Year"]  
year_sorted ← sort(year)
```

```
# Prepare a three-year sequence of information for each firm
```

```
function prepare_sequence(subset, include_labels):  
    seq_len = 3 # Three years' worth of information per firm  
    sequences ← []  
    labels ← []  
    firms = unique(subset['Firm_Name']) # Preserve original order  
    for firm in firms:  
        #Preserving order to ensure that the model learns from evolving patterns  
        firm_data = subset[subset['Firm_Name'] == firm].sort('Year')  
        values = firm_data[subset]  
        for j in range(len(firm_data)):  
            str_ind = max(0, j - seq_len + 1)  
            seq = values[str_ind:j+1]  
            # Left padding with zeros to ensure a consistent three-year sequence  
            if len(seq) < seq_len:  
                seq = pad(seq, ((seq_len - len(seq), 0), (0, 0)))  
            sequences.append(seq)  
            if include_labels:  
                labels.append(firm_data.iloc[j]['Ground_truth_labels'])  
    if include_labels:  
        # FOR TRAINING  
        RETURN (sequences, labels)  
    else:  
        # FOR TESTING  
        RETURN sequences
```

```

#Call the three-year sequence function, i.e., prepare_sequence
three_year_sequence_TRAIN ← prepare_sequence(train_setpreprocessed)
three_year_sequence_TRAINaugmented ← apply_SMOTE(three_year_sequence_TRAIN)

# Define LSTM and MLR models
modelLSTM
modelMLR

# Train the LSTM and MLR models

train_model(modelLSTM, three_year_sequence_TRAINaugmented)
train_model(modelMLR, three_year_sequence_TRAINaugmented)

three_year_sequence_TEST ← prepare_sequence(test_setpreprocessed) # SMOTE only applied
to trainingset

# Test the LSTM and MLR models

predicted_labelsLSTM← predict(modelLSTM, three_year_sequence_TEST)
predicted_labelsMLR← predict(modelMLR, three_year_sequence_TEST)

# Evaluate predictions using predicted and ground truth labels

ScoresLSTM← calculate_scores(predicted_labelsLSTM, labelsgnd)
ScoresMLR← calculate_scores(predicted_labelsMLR, labelsgnd)

```

Chapter 4

Interactional metadiscourse markers in CEO letters to shareholders: A comparison of failed and active firms

4.1 Introduction

Corporate annual reports, which contain information about the firm's financial performance, along with narrative texts authored by the firm's management, have consistently served as a primary means of communication (Brennan and Merkl-Davies, 2018). However, the rhetorical nature of the information provided within these reports is complex (Conaway and Wardrope, 2010). One of the primary narrative sections included in corporate annual reports is the CEO letter to shareholders, which is an essential rhetorical vehicle employed by CEOs to communicate their attitude, values and behaviours (Amernic et al., 2010, p. 5). Various studies, drawing from both linguistics and accounting and finance research, have consistently emphasised that CEO letters do not contain a neutral description of a firm's performance outcomes but are instead highly rhetorical products, strategically crafted to project ethos, establish credibility, persuade shareholders, and manage their perceptions (Hooghiemstra, 2010; Hyland, 2005, p. 74). These letters to shareholders, therefore, serve as an effective medium for assessing the CEO's intentions, as they can play a key role in demonstrating how CEOs interpret significant opportunities and risks, and how they strategise and respond to them.

The persuasive and rhetorical nature of CEO letters is especially prominent during periods of financial distress and uncertainty (Segars and Kohut, 2001). That is because during such periods, impression management may be used to legitimise, alleviate concerns, and secure stakeholders' support through various ways, including assertive/defensive strategies, rhetorical framing, attribution, and selectivity (Edgar et al., 2018). In other words, the

rhetorical choices CEOs use to express doubt and certainty, convey their attitudes toward the information, and establish an authorial persona may influence how stakeholders perceive the firm and evaluate the CEO's competence. As Hyland (2005, p. 58) argues, a text's persuasive force is achieved through the writer's expressions of stance, framing of propositions, readers' engagement, and the establishment of relationships with interlocutors. Similarly, recent studies investigating the metadiscourse features of corporate narratives reveal that CEOs strategically employ epistemic stance markers in their statements to construct persuasive discourse that engages stakeholders, projects a positive image of the firm, and mitigates uncertainties (e.g., Liu and Liu, 2023).

Despite the well-established importance of CEO letters in the literature, a critical gap remains in the understanding of whether CEOs of failed firms use stance and engagement metadiscourse markers differently from those of CEOs of active firms. Linguistic studies on metadiscourse have established the importance of stance and engagement expressions in academic, professional and personal writing (Hyland, 2005, p. 14). However, few studies have examined whether there are differences in stance and engagement markers between CEO letters from failed and active firms. This gap is notable given that interactional metadiscourse resources, as conceptualised by Hyland (2005, pp. 37-54, 2017), reflect how writers engage readers, build personal authority and credibility, and manage interpersonal relationships through text, which are linguistic devices that may significantly differ between firms facing financial distress and those demonstrating stability. Understanding these variations can therefore provide important insights into how managerial communication styles evolve in contexts of organisational success and failure.

In particular, a substantial body of research investigates multiple linguistic aspects of various annual report narratives, including CEO letters, to determine whether they enable better discrimination between bankrupt and non-bankrupt firms (e.g., Lohmann and Ohliger, 2020; Lopatta et al., 2017). There are also studies in the literature that compare differences in the use of textual expressions and keyword frequencies across narrative disclosures of bankrupt

and non-bankrupt firms (e.g., Yang et al., 2018). However, few studies have examined the similarities and differences in the use of interactional metadiscourse features between CEO letters of failed and active firms. This is especially important, as it would reveal how hedges, boosters, attitude markers, self-mentions, and engagement markers are distributed across both corpora and whether there are statistical differences in their usage. Such an examination, therefore, improves understanding of which linguistic devices are relied upon more during periods of distress compared to periods of stability, as well as the communicative priorities of CEOs in each context.

Another important factor that necessitates analysis is the rhetorical and communicative functions that interactional metadiscourse resources serve, and whether they differ across CEO letters of failed and active firms. For example, CEOs of both failed and active firms may be drawing on similar interactional devices; however, the rhetorical functions these devices serve may differ between the two corpora. Thus, depending on CEOs' objectives and the interpretations they wish the readers to draw from the text, the same interactional resources may be used in the CEO letters of both failed and active firms, but for different rhetorical and communicative purposes. This arises from prior research observations that while managers generally use narrative disclosures to persuade the audience, those of poorly performing firms with heightened reputational threats are more likely to use their discourse reactively to repair perceived legitimacy (e.g., Edgar et al., 2018), while those of firms facing less organisational pressures are more inclined to employ a more assertive language in their discourse to project leadership, make positive performance more obvious, and further enhance their legitimacy (e.g., Berkin et al., 2025). Consequently, the same interactional metadiscourse features may function differently across CEO letters of failed and active firms, serving different rhetorical purposes depending on the firm's performance, exposure to legitimacy challenges and the overall communicative goals the CEO seeks to achieve.

Moreover, several studies have investigated CEO letters from a metadiscourse perspective, including the use of metadiscourse resources across Chinese and U.S. firms (e.g., Lee, 2020), Chinese and Western banks (e.g., Huang and Rose, 2018) and UK and U.S. firms (e.g., Aerts and Yan, 2017). However, again, comparisons of the contextual use and rhetorical functions of interactional resources across CEO letters of failed and active firms remain largely underexplored in the literature. This focus is particularly crucial, given Hyland's (1998) assertion that, within the domain of corporate reporting, it would be "rewarding to examine the use of CEOs' metadiscourse in high and low performance companies, good and bad years, or different commercial sectors". The current study thus directly responds to this call and addresses the identified need by exploring CEOs' use of interactional metadiscourse resources across firms with different levels of financial conditions and legitimacy pressures.

These gaps, therefore, motivate my investigation of the following research questions:

RQ1: Do CEO letters of failed and active firms differ in their use of stance and engagement markers?

RQ2: Do the communication functions of the identified stance and engagement markers in the CEO letters of failed firms differ from their active counterparts?

By addressing these questions, the present study contributes to the extant literature on annual report narratives, and CEO letters in particular, by providing important and valuable insights into how CEOs employ communication and linguistic styles in response to organisational performance and pressures. Specifically, it examines whether the CEO letters of failed firms differ from those of active firms in their reliance and contextual use of interactional resources to persuade readers and establish authority, integrity, and credibility. The present study, therefore, sheds light on how managerial communication styles, and in

particular their choice of interactional linguistic devices, evolve in contexts of organisational success and failure.

In this study, I employ a comparative corpus-based research methodology to examine the use of interactional metadiscourse features in CEO letters, comparing those of UK failed and active firms between 2018 and 2022. Drawing on Hyland's framework of interactional metadiscourse features (2005, pp. 52-54), I compile two corpora to investigate how CEOs of failed firms employ hedges, boosters, attitude markers, self-mentions, and engagement markers in their discourse to express their stance and engage with stakeholders in comparison to those of active firms.

The quantitative analysis reveals statistically significant differences in the distribution and use of all five interactional metadiscourse resources across the CEO letters of failed and active firms. I find that CEOs' rhetorical choices appear to be shaped by firms' financial positions and threats to their legitimacy. In particular, CEOs of failed firms seem to rely more heavily on self-mentions and engagement markers. In contrast, the CEOs of active firms tend to use hedges, boosters and attitude markers more frequently.

Moreover, the findings of the qualitative analysis further reveal that although CEOs of both failed and active firms employ similar interactional metadiscourse resources, the communicative functions of these devices and the rhetorical purposes they serve vary significantly across the two corpora. For instance, CEOs of failed firms are more inclined to use reactive impression management strategies in their letters to persuade stakeholders, influence their perceptions of organisational performance, and restore perceived legitimacy. These strategies are reflected in: the CEOs extensive use of personalisation devices to soften negative disclosures and frame them as personal interpretations; increased focus on constructing a dialogic relationship with stakeholders to regain their confidence through the normalisation and rationalisation of reputational damage; and moderate use of hedges, boosters and attitude markers to frame organisational progress in a positive light, construct a personal ethos and repair legitimacy.

The CEOs of active firms, on the other hand, appear to employ proactive impression management strategies in their discourse to gain stakeholders' support, maintain legitimacy, and present themselves as competent, trustworthy, and honest. These strategies are evident in the CEOs' excessive use of hedges, boosters, and attitude markers to build both ethos and pathos, project a credible image of authority, decisiveness, and ethics, while also moderately personalising the discourse and involving stakeholders by directing their attention to evidence of strong firm performance to gain praise and persuade stakeholders of the firm's positive qualities. Collectively, these findings demonstrate how the metadiscourse markers of CEO letters are used for persuasion and reputation management, enabling CEOs to exert control over the information they present by expressing their stance and evaluating the information provided, thereby directing stakeholders' interpretation and appraisal of the subject matter.

These findings have empirical and practical implications. Empirically, this study fills an underexplored gap in the literature on corporate narrative disclosures by enhancing understanding of stance and engagement markers in CEO letters and providing evidence of statistically significant differences in the distribution and contextual use of these interactional devices across the CEO letters of both failed and active firms. The present study, therefore, contributes to the literature on CEO letters and metadiscourse by offering valuable insights into how CEOs' rhetorical choices and communicative practices are context-dependent and shaped by firms' performance and legitimacy challenges. This is particularly crucial for stakeholders, including investors and regulators, as it provides practical insights into how the non-propositional content in corporate narrative disclosures can be used to detect early signs of financial distress, potential future corporate failure, and differences in the underlying communicative objectives. Recognising these rhetorical and communicative-functional differences in the use of stance and engagement markers in CEO discourse could help stakeholders distinguish between reactive legitimacy-repairing discourse and proactive legitimacy-maintaining discourse. This, in turn, could enhance stakeholders' ability to identify

financially distressed firms at risk of failure from stable ones likely to remain operational. In addition, regulators may gain valuable insights into how CEOs use their discourse strategically to influence stakeholders' perceptions of organisational performance, manage societal expectations and respond to legitimacy challenges. Such insights could thus inform regulatory and narrative reporting guidelines that encourage clear and more transparent disclosures.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical background of the study. Section 3 discusses the analytical framework. Section 4 outlines the research methodology. Sections 5 and 6 present the results. Section 7 discusses the research findings and their implications. Section 8 concludes the paper.

4.2 Literature review

Over time, there has been increased interest in the textual and informational content of CEO letters, yielding diverse perspectives on their role in shaping stakeholders' perceptions. The debate in the literature, underpinned by several theoretical perspectives, centres on whether CEO letters serve as strategic tools to persuade stakeholders and influence their interpretation of performance results, or as sincere communicative actions intended to engage in dialogue with shareholders through sincere disclosures.

From an impression management perspective, narrative disclosures are viewed as strategic actions deliberately crafted to persuade stakeholders by influencing their perceptions and interpretations of the firm's performance (Brennan et al., 2009; Edgar et al., 2018). Various studies on impression management have provided empirical evidence demonstrating managers' use of rhetorical manipulation (e.g., Berkin et al., 2025), thematic manipulation (e.g., Brennan et al., 2010), structural manipulation (e.g., Boudt and Thewissen, 2019), and performance attribution (e.g., Moreno and Jones, 2021) strategies in corporate narratives to

construct a favourable image of the firm, particularly during periods of poor performance. For instance, in their study on how investors react to managerial self-inclusive language, Chen and Loftus (2019) find that when CEOs use singular personal pronouns to discuss firm performance, even when performance is negative, investors respond more positively to individual than collective self-inclusive language, regardless of performance news. Aerts and Yan (2017) examine the metadiscourse features of CEO letters from UK and U.S. corporations, documenting that CEOs tend to adopt an assertive stance, a defensive framing, or a logic-based rationalisation to shape stakeholders' perceptions, depending on the level of institutional scrutiny. Similarly, Lee (2020) examines the use of hedges and boosters in the CEO letters of Chinese and U.S. firms, providing empirical evidence of the strategic use of hedging and boosting as tools for impression management. Additionally, Leung et al. (2015) provide empirical evidence suggesting that financially distressed firms are more likely to conceal negative organisational outcomes in their annual report narratives to distract shareholders' attention from the firm's poor performance. Relatedly, Aerts' (2005) findings reveal that managers behave opportunistically when crafting their accounting narratives by disclosing selective information and manipulating its presentation, as an attempt to distort stakeholders' perceptions of the firm's performance.

Furthermore, legitimacy theory has also been widely employed to explain corporate reporting practices, with narrative disclosures hypothesised to play a key legitimising role by influencing stakeholders' perceptions of firms' legitimacy (Hummel and Schlick, 2016). That is because, according to the legitimacy theory, a social contract exists between firms and society; therefore, any legal, economic, or social sanctions can pose a threat to a firm's legitimacy and its survival (Dowling and Pfeffer, 1975). Consequently, accounting narratives are viewed as instruments used by managers to respond to public pressure, as well as to gain, maintain, and/or restore organisational legitimacy, enabling the firm to continue operating successfully (O'Donovan, 2002).

Many studies have applied impression management theory as a framework for analysing firms' narrative disclosures and communicative practices in response to legitimacy threats. For instance, Edgar et al. (2018) provide evidence demonstrating that firms employ impression management techniques in accounting narratives, particularly during periods of uncertainty, to alleviate concerns and restore organisational legitimacy. Similarly, Berkin et al. (2025) suggest that CEOs' language styles and the use of impression management strategies in their letters to shareholders are driven by distinct legitimacy concerns. In light of threats to organisational legitimacy, defensive and reactive impression management strategies are employed to repair reputation and prevent negative qualities from being ascribed to the reporting entity (Roussy and Rodrigue, 2018).

According to Suchman (1995), firms use assertive strategies to gain or maintain legitimacy, whereas defensive strategies are employed to restore compromised organisational legitimacy. Therefore, the employment of such strategies is context-dependent, and firms may implement a combination of assertive and defensive impression management strategies in their disclosures to gain, maintain, and/or restore organisational legitimacy (Elsayed, 2025). Such defensive impression management techniques are reactive and comprise tactics, such as; formulating a normalising account by providing excuses to avoid or deny accountability for a particular event; justifications to acknowledge responsibility for an event while dismissing its adverse implications; strategic restructuring mainly through dissociation and distancing oneself from adverse events or sources of illegitimacy; and apologies to admit responsibility while pledging to engage in more appropriate behaviour moving forward (Hassan, 2014; Ogden and Clarke, 2005; Suchman, 1995). Assertive impression management strategies, on the other hand, are proactive and include techniques such as; ingratiation to communicate to gain praise; self-promotion to persuade the audience of the firm's positive attributes; exemplification to represent an ethical and morally virtuous behaviour; entitlements to maximise accountability for events that improve managerial credibility; and enhancements to claim that past actions are of greater advantages than they

are perceived (Hassan, 2014; Ogden and Clarke, 2005; Suchman, 1995). Hence, the impression management and legitimacy theories complement one another in explaining narrative disclosure strategies. While impression management theory emphasises the rhetorical strategies used in accounting narratives to persuade stakeholders and manage their perceptions, legitimacy theory explains the circumstances that pose threats to firms' organisational legitimacy and existence, under which such impression management strategies are employed.

Although most studies in the accounting and finance literature view narrative disclosures as strategic tools used to influence and shape stakeholders' perceptions (e.g., Boudt and Thewissen, 2019; Huang et al., 2018), Yuthas et al. (2002) challenge this view and propose an alternative perspective grounded in the principles of the communicative action theory. The theory of communicative action, by Habermas (1984, 1987), asserts that societal actors rely on rational arguments to achieve mutual understanding. Opposed to strategic action, communicative action refers to the interactions in which actors in society are expected to engage in communication motivated by self-interest behaviour (Patelli and Pedrini, 2013). In other words, through the lens of communicative action, accounting narratives, such as CEO letters, are undertaken to achieve mutual understanding among members of society, rather than to mislead readers' interpretation and responses to performance results. Several studies in the literature offer empirical support for the theory of communicative action, suggesting that accounting narratives embody the discourse ethics of comprehensiveness, truthfulness, sincerity, and legitimacy (e.g., Dhanani and Connolly, 2015; Patelli and Pedrini, 2013). For instance, Yuthas et al. (2002) find that both poorly and well-performing firms engage in communicative action when communicating information about the management's trustworthiness and the firm's financial performance. Similarly, Dhanani and Connolly (2015) find that annual report narratives are characterised by communicative action, with truthful information disclosed to improve understandability. Patelli and Pedrini (2013) further argue that during periods of tough macroeconomic conditions, managerial incentives to

strategically manipulate information disclosed in routine vehicles, such as CEO letters, are low, as they could jeopardise organisational legitimacy. Instead, managers engage in communicative action by disclosing sincere information in accounting narratives to dialogue with stakeholders. However, several studies provide empirical evidence of various forms of impression management strategies employed in routine vehicles, particularly CEO letters (e.g., Berkin et al., 2025; Boudt and Thewissen, 2019), to influence stakeholders' perceptions and contain legitimacy threats.

Overall, despite the ongoing debate in the literature over whether CEO letters serve as strategic actions to influence stakeholders' perceptions and maintain organisational legitimacy or as communicative actions to achieve mutual understanding, they emerge as highly rhetorical creations crafted to achieve specific rhetorical purposes. Importantly, these rhetorical purposes are enacted through linguistic devices, such as interactional metadiscourse markers that writers or speakers use to express their stance toward the information they communicate and to engage with their audience (Hyland, 2005, pp. 49-53).

4.3 Analytical framework

Harris (1959) first introduced the term metadiscourse as a means to understand how writers use language to influence and shape readers' interpretations of a text. Subsequently, the concept of metadiscourse has become a subject for further development and refinement, reflecting scholars' efforts to capture how writers and speakers engage with their discourse to shape the audience's reception of their messages. Early contributions by Williams (1981), Kopple (1985) and Crismore and Farnsworth (1989) were instrumental in shaping the foundational understanding of metadiscourse by integrating both textual and interpersonal discourse features through which writers signal their presence in the text and guide readers' interpretation. These pioneering studies, however, are grounded in "pedagogic style guides" and "intuitive reflection" and are therefore criticised for offering an insufficiently robust

theoretical foundation for analysing real texts or for understanding writers' effective communication mechanisms (Hyland, 2005, p. 6).

Later, in their seminal work, Biber et al. (1999) conduct a corpus-based approach to examine the actual use of grammatical features across registers from American and British English. The innovation of their work lies in examining grammatical features in the Longman Grammar of Spoken and Written English (LGSWE), considering three interconnected aspects of usage that are important for corpus-based investigation: frequency information and register distribution, lexico-grammatical patterns, and grammatical/discourse factors. More importantly, they argue that many grammatical devices in English can be used to express the writer's or speaker's personal thoughts, feelings, attitudes, and judgments, and thus have developed a framework of stance markers, categorised by semantic function into epistemic, attitudinal, and style-of-speaking stance. While their framework has made significant empirical contributions to stance research, LGSWE has been criticised for treating certain adverbs, such as "evidently", as an epistemic adverbial of stance without considering that such adverbials may change meaning depending on where they are positioned in the sentence, nor does it mention the specific category of epistemic stance adverbials that "evidently" belongs to (Silver, 2003). Similarly, Ädel (2006, pp. 16-20) introduces the reflexive model in which metadiscourse is defined as "reflexive linguistic expressions" that relate to the text itself, including references to the writer-speaker as writer-speaker and the audience as audience. While the reflexive model avoids some of the vagueness in defining metadiscourse present in earlier frameworks, it does not include the stance and engagement markers included in the metadiscourse model proposed by Hyland (2005, pp. 52-54), which are essential for facilitating interaction and persuasion between the writer and the audience.

A key contribution of Hyland's (2005, pp. 48-54) taxonomy of metadiscourse lies in its distinction between interactive resources that assist with the readers' navigation and coherence within a text, and interactional resources that emphasise the writer's stance and engagement with the reader. By incorporating these interpersonal features of discourse,

Hyland (2005, pp. 52-54) highlights the social and persuasive nature of writing, establishing that texts are never neutral but are shaped by writers' rhetorical choices, which allow them to create an impression of authority, integrity, and credibility. This analytical approach to written discourse makes the framework particularly suitable for analysing the textual content of CEO letters, which serve as a medium through which CEOs communicate corporate performance and construct a persuasive logos, ethos and pathos in their texts (Aerts and Yan, 2017). This is especially relevant in contexts of financial distress, where CEOs of financially distressed firms may rely on persuasive rhetorical strategies to maintain shareholders' trust, while CEOs of financially stable firms may employ similar techniques to establish an ethos of credibility.

While Hyland's model (2005, pp. 50-54) distinguishes between interactive and interactional metadiscourse resources, this study focuses solely on interactional devices to investigate how CEOs of failed and active firms communicate and engage stakeholders through their discourse. The rationale is that interactional metadiscourse features are more personal and directly implicated in how writers interact with readers and, consequently, involve readers in the text, whereas interactive features primarily involve the management of information flow and the writer's arrangement of texts (Hyland, 2005, pp. 43-44). Therefore, by focusing on the interactional dimension of CEO letters, this study investigates how CEOs engage with stakeholders, navigate between doubt and certainty, express their attitudes and establish their corporate leadership identity during periods of both financial distress and stability, thus contributing to an enhanced understanding of the stance and engagement resources deployed in CEO letters.

Consequently, the analysis focuses exclusively on Hyland's model of interactional metadiscourse (2005, p. 49) to investigate how CEOs of failed and active firms communicate with shareholders through metadiscourse. As demonstrated in Table 4.1, the interactional metadiscourse features I analyse are hedges, boosters, attitude markers, self-mention and engagement markers. However, following Huang and Rose (2018), my study includes only

singular first-person pronouns, such as "I", "me", "my", and "mine", as they explicitly represent the CEO's presence in the text. The decision to exclude plural first-person pronouns aligns with the study's objective of capturing how CEOs of both failed and active firms use interactional markers in their metadiscourse to construct an ethos of credibility and project personal authority in their texts when addressing shareholders.

Table 4.1
Hyland's model of interactional metadiscourse markers

Category	Function	Examples
Interactional	To involve the reader in the text	Resources
Hedges	withhold commitment and open dialogue	might; perhaps; possible; about
Boosters	emphasise certainty or close dialogue	in fact; definitely; it is clear that
Attitude markers	express writer's attitude to proposition	unfortunately; I agree; surprisingly
Self-mentions	explicit reference to author	I; we; my; me; our
Engagement markers	explicitly build relationship with reader	consider; note; you can see that

This table presents the stance and engagement markers included in the metadiscourse model proposed by Hyland (2005, pp. 52-54).

4.4 Corpus and method

4.4.1 Data collection

The present study employs a corpus-based research method to investigate the CEO letters of UK FTSE 350 listed firms from 2018 to 2022 (see Appendix 4.1 for the list of firms included in the sample). I select this timeframe because it encompasses the most recent period for which substantial financial and textual data on both failed and active firms are available; several firms that failed after this year did not have annual reports released in the

year preceding their failure. Consequently, such firms could not be included in the analysis. Additionally, it captures a period characterised by uncertainty, including the economic impact of Brexit and the COVID-19 pandemic, ensuring the analysis reflects CEOs' discourse amid significant financial ambiguity. Furthermore, a corpus-based methodology is particularly suitable for this study, as it facilitates the identification and quantification of discourse features within corpus data, thereby enabling statistical analysis and comparisons across groups (McEnery and Hardie, 2011).

I particularly examine CEO letters due to their essentially rhetorical nature, as the metadiscourse employed in these letters allows CEOs to exercise control over the information they present through the employment of stance and evaluative expressions that assist them in directing stakeholders to how they should interpret the subject matter (Hyland, 1998). In accounting and finance literature, CEO letters have been perceived by some as tools for strategic impression management (e.g., Aerts and Yan, 2017; Boudt and Thewissen, 2019), while others view them as a form of communicative action aimed at dialoguing with shareholders (e.g., Patelli and Pedrini, 2013). From a linguistic perspective, CEO letters are "highly rhetorical creations", crafted to influence readers' perceptions through expressions of credibility, the resolution of doubt, and the avoidance of dispute (Hyland, 2005, p. 74). The non-propositional content of CEO letters, therefore, includes strategic persuasive metadiscourse features aimed at engaging with readers, gaining their acceptance and constructing an ethos of credibility and competence.

Regarding the composition of the two corpora, I apply three selection criteria. First, financial firms are excluded from both my datasets due to their different reporting practices (Schleicher and Walker, 2010). The underlying logic is that financial firms are risk-oriented corporations, exhibiting different types of risk compared to non-financial firms (Basel Committee on Banking Supervision, 2006). According to Beretta and Bozzolan (2004), firm industry can be a significant factor in explaining corporate disclosure practices. Corporate narrative disclosures may therefore be used differently to address public concerns,

depending on the distinct regulatory and societal expectations that apply to financial and non-financial firms. This suggests that financial and non-financial firms may also exhibit linguistic variation in their narrative disclosures, particularly in their use of rhetorical and metadiscourse features, which may be attributed to sector-specific reporting styles. The exclusion of financial firms, therefore, ensures that the observed differences in the distribution and use of interactional metadiscourse features more accurately reflect the variation in the discourse rhetorical strategies employed by CEOs of failed and active firms, rather than being influenced by sector-specific communication standards and rules.

Second, it is essential to recognise that delisting from the stock market does not necessarily imply bankruptcy. Therefore, only firms that have formally filed for insolvency proceedings and have been delisted for bankruptcy-related reasons, as per their status on the GOV.UK Companies House Register and were consequently delisted from the market between 2018 and 2022 are considered failed firms in this study.¹² Meanwhile, active firms are those that remain listed and operational during the same period.

Third, I calculate the Altman Z-score for each firm-year observation for both failed and active firms over a five-year window to ensure that the failed firms' corpus comprises financially distressed firms, while the active firms' corpus encompasses those with stable and healthy financial performance, a key methodological innovation of the present study. For instance, only failed firms with Z-scores indicating high bankruptcy risk or financial uncertainty ($Z < 2.99$) five years before their actual failure are included. For active firms, only those with firm-year observations whose Z-scores indicate strong financial health and a low risk of bankruptcy ($Z > 2.99$) are included. The application of the Altman Z-score as a financial screening criterion to classify firms by financial health and position thus facilitates a fair and

¹² In line with prior UK-based studies (e.g., Elsayed and Elshandidy, 2020; Tinoco and Wilson, 2013), a firm is declared legally insolvent (i.e., failed) if it has formally filed for insolvency proceedings and appears as "in administrative receivership, in administration, company voluntary arrangement, creditor's voluntary liquidation or in liquidation " on the GOV.UK Companies House Register (see also Christidis and Gregory, 2010).

rigorous examination of interactional metadiscourse resources across CEO letters from failed and active firms. In addition, the choice of a five-year window primarily ensures comparability between the failed firms corpus and the active firms corpus, as the number of firms that fail due to bankruptcy-related reasons is significantly lower than the number of active firms. Therefore, using a five-year window while maintaining the same Z-score criteria across all years for both groups increases the pool of eligible firm-year observations for failed firms and ensures that the number of observations for active firms remains within a comparable range, albeit slightly larger, producing a sufficient dataset for comparative analysis.

Consequently, the first corpus comprises CEO letters from failed firms that have formally filed for insolvency proceedings and been delisted from the stock market due to bankruptcy-related reasons between 2018 and 2022. Meanwhile, the second corpus, the reference corpus, comprises the CEO letters of a sample of firms that have remained listed and operational during the same period with strong financial performance.

I obtain all accounting and financial data from Thomson Reuters Eikon. Additionally, I manually collect annual reports (mainly PDFs) from various sources, including Companies House (2015) (for failed firms) and companies' websites (for active firms). I then convert the PDFs to plain text, removing non-narrative elements such as tables, figures, and legends.

4.4.2 Procedure for the extraction and analysis of metadiscourse

Each corpus has been processed separately using Wmatrix7, which provides part-of-speech tagging (PoS) via the CLAWS tool, assigning grammatical categories to words. LancsBox has also been used to generate word counts for each corpus and for keyword-in-context (KWIC) searches, facilitating the qualitative investigation of stance and engagement within their surrounding co-text.

Furthermore, to examine whether there are significant differences in the use of interactional metadiscourse markers across CEO letters of failed and active firms, I conduct both the Chi-square (χ^2) test and the Log-Likelihood (LL) test, which are widely employed in corpus linguistics (Rayson and Garside, 2000). The decision to conduct both tests stems from the longstanding debate regarding which statistical assumption is proven more reliable for corpus-based frequency comparisons. Hence, while the Chi-square test has been widely employed in various corpus linguistic studies (e.g., Chen and Li, 2023; Huang and Rose, 2018), other scholars, such as Dunning (1993) argue that the assumption of a normal distribution should not be relied upon when conducting a statistical textual analysis, and suggest using the Log-Likelihood test as an alternative test. The present paper, therefore, conducts both tests to determine whether the results obtained with the χ^2 test align with those of the LL test, ensuring the statistical validity of my findings.

4.5. Results

4.5.1 Overall distribution of interactional metadiscourse in failed and active firms

Table 4.2 presents the overall distribution of interactional metadiscourse markers in the two corpora, failed and active firms. The total items for stance markers represent the sum of the total frequencies of hedges, boosters, attitude markers, and self-mentions, whereas the total items for engagement correspond to the total frequency of engagement markers only. The percentage of the total reflects the proportion of each category (stance or engagement) relative to the overall number of interactional metadiscourse items in the corpus. Normalised frequencies per 10,000 words correspond to the total frequency of each category divided by the total corpus size and multiplied by 10,000 to allow for comparability across corpora of differing sizes.

As demonstrated in Table 4.2, the results reveal that stance markers are employed more frequently in both corpora, with 68.89% in failed firms and 75.06% in active firms.

Engagement markers represent the remaining share, accounting for 31.11% in failed firms and 24.94% in active firms. Additionally, the distribution of frequencies, normalised per 10,000 words, is 102.58 for stance markers and 46.32 for engagement markers in failed firms, compared to 110.96 and 36.87, respectively, in active firms. The increased use of stance over engagement resources in both corpora is consistent with the findings of previous studies (e.g., Chen and Li, 2023; Huang and Rose, 2018), indicating that although CEOs employ both features in their letters, they tend to rely more on hedges, boosters, self-mentions and attitude markers to create an impression of authority, integrity and credibility, while simultaneously engaging shareholders.

Consistent with Rayson et al. (2004), I assess whether the observed differences between the two corpora are statistically significant by conducting the Chi-square (χ^2) and the Log-Likelihood (LL) tests using the UCREL Significance Test System (Lancaster University, n.d.). The results indicate that the difference in usage of stance features between failed and active firms is statistically significant ($\chi^2 = 4.45$, $p = 0.035$; $LL = 4.56$, $p = 0.033$), as well as in engagement features ($\chi^2 = 14.96$, $p = 0.000$; $LL = 14.97$, $p = 0.000$). These outcomes reveal that CEOs of failed and active firms exhibit distinct patterns in their use of stance and engagement markers, suggesting that these features may be employed strategically in response to the firm's financial condition. These findings align with the general perspective in discourse studies that metadiscourse is embedded and integrated within particular social, cultural and institutional contexts (e.g., Faigley, 1986, p. 535; Hyland, 2005, p. 14), while also resonating with accounting and finance research, highlighting the strategic aspects of accounting narratives (e.g., Leung et al., 2015; Moreno and Jones, 2021). Additionally, as shown in Table 4.2, the LL test yields results consistent with those of the χ^2 test, enhancing the statistical validity of my findings and further supporting the notion that CEOs of failed firms employ interactional metadiscourse markers differently than those of active firms.

Table 4.2

Overall distribution of interactional metadiscourse in failed and active firms

Category	Failed firms			Active firms			Chi-square test		Log-likelihood test	
	Total items	% of total	Per 10,000 words	Total items	% of total	Per 10,000 words	χ^2	p-value	LL	p-value
Stance	1156	68.89	102.58	1956	75.06	110.96	4.45	0.035**	4.56	0.033**
Engagement	522	31.11	46.32	650	24.94	36.87	14.96	0.000***	14.97	0.000***

This table presents the overall distribution of stance and engagement markers across the CEO letters of failed and active firms. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively (two-tailed test).

4.5.2 Sub-categorical distribution of interactional metadiscourse in failed and active firms

Table 4.3 presents the distribution of each sub-category of interactional metadiscourse across the two corpora, failed and active firms. As demonstrated in Table 4.3, significant statistical differences are observed in the distribution of all five interactional metadiscourse features including, hedges ($\chi^2 = 8.63$, $p = 0.003$; LL = 8.93, $p = 0.003$), boosters ($\chi^2 = 8.13$, $p = 0.004$; LL = 8.43, $p = 0.004$), attitude markers ($\chi^2 = 13.38$, $p = 0.000$; LL = 14.27, $p = 0.000$), self-mentions ($\chi^2 = 46.04$, $p = 0.000$; LL = 45.30, $p = 0.000$), and engagement markers ($\chi^2 = 14.96$, $p = 0.000$; LL = 14.97, $p = 0.000$). These outcomes reveal that while CEOs of both failed and active firms heavily rely on interactional metadiscourse features to shape stance and construct relationships with readers, they employ different rhetorical strategies in their letters to shareholders.

Table 4.3

Sub-categorical distribution of interactional metadiscourse in failed and active firms

Category	Failed firms			Active firms			Chi-square test		Log-likelihood test	
	Total items	% of total	Per 10,000 words	Total items	% of total	Per 10,000 words	χ^2	p-value	LL	p-value
Hedges	415	24.73	36.83	777	29.82	44.08	8.63	0.003***	8.93	0.003***
Boosters	364	21.69	32.30	686	26.32	38.91	8.13	0.004***	8.43	0.004***
Attitude markers	115	6.85	10.20	271	10.40	15.37	13.38	0.000***	14.27	0.000***
Self-mentions	262	15.61	23.25	222	8.52	12.59	46.04	0.000***	45.30	0.000***
Engagement markers	522	31.11	46.32	650	24.94	36.87	14.96	0.000***	14.97	0.000***

This table presents the sub-categorical distribution of interactional markers across the CEO letters of failed and active firms. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively (two-tailed test).

Furthermore, as illustrated in Table 4.3, engagement markers are the most frequently used interactional metadiscourse features in the discourse of failed firms, followed by hedges, boosters, self-mentions and attitude markers. Meanwhile, hedges are the most frequently used interactional metadiscourse features in the discourse of active firms, followed by boosters, engagement markers, attitude markers and self-mentions.

In terms of hedges, CEOs of active firms use them more frequently than those of failed firms, with rates of 29.82% and 24.73%, respectively. This observed higher frequency of hedges in the active firms corpus suggests that the CEOs of active firms are more inclined to use hedges in their letters, allowing them to signal caution and withhold complete commitment by presenting information as opinions rather than facts, thereby maintaining credibility and protecting themselves if circumstances change. Similarly, boosters appear more frequently in the discourse of active firms, where they account for 26.32%, compared to the discourse of failed firms, where they account for 21.69%. This observation suggests that CEOs of

active firms are more likely to use boosters in their communications to express certainty and project solidarity with the audience by closing down potential alternatives with a single, confident voice. Attitude markers are also more frequently employed in the CEO letters of active firms, with an observed rate of 10.40%, compared with 6.85% in the CEO letters of failed firms. This outcome demonstrates that CEOs of active firms tend to adopt a more affective and evaluative stance in their discourse, thereby strengthening the persuasiveness of their arguments, compared to CEOs of failed firms.

On the other hand, the frequency of self-mentions in the discourse of failed firms (15.61%) is considerably higher than that of active firms (8.52%). The observed higher frequency of self-mentions in the CEO letters of failed firms suggests that CEOs of failed firms tend to represent themselves more prominently in the text and consciously adopt a particular stance and a contextually situated authorial persona, compared to those of active firms.

Engagement markers are also more frequent in the failed firms corpus (31.11%) than in the active firms corpus (24.94%). This finding demonstrates that CEOs of failed firms are more likely to rhetorically position the audience and draw them into the discourse to establish solidarity than CEOs of active firms.

Taken together, these findings reveal that CEOs of both failed and active firms employ different rhetorical strategies in their communications with the shareholders. The more frequent use of self-mentions and engagement markers in the discourse of failed firms suggests that CEOs of failed firms tend to focus more on self-representation and direct reader involvement in their letters to shareholders than those of active firms. Meanwhile, the more frequent use of hedges, boosters, and attitude markers in the discourse of active firms suggests that CEOs of active firms tend to focus more on balancing caution, certainty, and evaluative stance in their letters to shareholders than those of failed firms.

4.6. Ranked frequencies of interactional metadiscourse in failed and active firms

4.6.1 Hedges

According to Hyland (2005, p. 14), hedges are linguistic devices that demonstrate the writer's choice to recognise diverse voices and opinions, allowing them to “withhold complete commitment to a proposition”. Hyland (1998) explains that writers must calculate the weight they want to give to an assertion, considering the level of accuracy and reliability they wish it to carry, while also potentially seeking protection in case the assertion is proven incorrect. Including hedges within the text, therefore, facilitates the presentation of the writer's statements and information provided as an opinion instead of a definitive fact. This is illustrated in example (1), where the modal verb “may” functions epistemically, allowing the CEO to express caution in their argument by framing the potential reduction in planning consents as a possibility grounded in the CEO's plausible reasoning, rather than as certain knowledge. Similarly, the epistemic modal verb “should” is used to project anticipated higher returns as an expectation rather than as guaranteed results. This highlights how CEOs of financially distressed firms strategically employ hedges in their discourse to mitigate the force of negative claims, framing them as possibilities rather than definitive facts, enabling them to mitigate their direct responsibility for disappointing results while still maintaining a persuasive stance. In example (2), the CEO employs the epistemic adjective “possible” and the modal verb “may” as hedging devices that jointly function to balance acknowledging uncertainty with cautious optimism, allowing them to protect their credibility and maintain investor confidence. The hedge “possible” conveys the tentativeness of the claim, signalling to readers that the scenario presented is one possible outcome among others. The hedge “may” further reinforces this epistemic stance by emphasising the uncertainty surrounding the “feasible outcome”, enabling the CEO to mark their statement as provisional and withhold complete commitment to their proposition. While “may” primarily conveys an epistemic function in this example as it expresses uncertainty regarding the form of the

“feasible outcome”, it also functions deontically, in that it gestures at what the outcome can look like under given conditions. This highlights that the CEOs of active firms employ hedges in their discourse not merely to acknowledge uncertainty but also to strategically shape expectations by constructing provisional, forward-looking scenarios open to negotiation, which in turn allows them to preserve their credibility while shifting responsibility onto external factors.

- (1) A consequence of this may be that we achieve fewer first time planning consents, but the returns generated should be higher. (McCarthy and Stone plc, 2018 – Failed firm)*
- (2) However, it is possible to illustrate what a feasible outcome may look like and where it will be derived from, assuming a supportive economy and no significant downturn in any of our major markets, and that is what we sought to disclose. (Hays plc , 2022 – Active firm)*

Table 4.4

The ranked frequencies of the top ten hedges

Failed firms				Active firms			
Hedges	PoS tag	Total items	Per 10,000 words	Hedges	PoS tag	Total items	Per 10,000 words
would	epistemic modal verb	63	5.59	would	epistemic modal verb	87	4.94
should	epistemic modal verb	52	4.61	around	approximator	85	4.82
about	approximator	34	3.02	about	approximator	76	4.31
around	approximator	31	2.75	should	epistemic modal verb	62	3.52
possible	epistemic adjective	29	2.57	possible	epistemic adjective	57	3.23
could	epistemic modal verb	19	1.69	may	epistemic modal verb	37	2.10
may	epistemic modal verb	18	1.60	could	epistemic modal verb	27	1.53
likely	epistemic adjective	15	1.33	likely	epistemic adjective	27	1.53
feel	epistemic verb	10	0.89	might	epistemic modal verb	23	1.30
largely	epistemic adverb	10	0.89	often	epistemic adverb	23	1.30

This table presents the top ten hedges used across the CEO letters of failed and active firms.

Furthermore, Table 4.4 presents the top ten hedges across both failed and active firms, ranked by frequency. In the failed firm's corpus, "would" appears as the most employed hedge (5.59 per 10,000 words), while "should" (4.61 per 10,000 words) and "about" (3.02 per 10,000 words) appear as the second and third frequently used hedges. Similarly, "would" (4.94 per 10,000 words) emerges as the most used hedge in the active firm's corpus, followed by "around" (4.82 per 10,000 words) and "about" (4.31 per 10,000 words). As demonstrated in Table 4.4, epistemic modal verbs such as "would", "may", "could", "should" and "might" frequently appear across the top ten hedges in both corpora, highlighting the central role that these linguistic forms play in expressing caution, certainty and plausible reasoning in a statement. Examples (1) and (2) provided earlier are representative cases of such. In addition, other linguistic devices, such as the approximators "around" and "about," also rank highly in both corpora, suggesting that CEOs rely on such hedging devices to

cautiously present claims, reduce the weight of their assertions, and withhold complete commitment to a proposition. Overall, this analysis reveals that although both corpora exhibit similar patterns in their use of hedging devices, the contexts in which they are employed differ between failed and active firms.

4.6.2 Boosters

Boosters help writers assert their claims with certainty while building rapport and demonstrating involvement with the topic and their audience (Hyland, 1999a). Their employment within a discourse, therefore, enables the writer to highlight certainty and establish their presence. According to Hyland (2005, pp. 78-80), boosters serve to convey an image of confidence, decisiveness, and authority, helping writers instil confidence and trust among readers through an impression of certainty and conviction in their expressed views. In example (1), the epistemic verb “demonstrate” functions as a booster, enabling the CEO to present the firm’s financial results as definitive evidence of the business model’s resilience. In this example, the booster “demonstrate” performs a persuasive rhetorical function to manage impressions at a time when underlying risks and certainties were present, as indicated by “against that background”. This suggests that CEOs of failed firms strategically employ boosters in their letters to persuade stakeholders and instil trust during periods of uncertainty, aligning themselves with a confident stance to close down alternatives, restore organisational legitimacy, and construct an ethos of competence and authority. On the other hand, the booster “demonstrate” in example (2) performs a different rhetorical function, despite it being used to project epistemic certainty, as in example (1). Hence, in example (2), “demonstrate” is employed proactively to emphasise certainty and construct rapport by reinforcing solidarity with the shareholders by presenting the firm’s success and financial strength as a matter of fact rather than opinion. This indicates that, unlike CEOs of failed firms who employ boosters defensively, CEOs of active firms use

boosters in their discourse to highlight corporate success and financial strength, thereby projecting a credible image of authority, decisiveness, and conviction.

(1) Against that background these results demonstrate that we have a resilient business model. (Debenhams plc, 2017 – Failed firm)

(2) Persimmon's 2022 performance demonstrates our capabilities to deliver both strong financial performance and consistent build quality and customer service for the first time in our history. (Persimmon plc, 2022 – Active firm)

Table 4.5
The ranked frequencies of the top ten boosters

Failed firms				Active firms			
Boosters	PoS tag	Total items	Per 10,000 words	Boosters	PoS tag	Total items	Per 10,000 words
believe	epistemic verb	47	4.17	believe	epistemic verb	100	5.67
clear	epistemic adjective	41	3.64	clear	epistemic adjective	55	3.12
know	epistemic verb	33	2.93	always	frequency adverb	54	3.06
established	epistemic verb	26	2.31	find	epistemic verb	51	2.89
always	frequency adverb	19	1.69	sure	epistemic adjective	33	1.87
clearly	epistemic adverb	19	1.69	think	epistemic verb	32	1.82
must	epistemic modal verb	18	1.60	established	epistemic verb	30	1.70
certain	epistemic adjective	11	0.98	never	epistemic adverb	30	1.70
of course	epistemic phrase	10	0.89	demonstrated	epistemic verb	25	1.42
demonstrated	epistemic verb	9	0.80	clearly	epistemic adverb	22	1.25

This table presents the top ten boosters used across the CEO letters of failed and active firms.

Moreover, Table 4.5 presents the top ten boosters in both failed and active firms, ranked by frequency. As illustrated in Table 4.5, epistemic verbs are the most frequently used boosters across both corpora. In the failed firm's corpus, "believe" emerges as the most used booster (4.17 per 10,000 words), while "clear" (3.64 per 10,000 words) and "know" (2.93 per 10,000 words) appear as the second and third most employed boosters. In terms of the active firm's corpus, "believe" (5.67 per 10,000 words) also occupies the first place, followed by "clear" (3.12 per 10,000 words) and "always" (3.06 per 10,000 words). Among the top ten boosters across both corpora, epistemic verbs such as "believe", "know", "established", "find", "think", and "demonstrated" are widely used, allowing CEOs to project an image of authority and conviction. Similarly, other linguistic forms, including epistemic adjectives and adverbs, rank highly in both corpora, indicating that CEOs frequently use such boosters to present their proposition with confidence while also emphasising involvement with the audience. Overall, these findings suggest that while CEOs of both failed and active firms tend to use similar boosters in their discourse, the contexts in which these terms are used differ between the two corpora.

4.6.3 Attitude markers

Attitude markers represent the writer's attitude to textual information, expressing surprise, agreement, relevance, disappointment (Hyland, 1999b). Such evaluative terms and stances play a significant role in strengthening the persuasiveness of the writer's arguments by claiming solidarity with the audience and are signalled metadiscoursally through attitude verbs, sentence adverbs and adjectives (Hyland, 2005, p. 70). This is demonstrated in the two examples below, in which CEOs use adjectives to express their attitudes toward propositions. In example (1), the attitude adjective "important" functions as an attitude marker, enabling the CEO to express their affective evaluation of the firm's chosen strategy. In particular, the attitude marker "important" is used to frame the firm's strategic direction

positively and persuade stakeholders to align with the CEO's appraisal of the "milestone" achieved through the strategic path being pursued. This reveals that CEOs of failed firms strategically employ attitude markers to strengthen the persuasiveness of their argument and shape perceptions by legitimising strategic success and affirming management decision-making even when underlying risks are present. In example (2), however, the attitude marker "important" performs an evaluative and a directive function by allowing the CEO to express their personal judgment by acknowledging the firm's progress, but at the same time steering stakeholders' attention to what should also be prioritised. This suggests that the CEOs of active firms employ attitude markers to reinforce solidarity with stakeholders and strengthen credibility by constructing an ethos of responsibility and accountability.

(1) This is an important milestone and indicates that our strategy for the business is the right one. (Johnston Press plc, 2013 – Failed firm)

(2) While we acknowledge our progress, it is more important to recognise that there is still much to do to improve diversity within Kainos and in the wider technology industry. (Kainos Group plc, 2021 – Active firm)

Table 4.6

The ranked frequencies of the top ten attitude markers

Failed Firms				Active firms			
Attitude markers	PoS tag	Total items	Per 10,000 words	Attitude markers	PoS tag	Total items	Per 10,000 words
important	attitude adjective	67	5.95	important	attitude adjective	66	3.74
appropriate	attitude adjective	14	1.24	even	attitude adverb	59	3.35
importantly	attitude adverb	14	1.24	appropriate	attitude adjective	45	2.55
essential	attitude adjective	9	0.80	essential	attitude adjective	26	1.47
disappointing	attitude adjective	6	0.53	importantly	attitude adverb	13	0.74
fortunate	attitude adjective	3	0.27	dramatically	attitude adverb	11	0.62
unfortunate	attitude adjective	1	0.09	amazing	attitude adjective	8	0.45
unfortunately	attitude adverb	1	0.09	appropriately	attitude adverb	6	0.34
				understandably	attitude adverb	6	0.34
				remarkable	attitude adjective	3	0.17

This table presents the top ten attitude markers used across the CEO letters of failed and active firms.

As demonstrated in Table 4.6, attitude adjectives emerge as the most frequently employed attitude markers in both corpora. In the failed firm's corpus, "important" (5.95 per 10,000 words) appears as the most used attitude marker, while "appropriate" (1.24 per 10,000 words) and "importantly" (1.24 per 10,000 words) are the second and third frequently used attitude markers. For the active firm's corpus, "important" (3.74 per 10,000 words) ranks first, followed by "even" (3.35 per 10,000 words) and "appropriate" (2.55 per 10,000 words).

Across the top ten attitude markers in both corpora, attitude adjectives and adverbs are heavily used, indicating their key role in expressing CEOs' attitudes and positions towards textual information. Taken together, these results reveal that CEOs of both failed and active firms tend to heavily rely on adjectives and adverbs as attitude markers to express their attitudes toward propositions. However, the contexts in which these attitude markers are

employed differ across the two corpora, reflecting different rhetorical strategies and the particular stakeholder's interpretation of the textual information presented, each seeks to achieve.

4.6.4 Self-mention

Self-mention refers to the extent of explicit authorial presence within a text, quantified by the frequency of first-person pronouns and possessive adjectives (Hyland, 2005, p. 53).

Although all writings convey a representation of the self of the writer, first-person pronouns are the most powerful forms of self-representation (Ivanič and Camps, 2001). The subsequent examples are typical instances of such. For instance, in example (1), the CEO uses the singular first-person pronoun "I" with the epistemic verb of judgement "believe" that functions as a booster to provide an expression of personal belief, allowing them to strengthen their authorial presence in the text and directly align themselves with the view expressed. In particular, the CEO takes epistemic responsibility while still projecting authority by grounding the explanation of "disappointing performance" in their own personal judgment, in an attempt to shift responsibility for negative outcomes away from themselves and onto the "three factors" said to have contributed to it. This suggests that CEOs of failed firms use self-mentions to strategically soften negative disclosures by offering expressions of personal belief, thereby constructing a personal ethos of competence and authority while reducing potential reputational damage. Meanwhile, in example (2), the use of the self-mention "I" highlights the CEO's high degree of authorial presence within the statement as they express a performative commitment, positioning themselves as personally responsible for reporting "open and honest" information. In this example, the self-mention "I" builds a personal ethos by allowing the CEO to project themselves into their text, presenting a competent, credible, commanding, and honest persona. This demonstrates that the CEOs of active firms

strategically employ self-mentions to present an authoritative authorial stance in their texts, thus strengthening their credibility and signalling solidarity with stakeholders.

(1) I believe that three factors contributed to this disappointing performance.

(Debenhams plc, 2014 – Failed firm)

(2) I promise to report progress on those in our normal open and honest way. (Workshop

Group plc, 2022- Active firm)

Table 4.7

The ranked frequencies of the top ten self-mentions

Failed firms				Active firms			
Self-mentions	PoS tag	Total items	Per 10,000 words	Self-mentions	PoS tag	Total items	Per 10,000 words
I	first-person pronoun	195	17.30	I	first-person pronoun	192	10.89
my	first-person possessive adjective	49	4.35	my	first-person possessive adjective	20	1.13
me	first-person pronoun	18	1.60	me	first-person pronoun	10	0.57

This table presents the top ten self-mentions used across the CEO letters of failed and active firms.

Furthermore, as illustrated in Table 4.7, personal pronouns and possessive adjectives are the most used forms of self-representation in both corpora. In the corpus of failed firms, “I” appears to be the most frequently used self-mention (17.30 per 10,000 words), while “my” (4.35 per 10,000 words) and “me” (1.60 per 10,000 words) are the second and third most frequently employed self-mentions. In terms of the active firms corpus, “I” also ranks first (10.89 per 10,000 words), followed by “my” (1.13 per 10,000 words) and “me” (0.57 per 10,000 words). The extensive use of “I” in both corpora, compared with other self-mentions,

highlights the power and key role of first-person pronouns in allowing CEOs to construct a personal ethos of competence and authority. Similarly, the frequent use of “my” and “me” also reveals how CEOs tend to employ possessive pronouns and adjectives to demonstrate their direct involvement and alignment with corporate outcomes. Overall, these findings suggest that CEOs of both failed and active firms rely on self-representations to convey a particular stance and emphasise their authorial presence in their statements. However, the rhetorical function these self-mentions serve varies across the two corpora.

4.6.5 Engagement markers

Engagement markers are linguistic devices that directly address readers, either to attract their attention or involve them as participants in the discourse (Hyland, 2005, p. 53). Hence, they allow writers to emphasise or minimise the presence of readers in the text. For instance, employing engagement markers like reader pronouns in discourse enables writers to establish a writer-reader relationship by drawing readers into the discourse, allowing them to construct an effective line of reasoning and align their goals with those of the readers (Fu and Hyland, 2014; Hyland, 2005, p. 82). This is illustrated in the following examples. For instance, in example (1), the reader pronoun “you” functions as an engagement marker by directly addressing the stakeholders and involving them as discourse participants. In this example, the CEO employs the engagement marker “you” to mitigate the distance between themselves and stakeholders by presenting stakeholders as informed participants in the firm's challenges, reflecting a rhetorical strategy aimed at restoring organisational legitimacy. This suggests that CEOs of failed firms strategically employ engagement markers to establish a writer-reader relationship by involving stakeholders in the discourse during periods of corporate vulnerability, which in turn enables them to manage negative impressions and reputational risks. The engagement marker “you” in example (2), on the other hand, directs the stakeholder’s attention to evidence of strong financial health, as it is

used alongside the directive verb “see”, which also serves as an engagement marker. In particular, the CEO employs the engagement markers “you” and “see” to pull readers into the discourse and guide them toward a particular interpretation of the presented evidence, thereby reinforcing credibility and legitimacy. This demonstrates that the CEOs of active firms employ engagement markers in their discourse to rhetorically position stakeholders and align them with the CEO’s positive interpretation of the firm’s performance.

- (1) You are no doubt aware that we’ve faced some recent criticism, particularly in the UK, for events in which we should have provided a higher degree of care and support to our customers. (Thomas Cook Group plc, 2015 – Failed firm)
- (2) You can see from these results that our business and our Warhammer Hobby are in good shape. (Games Workshop Group plc, 2018 – Active firm)

Table 4.8

The ranked frequencies of the top ten engagement markers

Failed firms				Active firms			
Engagement markers	PoS tag	Total items	Per 10,000 words	Engagement markers	PoS tag	Total items	Per 10,000 words
key	directive noun	60	5.32	develop	directive verb	73	4.14
see	directive verb	58	5.15	order	directive verb	65	3.69
order	directive verb	38	3.37	set	directive verb	59	3.35
set	directive verb	32	2.84	need to	obligation modal	53	3.01
review	directive verb	31	2.75	you	reader pronoun	41	2.33
develop	directive verb	31	2.75	pay	directive verb	34	1.93
need to	obligation modal	27	2.40	review	directive verb	29	1.65
use	directive verb	20	1.77	allow	directive verb	25	1.42
go	directive verb	19	1.69	add	directive verb	22	1.25
turn	directive verb	16	1.42	measure	directive verb	21	1.19

This table presents the top ten engagement markers used across the CEO letters of failed and active firms.

As presented in Table 4.8, various lexical elements and devices serve as engagement markers in both corpora. In the corpus of failed firms, the terms “key” (5.32 per 10,000 words), “see” (5.15 per 10,000 words), and “order” (3.37 per 10,000 words) rank as the top three most frequently used engagement markers. Meanwhile, in the corpus of active firms, “develop” appears to be the most frequently used engagement marker (4.14 per 10,000 words), followed by “order” (3.69 per 10,000 words) and “set” (3.35 per 10,000 words). Across the top ten engagement markers in both corpora, the heavy reliance on directives including, “see”, “set”, “review”, “develop”, “use”, “go”, “turn”, “pay”, “allow” and “measure” and obligation modals such as, “need to” indicate they key role these devices play in rhetorically positioning the readers and pulling them into the discourse. Meanwhile, the use

of the personal pronoun “you”, which is frequently employed by CEOs of active firms, underscores its pivotal role in constructing a writer-reader relationship and fostering a sense of shared responsibility and engagement. Taken together, these findings suggest that CEOs of both failed and active firms strategically use engagement markers to bring shareholders into the discourse, guide their interpretation of presented information and seek alignment. However, how these engagement markers are employed within the discourse, and the rhetorical functions they perform, vary across the two corpora.

4.7. Discussion and implications

The results reveal statistically significant differences in the distribution of interactional metadiscourse resources across both failed and active firms, providing valuable insights into the rhetorical strategies and communicative practices of CEOs when addressing shareholders during periods of distress and stability. At the overall distributional level, I find that CEOs of failed firms more frequently employ engagement markers, while stance features are more prevalent in the CEO letters of active firms. This finding suggests that the CEOs of failed firms tend to prioritise positioning and engaging stakeholders by relying heavily on engagement markers to draw them into the discourse and to mark solidarity in the face of financial difficulties. Whereas the CEOs of active firms appear to rely more on stance features to signal authority, prudence and credibility. Prior studies have suggested that the extensive use of hedges undermines the communicator's authority (Aerts and Yan, 2017), an overly assertive language can increase litigation risks when firms are facing reputational threats (Rogers et al., 2011), and excessive positive evaluative language without supporting evidence may be penalised by sophisticated investors (Tan et al., 2014), which may help account for the lower frequency of stance employed in the CEO letters of failed firms compared to those of active firms.

At the sub-categorical distributional level, self-mentions and engagement markers appear more frequently in the CEO letters of failed firms. In contrast, hedges, boosters and attitude markers dominate in the CEO letters of active firms. This observation highlights the distinct ways in which CEOs craft their communications during periods of financial uncertainty and financial stability, suggesting that financial conditions and organisational pressures shape CEOs' rhetorical choices. The analysis also reveals that, although similar interactional devices appear in both corpora, their contextual deployment and rhetorical function differ significantly.

A first area of divergence in the sub-categorical distribution of interactional metadiscourse features across both corpora concerns the use of hedges, with CEOs of active firms using them more frequently than those of failed firms. The observed lower use of hedges by the CEOs of failed firms suggests a rhetorical reluctance to emphasise uncertainty, as their employment can signal a lack of confidence in the CEOs' statements (Aerts and Yan, 2017), consistent with reactive impression management strategies aimed at persuading stakeholders and managing their interpretation of performance (Lee, 2020). Findings from the qualitative analysis further support this interpretation, in which hedges are employed in CEO letters of failed firms (e.g., "*A consequence of this may be that we achieve fewer first time planning consents, but the returns generated should be higher*") to soften the negative outcome by formulating a normalising account and balancing the disclosure of potential risk with reassurance, reflecting a reactive response aimed at persuading stakeholders, managing their future expectations, and mitigating reputational threats. In contrast, hedges are employed more proactively by CEOs of active firms (e.g., "*However, it is possible to illustrate what a feasible outcome may look like and where it will be derived from, assuming a supportive economy and no significant downturn in any of our major markets...etc*") to not only signal uncertainty but to strategically shape readers' expectations by constructing forward-looking scenarios that are open to negation, allowing them project credibility and claim protection in case circumstances change.

A similar pattern is observed in the use of boosters across both corpora, with boosters appearing more frequently in the CEO letters of active firms than in those of failed firms. The relatively lower usage of boosters in the CEO communications of failed firms suggests that CEOs of failed firms tend to be more cautious when making strong assertions, reflecting style-based impression management strategies through the careful selection and use of certainty expressions, seeking to repair legitimacy (Berkin et al., 2025). This is consistent with the insights from the qualitative analysis of boosters in the CEO letters of failed firms (e.g., “*Against that background these results demonstrate that we have a resilient business model*”), where boosters are strategically employed reactively to project stability and reassure stakeholders amidst uncertainty to restore legitimacy. In comparison, the CEO communications of active firms (e.g., “*Persimmon’s 2022 performance demonstrates our capabilities to deliver both strong financial performance and consistent build quality ...etc.*”) employ the same booster, which in this case is strategically used to persuade stakeholders of the firm's positive attributes proactively by presenting the firm's strong performance as a matter of fact rather than opinion, facilitating the projection of a credible image of authority and, consequently, maintaining legitimacy.

The distribution of attitude markers also reveals significant differences in their use across both corpora, with CEOs of active firms using them more frequently in their letters to shareholders than those of failed firms. The lower deployment of attitude markers in the failed firms corpus suggests that the CEOs of failed firms tend to avoid over-expressing their affective and evaluative stances toward firm performance. Instead, they selectively and carefully employ attitude markers to frame progress in a positive light, reflecting a reactive impression management strategy intended to strengthen the persuasiveness of the communicated message and restore threatened legitimacy (Aerts and Yan, 2017). For instance, attitude markers in CEO letters of failed firms (e.g., “*This is an important milestone and indicates that our strategy for the business is the right one*”) function to frame a specific corporate event as an “important milestone”, inviting stakeholders to share the CEO's

positive appraisal of the firm's strategic direction. This demonstrates the rhetorical role of the attitude marker "important" in persuading stakeholders and shaping their perceptions of organisational legitimacy and credibility. On the other hand, in the CEO communications of active firms (e.g., "*While we acknowledge our progress, it is more important to recognise that there is still much to do to improve diversity...etc*") the same attitude marker is used to express the CEO's affective stance and personal judgment on what should also be prioritised, aligning the firm with stakeholders' and broader social expectations. This thus reflects proactive impression management strategies, and exemplification in particular, aimed at projecting an image of a moral and trustworthy leader, constructing an ethos of responsibility and sustaining organisational legitimacy.

The employment of self-mentions, however, exhibits a contrasting pattern from that observed in hedges, boosters and attitude markers, as CEOs of failed firms appear to rely more heavily on marking authorial identity in their discourse than those of active ones. The extensive use of first-person pronouns by CEOs of failed firms suggests that they tend to prioritise making their messages more persuasive through personalisation efforts to build a personal ethos (Hyland, 1998), as the omission of personal pronouns in contexts of poor performance often leads to a disconnection between the communicator and the conveyed message (Aerts and Yan, 2017). This, therefore, reflects a style-based impression management strategy aimed at persuading stakeholders and mitigating reputational damage by using personalisation devices to soften negative disclosures and frame them as personal interpretations. For instance, the first-person pronoun in the CEO letters of failed firms (e.g., "*I believe that three factors contributed to this disappointing performance.*") is used alongside an epistemic verb of judgment, enabling the CEO to explain "disappointing performance" and ground it in their own personal belief, reflecting a reactive strategy intended to dissociate themselves from negative performance and deny accountability for it by shifting the blame into other factors to restore organisational legitimacy. Meanwhile, the employment of the first-person pronoun in CEO communications of active firms (e.g., "*I promise to report*")

progress on those in our normal open and honest way.") indicates that CEOs of active firms tend to adopt proactive impression management strategies through exemplification to demonstrate ethical behaviour, allowing them to express an authoritative authorial stance in their statement and present themselves as a competent, credible, and honest persona, which in turn marks solidarity with stakeholders and maintains legitimacy.

Interestingly, a similar pattern is evident in the use of engagement markers, which are more prevalent in the CEO letters of failed firms than in those of active firms. The overreliance on engagement markers by CEOs of failed firms suggests that they tend to focus on engaging with stakeholders, consistent with reactive and defensive impression management techniques directed toward restoring perceived legitimacy (Berkin et al., 2025), providing them with the means to construct a dialogic relationship with stakeholders and regain confidence by normalising and rationalising reputational damage. The qualitative analysis of the employment engagement markers across both corpora further supports this interpretation. For instance, in the CEO letters of failed firms (e.g., "You are no doubt aware that we've faced some recent criticism...etc.") the reader pronoun "you" functions as an engagement marker to mitigate the distance between the CEO and stakeholders by addressing them as informed discourse participants of the firm's "recent criticism". This reflects a strategic action aimed at managing negative impressions and at restoring organisational legitimacy reactively by directly involving stakeholders in the discourse and acknowledging accountability for an event while dismissing its negative implications. Whereas, the same engagement marker in the CEO communications of active firms (e.g., "You can see from these results that our business and our Warhammer Hobby are in good shape.") is used alongside another directive engagement marker "see" to rhetorically position stakeholders and direct their attention to the evidence of strong financial performance, suggesting that CEOs of active firms employ proactive impression management strategies to persuade audience of the firm's positive qualities, gain praise and sustain legitimacy.

Taken together, the analysis reveals that the CEOs of failed firms appear to employ reactive impression management strategies in their discourse to persuade stakeholders when their organisations' survival is threatened. On the other hand, the CEOs of active firms seem to employ proactive impression management strategies to persuade stakeholders, maintain legitimacy and present a confident, trustworthy, and positive image. These findings highlight that although CEOs of both failed and active firms employ similar interactional metadiscoursal features in their discourse, the rhetorical functions these linguistic devices perform and the contexts in which they are deployed vary across the two corpora.

The findings of the present study contribute to the literature on CEO letters and metadiscourse in several ways. First, this study's findings enrich theoretical understanding by highlighting the strategic use of reactive and proactive impression management strategies in CEO letters. This contributes to the extant literature on the legitimacy and impression management theories by demonstrating how CEOs manage stakeholders' perceptions during periods characterised by different financial performance and organisational pressures. In addition, the findings highlight how different impression management strategies are employed depending on whether the rhetorical objective and purpose of the communicated information is to maintain or restore organisational legitimacy.

Secondly, this research contributes to improving the understanding of how CEO communication and linguistic styles evolve in contexts of organisational success and failure. This is particularly critical for stakeholders as it sheds light on how stance and engagement markers are employed across firms with varying levels of financial distress and exposure to legitimacy threats. In particular, the results reveal subtle rhetorical and communicative differences between financially distressed and healthy firms, which may significantly assist stakeholders' ability to interpret CEO letters and better evaluate firms' performance through analysing CEOs' linguistic styles and whether the information is disclosed reactively or proactively. Hence, an improved understanding of this matter could significantly help stakeholders distinguish between distressed firms on the verge of failure and financially

healthy firms. The findings could also provide regulators with important insights into how CEOs use their discourse to strategically shape stakeholders' perceptions of firm performance and manage societal expectations and organisational legitimacy. Such insights could therefore inform regulatory and narrative reporting guidelines that encourage clearer and transparent disclosures.

Finally, the present study's findings offer several promising avenues for future research. Investigating how the rhetorical functions of interactional metadiscourse resources within CEO narratives of failed firms evolve, particularly as they transition through stages of stability, financial distress, and eventual failure, can be an avenue for future research. Another interesting avenue for future research could be to compare CEO letters with other narrative disclosures to assess whether interactional devices function differently across communication outlets. Further, investigating the impact of interactional metadiscourse markers in CEO narratives on capital markets, such as stock price reactions and credit ratings, is a promising avenue for future research. Collectively, these avenues would further improve the understanding of the rhetorical function of interactional metadiscourse in corporate communications, while highlighting the present study's contribution in laying the groundwork for future research.

4.8 Conclusion

The present study adopts a corpus-based approach to investigate the use of interactional metadiscourse resources in CEO letters of UK FTSE 350-listed firms from 2018 to 2022, comparing those of firms that failed with those of firms that remained financially stable and operational during that period. Interactional metadiscourse resources, including hedges, boosters, attitude markers, self-mentions, and engagement markers, play a crucial role in the construction of CEO letters, enabling CEOs to express their stance and address shareholders. Although these linguistic devices are prevalent in both failed and active firms, their frequencies and rhetorical functions differ significantly.

The outcomes reveal that the employment of all five interactional metadiscourse features across CEO letters of failed and active firms varies significantly in their distribution and rhetorical functions. Interestingly, I find that CEOs of failed firms focus more on projecting themselves in their discourse and on engaging stakeholders, relying heavily on self-mentions and engagement markers. In particular, I find that CEOs of failed firms tend to employ reactive impression management strategies to persuade stakeholders, manage their interpretations of performance outcomes, and restore perceived legitimacy.

The CEOs of active firms, by contrast, appear to focus more on using hedges, boosters, and attitude markers. I find that the CEOs of active firms are more inclined to employ proactive impression management strategies to gain stakeholders' support, maintain legitimacy, and construct a personal ethos of credibility and competence. These findings, therefore, suggest that although similar interactional resources may have been employed in both corpora, their rhetorical functions and contextual uses differ significantly depending on the firm's financial position and organisational pressures.

Appendix

Appendix

A4.1 List of sampled firms

Failed firms	Active Firms
1 Carpetright plc	1 Next plc
2 Cineworld Group plc	2 Burberry Group plc
3 Debenhams plc	3 Auto Trader Group plc
4 Avocet Mining plc	4 Games Workshop Group plc
5 Johnston Press plc	5 Greggs plc
6 NMC Health plc	6 Hays plc
7 McCarthy and Stone plc	7 Howden Joinery Group plc
8 Thomas Cook Group plc	8 Johnson Matthey plc
9 Interserve plc	9 Persimmon plc
10 Georgia Healthcare Group plc	10 Renishaw plc
11 Petropavlovsk plc	11 WH Smith plc
12 Studio Retail Group plc	12 4imprint Group plc
	13 Bellway plc
	14 Computacenter plc
	15 Cranswick plc
	16 Dunelm Group plc
	17 Hilton Food Group plc
	18 Kainos Group plc
	19 PageGroup plc
	20 Softcat plc

This table lists the sampled firms.

Chapter 5

Conclusion

The main aim of the current thesis is to examine the information content of CEO letters in annual reports of UK firms. CEO letters are important vehicles that CEOs use to qualitatively describe firms' performance and prospects, communicate their attitudes and values, and thus enact their leadership (Amernic et al., 2010, pp. 31-32). Such personal, signed narratives are an essential source of information for investors, whom they use and rely on when making investment decisions (Kiattikulwattana, 2019). Therefore, the information contained in CEO letters is influential, as narrative disclosures play a key role in shaping how various users interpret, assess, and respond to organisational performance (Michelon et al., 2022). In this thesis, I particularly investigate the informative, predictive, and rhetorical roles of the language employed in CEO letters to improve understanding of how their information content can signal underlying performance and influence reader interpretation. Drawing on quantitative and qualitative methods, I conduct three interconnected studies that investigate the relationship between information in CEO letters and firm performance.

In the first study, Chapter 2, I examine the relationship between the tone of CEO letters and firm performance. In addition, I investigate whether this association is moderated by CEO power. The results reveal that the tone of CEO letters and firm performance are significantly and positively related, suggesting that the tone conveyed in CEO letters reflects actual financial performance. However, interestingly, I find that this relationship becomes less pronounced as CEO power increases, suggesting that greater CEO power provides stronger support for impression management theory. Hence, when CEOs are more powerful, the tone of CEO letters may be more subject to impression management.

Moreover, in the second study, Chapter 3, I investigate the predictive value of textual information in CEO letters and examine whether it improves the understanding of firms' multi-horizon bankruptcy risks. The results reveal that including textual features derived from CEO letters, along with accounting, market, and macroeconomic data, significantly enhances the model's predictive performance, underscoring the explanatory power of CEO letters in predicting corporate failure. More importantly, I find that although all textual features derived from CEO letters contribute to the model's predictive performance and reliability, their informativeness and importance vary with the prediction horizon. For instance, among all textual features, the readability of CEO letters appears to contribute most to improving the detection of imminent bankruptcy risks. The latent representative topics in CEO letters, on the other hand, appear to be the textual features that contribute most to enhancing the identification of medium-term bankruptcy risks. Meanwhile, CEO turnover appears to contribute most to enhancing the prediction of longer-term bankruptcy risk, compared with other text-based features.

In the third study, Chapter 4, I examine the use of interactional metadiscourse markers in CEO letters, comparing those of failed and active firms. In particular, I investigate whether there are statistically significant differences in the use of stance and engagement markers across CEO letters of failed and active firms. I also analyse whether the communicative and rhetorical functions of these interactional devices differ across both corpora. The results reveal that CEOs of failed firms seem to rely more on personalising their discourse and engaging stakeholders through extensive self-mentions and engagement markers. In contrast, CEOs of active firms appear to use hedges, boosters and attitude markers more frequently in their discourse. More importantly, although CEOs of both failed and active firms draw on similar interactional devices, the communicative functions and rhetorical purposes that these devices serve vary across contexts. For instance, CEOs of failed firms seem to adopt reactive impression management techniques in their letters to persuade stakeholders, influence stakeholders' interpretation of performance outcomes, and restore legitimacy.

Meanwhile, the CEOs of active firms appear to employ proactive impression management strategies to persuade stakeholders, gain their support, project credibility, and maintain legitimacy.

Moreover, the findings of this thesis have significant practical implications, as follows. First, the results highlight for various stakeholders that the interpretive value and credibility of the tones employed in CEO letters appear to vary with the level of CEO power. Therefore, recognising that when CEOs are more powerful, the tone of the CEO letters may be more subject to impression management, helps various stakeholders, including regulators and investors, better interpret tone signals by accounting for the influence of CEO power and thus avoid misjudging a firm's performance.

Second, the findings underscore the key role of textual information in CEO letters in complementing financial ratios and thus in assessing multi-horizon bankruptcy risks. Such findings can inform different stakeholders interested in the early detection of financial distress about the incremental information that CEO letters offer beyond financial ratios, which they can incorporate into their evaluations. It also provides investors, banks and regulators with valuable insights into the informativeness and individual contributions of different textual features of CEO letters across multiple prediction horizons, which they can prioritise based on the prediction horizon of their interest to enable more informed decision-making. For example, investors interested in identifying firms likely to remain active may benefit significantly from tracking CEO turnover to improve capital allocation and investment decisions. In addition, banks, investors, and regulators could gain valuable insights by evaluating the readability of CEO letters to assess imminent bankruptcy risks, identifying and interpreting representative topics embedded in CEO letters to predict medium-term bankruptcy risks, and tracking CEO turnover to detect early warning signs of emerging distress. Therefore, the research findings on the predictive ability of CEO letters could greatly assist stakeholders in making more informed investment decisions, strengthening risk assessments and enabling timely interventions.

Third, the findings on differences in the contextual use of interactional metadiscourse markers across CEO letters from failed and active firms provide important insights into how CEO communication styles evolve in contexts of organisational success and failure. Thus, an improved understanding of how CEOs use their discourse to communicate firms' performance, convey their actions, and respond to organisational pressures helps stakeholders better interpret and evaluate the information in CEO letters. Regulators may also gain insights into how managers strategically use narrative disclosures, such as CEO letters, to manage societal expectations and mitigate legitimacy threats, and subsequently introduce regulatory and narrative reporting guidelines that encourage clearer and more transparent disclosures.

Despite the importance and practical relevance of the findings, the current research has several limitations that prompt further investigations. First, this thesis focuses solely on CEO letters, given the key role that the information they contain plays in how various corporate users interpret, assess, and react to corporate performance. Future studies may investigate the information content of other corporate communication outlets, which may also provide valuable insights into firms' performance and their likelihood of bankruptcy. Future research can also compare the information value of CEO letters with that of other narrative disclosures to determine which narratives are more important.

Second, when examining the relationship between the tone of CEO letters and financial performance, and the moderating impact of CEO power on this relationship, I use return on assets (ROA), one of the most widely used current firm performance metrics in accounting and finance studies examining corporate disclosure tone (e.g., Boudt and Thewissen, 2019; Martikainen et al., 2023; Mousa et al., 2022). Future research could extend this analysis by including future ROA and investigating the moderating effect of CEO power on the association between CEO letter tones and future ROA, to determine whether powerful CEOs' tones are primarily used for impression management or reflect a genuine managerial perspective on future performance.

Third, I investigate the differences in the use of interactional metadiscourse features in CEO letters, comparing those of failed and active firms to examine how managerial communication styles differ in contexts of financial distress and stability. However, I do not account for how the rhetorical functions of interactional linguistic devices evolve as they transition from stability to distress to failure. Therefore, studying how the communicative and linguistic styles of CEOs of failed firms evolve throughout time can be a promising avenue for future research.

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